SYNTHMAVEN

3.5.1 · FM-1 user guide

Melodee-based firmware · USB web and iPad editor

Programs, layers, microtuning, controllers and the complete engine and sequencer reference.

Back up before flashing. Installation resets user settings and musical storage. Bluetooth is not included in this USB-first build.

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About

SynthMaven 3.5.1 is a Melodee-based synthesizer and sequencer for the M-VAVE FM-1. It combines Melodee's panel graphics, sequencer, USB audio and native instrument editing with SynthMaven HYBRID/Vital translations, DX7 sounds, four-part programs, global microtuning, scenes, setlists and MIDI Learn.

Melodee is by Kerem Kilic (Ellic Studio), based on Felucca by Leo Kuroshita / Hügelton Instruments. SynthMaven also retains applicable Clipper/3dSam-derived work and third-party notices. See the release licence archive for the source attributions. Report this build's problems to SynthMaven. This project is not endorsed by M-VAVE or the upstream authors.

Programs, layers and tuning

A **sound preset** is one instrument sound. A **program** contains four sounds and their mix, split ranges and scenes. An **editor library entry** is a computer-side sound file; importing it does not automatically save a program. Melodee sequencer projects are a separate storage format. Do not restore a SynthMaven 3.0 whole-device backup into 3.5.1; keep it for rollback with its matching firmware.

Play and change sounds

1. Press HOME. Turn ALGORITHM to change the selected part's engine.

2. Turn PRESETS to browse that engine's sounds, wrapping at the end of its bank.

3. Turn SELECT to choose one of the four parts. With layering off, play the selected part. With layering on, notes reach every unmuted part whose split range includes the note.

4. Turn KNOB 1-4 for the displayed sound controls. MASTER controls both physical output and USB recording level.

The four parts share the processor, memory and voice resources. An engine's maximum voice count is not a separate guaranteed allocation for every part. Mono backing parts replace their previous note when you play a chord; complex patches and effects can reduce available polyphony.

Four-layer Home and independent drum Home

HOME switches between the four-layer program overview and the independent drum machine. Layer colors are L1 red, L2 pink, L3 light blue and L4 gray, with translucent card backgrounds and a white selection outline. Sequencer tracks use the same order. Layer numbers remain visible so color is not the only identifier.

On drum Home, SELECT or ALGORITHM chooses one of eight lanes. PRESETS or KNOB 1 selects a step. Turn KNOB 2 right to enable that step, or left to clear it. KNOB 3 sets the selected lane level; KNOB 4 changes tempo. PLAY starts or stops the independent drums. All four sound layers remain available to the keyboard. Stop playback before saving or loading programs.

Shared touch editor and playback

1. Connect the FM-1 by USB in COMPUTER mode. In SM FM1 on iOS, choose USB cable in the connection wizard. A page marked Mock device is a demonstration and cannot play the FM-1.

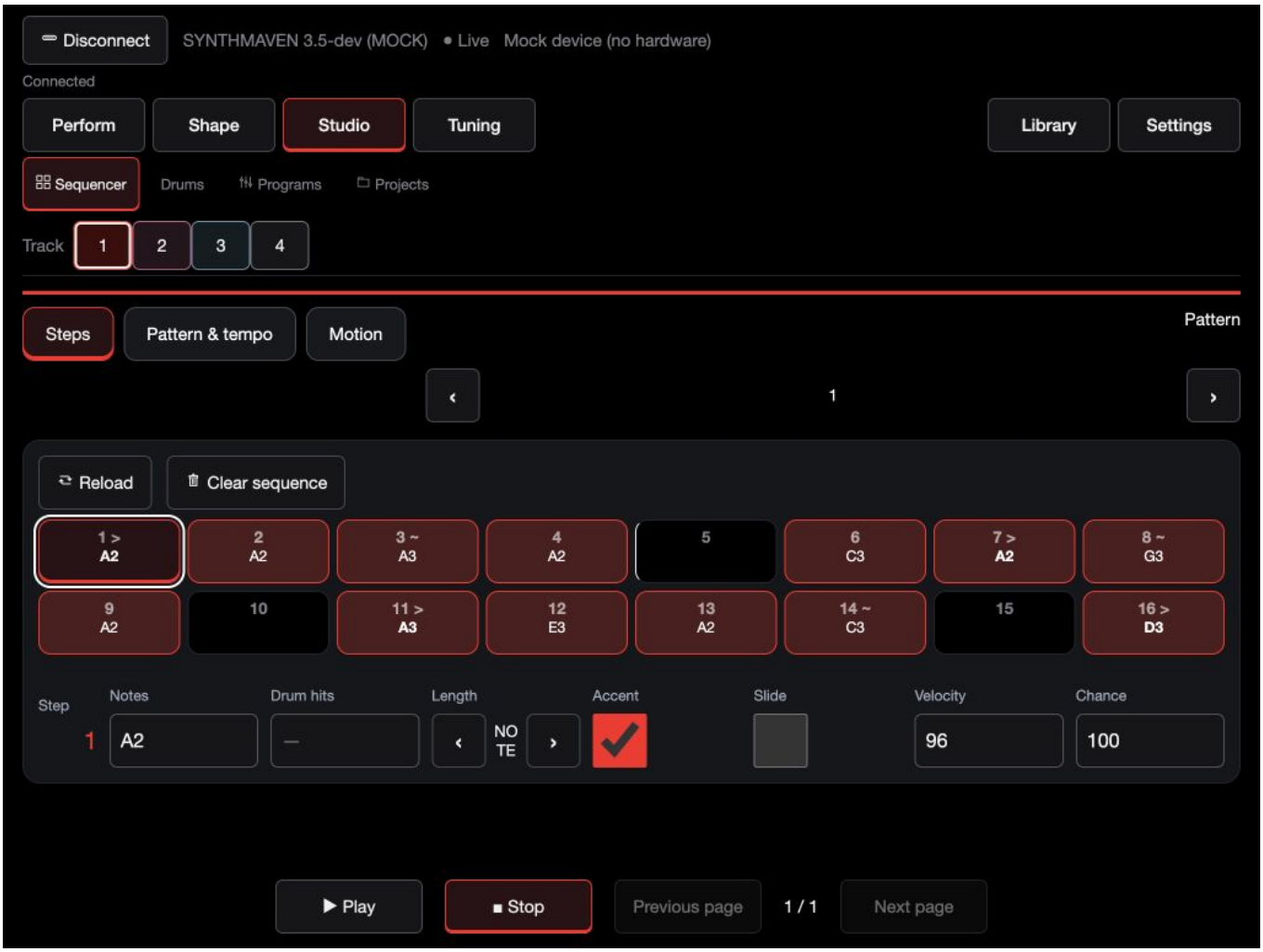
2. Open Studio → Sequencer, choose a sound part and enter a pattern. Select a numbered step to edit its notes, velocity, timing, accent, slide and chance. Play sends a start request to the connected FM-1; Stop releases sequencer playback. An empty pattern makes no sound. Check MASTER and the part's mute/level if a populated pattern is silent.

3. Open Studio → Drums for the independent eight-lane pattern. Play and Stop here control drums independently of the melodic sequencer. Tap a step to toggle it; choose a lane to adjust level or velocity. Audition plays that lane on the FM-1. Lanes 1-4 / 5-8 and step banks keep the pads touch-sized.

4. Perform → Layers contains the four-strip mixer: select the part, turn its pan knob, move its level fader or mute it. Expand Layering & keyboard splits for inclusive lower and upper note limits.

5. The editor uses screen pages instead of scrolling. Use Previous page / Next page to reach additional controls. Play and Stop remain available in the bottom navigation for Sequencer and Drums. Paging, hiding panels and color changes never alter a sound.

The iOS home-screen name is **SM FM1**. The app icon and FM-1 startup image use the Hybrid artwork; the editor itself has no logo. The web and iOS interfaces share the same controls and USB protocol. Bluetooth is not included.



Sequencer steps and aligned controls. Preview shown in mock mode.

Layer switch and mixer



Hold GLO for two seconds.

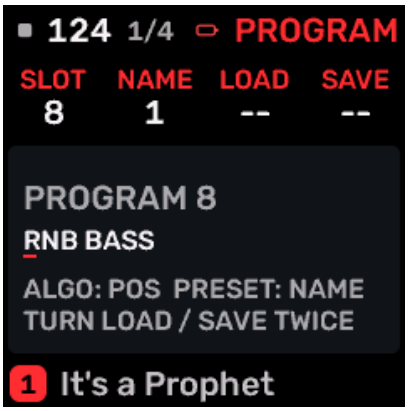


Hold EDIT for two seconds.

Hold GLO for **two seconds** to toggle layering. The screen shows **Layered ON** or **Layered OFF**; this gesture does not latch a held-button mode. The GLO pages also contain the layer switch.

Hold EDIT for **two seconds** to open and retain LAYER MIX. KNOB 1-4 control the four part volumes. SELECT chooses the part; PRESETS controls its pan; ALGORITHM sets its lower split point from A1 to A6. The editor exposes both lower and upper inclusive split limits. Press HOME to leave the mixer. Undo/redo use EDIT with OCT-/OCT+.

Save or load a named program



Eight named program slots.

1. Press SAVE to open PROGRAM.

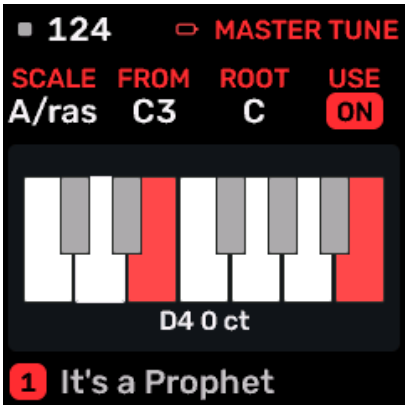
2. Turn SELECT or KNOB 1 to choose P1-P8; the panel shows the slot name.

3. Use KNOB 2 or PRESETS to change the highlighted name character; ALGORITHM moves the character position. Turn KNOB 3 right twice within 1.5 seconds to LOAD. Turn KNOB 4 right twice within 1.5 seconds to SAVE the four-part program under the edited name. The first turn only arms the action; turning left cancels it.

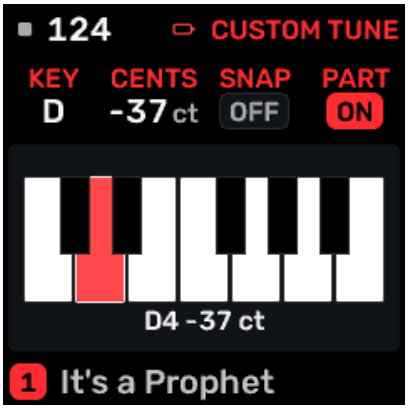
4. The editor provides the optional tuning override when saving. Device saves follow global tuning by default.

A fresh installation starts on P1 with layering enabled. The eight factory programs are the owner's final FM-1 edits, captured with all active native patches, part parameters, effects, splits and scenes: P1 DX7 STAGE, P2 E. Organ, P3 RHODES EP, P4 DISCO LEAD, P5 VINTAGE LEAD, P6 LEAD GUITAR, P7 ARAB STRINGS and P8 RNB BASS. The bank is a verified copy of the latest saved device programs, including the updated P1 and the already named P2 E. Organ. Saving replaces that slot's factory fallback. Autosave remains available in GLO → USB / SAVE and uses deferred, stopped-and-idle saving. Always save deliberately before switching off.

Global chromatic microtuning



Tap SEL for master tuning.



Hold SEL for three seconds for custom tuning.

Tap SEL (SCL in this guide's upstream labels). SELECT or KNOB 1 chooses the scale; KNOB 2 chooses the lower octave boundary C0-C8; KNOB 3 chooses the root; KNOB 4 enables master tuning. For C3, notes below C3 remain in equal temperament. Scale names are grouped as **Western/** and **Arabic/**. The 14 Western scales from SynthMaven Oriental include major and minor modes, pentatonics, blues, whole tone and diminished; they use equal-tempered notes with scale-aware quantization. The 46 maqam settings apply offsets to their original chromatic keys rather than renumbering the keyboard into scale degrees.

Hold SEL for three seconds to open CUSTOM TUNE. KNOB 1 selects the key, KNOB 2 changes its offset in **10-cent steps**, KNOB 3 controls Quantize all, and KNOB 4 controls Apply master tuning for the selected part. The latter is enabled for all four parts by default. Global tuning survives ordinary program changes; a program saved with an explicit tuning override can replace it.

In the editor, tap a key to select it, then use the horizontal wheel from -100 to +100 cents in **1-cent steps**. Double tap an untuned key to set -50 cents; double tap an adjusted key to reset it to zero. The selected key displays its current cents. Adjusted keys are red, ordinary white/black keys retain their colors, and keys outside the scale are gray. Quantize all applies the selected root/scale to all four parts, including incoming MIDI notes.

Chords, scenes and setlists

Hold ENV for **three seconds** to open CHORD; a short press opens the envelope pages. Chords use the selected scale. When master microtuning applies, incompatible chord choices are hidden and open-fifth / neutral harmony choices remain available. Below the octave boundary, or for a part bypassing master tuning, ordinary equal-tempered harmony remains available.

A program contains four named scenes: compact snapshots of levels, pans, mutes, effect sends, HYBRID macros and tuning. Capture a scene in the editor's Programs & scenes section, then save the program to retain it. A setlist is an ordered list of up to 32 references to P1-P8; repeated references are allowed. Assign Scene or Setlist Next/Previous using MIDI Learn.

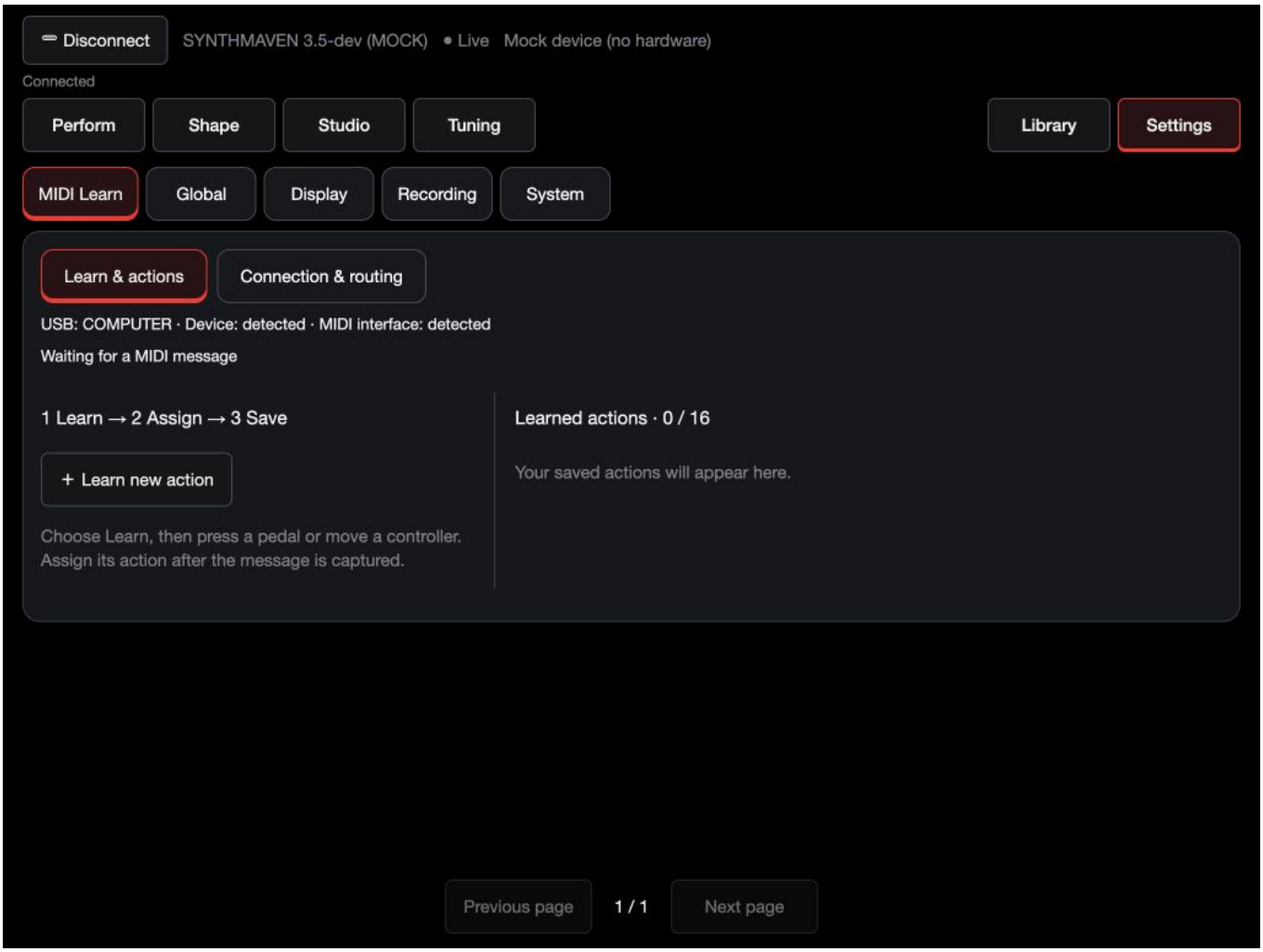
On HOME, hold OCT- for three seconds to load the previous program, or OCT+ for three seconds to load the next. Each hold switches once; release before repeating. Short taps change octave. Holding both retains the existing octave-reset/update gesture. In menus and dialogs, OCT-/OCT+ retain their back/confirm functions.

USB editor and MIDI Learn

On the FM-1 display, category prefixes are abbreviated W/ and A/ to preserve space for scale names. The editors retain Western/ and Arabic/.

The web editor and SM FM1 iPad app share the same dark, touch-friendly workspace. The iPad app supports landscape orientation. Perform provides quick sound, layer and tuning controls; Shape edits sound engines; Studio contains the sequencer, independent drums and Programs. Transport controls are at the bottom. Section tabs and page buttons keep controls accessible without scrolling.

Programs contains program save/load, scenes and setlists. The layer mixer and keyboard splits live in Perform → Layers. Library separates Device sounds, My library and User slots. Settings separates MIDI Learn, global controls, Display, Recording and System. Changing a view never changes sound settings.



Learn first, assign the captured message, then save. Preview shown in mock mode.

1. Set HOME-held menu → USB ROLE to COMPUTER.

2. Connect a USB data cable to Chrome/Edge on a computer, or to the iPad running SM FM1 Editor. Use a powered hub when connecting a controller and FM-1 together.

3. Tap Connect. The iOS USB wizard selects matching MIDI input/output ports together. A port may still be named Melodee FM-1; the editor verifies firmware capabilities.

4. Perform opens first. Program Up/Down and Scene Up/Down show their current names. Quick controls, Layers and Tuning switch the lower panel without hiding the program/scene controls.

Quick controls follow the selected part. FM6 uses its own six operator envelopes: its quick panel exposes native modulator envelope timing and velocity response instead of inactive generic ADSR controls. Use the 6-OP FM page for individual operator envelopes and DX7 import. HYBRID/Vital imports belong to sound editing, not the Perform screen.

1. Open Settings → MIDI Learn. In Connection & routing, select Controller input if routing an external controller through your computer or iPad.

2. In Learn & actions, tap Learn new action, then press the pedal or move a controller. The editor displays the captured message.

3. Assign an action such as Patch Up, Program Next or Scene Next. A captured CC can also control a sound parameter. Choose the target part and repeat suppression.

4. Tap Save action to commit the assignment to the FM-1. Operate the control again; the status shows the received message and whether it matched.

5. Add up to 16 assignments. The learned-action grid shows four at a time; use Previous actions and Next actions to access the rest without scrolling.

6. Relearn captures a replacement message while preserving the saved mapping until Save. Cancel keeps the old mapping. Unlearn removes that assignment.

7. Duplicate suppression accepts 0-2000 ms. A switch release rearms an intentional tap; zero disables suppression.

For a controller plugged directly into the FM-1, select **PEDAL** USB role and use a powered adapter or hub. Status distinguishes USB device detection from a usable MIDI interface. The same USB port cannot simultaneously host a pedal and connect as a computer peripheral: return to **COMPUTER** for editing or updates. Controller identity selects a supported profile when recognized; unknown devices retain generic MIDI. MCU/HUI mode and virtual MIDI port can be overridden in the editor. Device-specific modes still depend on the controller's own configuration.

USB host and powered-hub setup

The device menu calls this **USB ROLE: PEDAL**; it is USB MIDI host mode for keyboards, pads and pedals. It is not limited to foot pedals. COMPUTER is the device/peripheral role used by the editor and installer. Open GLO → USB / SAVE (7/10), then turn knob 3 for USB ROLE. Host mode is not retained across restart.

Controller directly into the FM-1

1. Stop playback and disconnect the FM-1 from the computer or iPad.

2. Power a compatible USB data hub using its own supply. Connect the MIDI controller to a downstream USB port on that hub.

3. Connect the hub's upstream connection to the FM-1 through a suitable USB host/OTG adapter. A charging-only splitter is not a data hub. The firmware cannot supply USB VBUS power; the controller needs its own power or the hub's supply. Follow the hub manufacturer's upstream-power requirements.

4. On the FM-1, open GLO → USB / SAVE (7/10), then turn knob 3 to change COMPUTER to PEDAL. Play a note or press the pedal. Use a class-compliant USB MIDI 1.0 controller; devices requiring a computer driver are not supported.

5. To edit or update again, change USB ROLE back to COMPUTER, disconnect the host adapter and reconnect the FM-1 to the computer/iPad with a USB data cable. Reopen the editor connection.

One hub is supported. Hub power does not guarantee compatibility: USB-C role negotiation and the adapter wiring still matter. A USB-C to USB-A host/OTG adapter with a suitable data hub can avoid some USB-C role conflicts. Device detection is distinct from finding a usable MIDI interface. If there is no response, check controller power, data cable, role, MIDI mode and channel; reconnect or toggle the role after correcting the setup.

Editor and controller together on an iPad or computer

1. Connect the powered hub upstream to the iPad/computer. In this arrangement the iPad/computer is the host.

2. Connect both the FM-1 and controller to downstream hub ports. Keep the FM-1 in COMPUTER mode.

3. Open SM FM1 or the web editor, connect the FM-1 MIDI input/output, then choose the controller input under Settings → MIDI Learn → Connection & routing.

4. Use Learn → Assign → Save to create mappings. Confirm incoming-message and match feedback. Save the program separately after sound edits.

The FM-1 cannot be USB host and an editor-connected USB peripheral at the same time. Direct-host mappings can be prepared in COMPUTER mode, then tested in PEDAL mode. Bluetooth is excluded from 3.5.1.

Sound editing, filters and modulation

1. Choose the part in Perform, then open Shape → Sound. Use the left/right engine and preset arrows to choose a starting sound.

2. Tone, Filter, Envelopes and Modulation use two adjacent cards. Page buttons within a card expose more controls without changing the selected part.

3. On Filter, adjust the model, cutoff, resonance and envelope amount where supported. Drag the filter preview to change cutoff/resonance. The graph is a response sketch, not a measurement of every engine's filter.

4. On Envelopes, change attack, decay, sustain and release and inspect the parameter preview. FM6 uses native operator envelopes; use 6-OP FM for individual operators. The graph is not streamed audio.

5. On Modulation, choose a template such as Velocity → brightness, Pressure → vibrato, Wheel → vibrato or Envelope swell. A free route is required; occupied routes are preserved. HYBRID templates are staged in its native draft: apply the draft to the part.

6. Open Shape → Effects for the effect rack. FX Send pages are Send FX 1, Send FX 2, Send FX 3, Routing and Sends. Open Shape → Motion for paged LFO and arpeggiator controls.

7. Save the edited sound to a user slot or library entry, or save the four-part program in Programs. Browsing and editing do not replace an explicit program save.

Custom scales and Scala files

1. Open Tuning and select the desired root and starting scale. Tap a keyboard key, then move the fine-tuning wheel in 1-cent steps. Double-tap toggles an untuned key to -50 cents or resets a tuned key to zero.

2. Enter a scale name in the custom-scale library and save it. Named editor scales are stored locally in that browser/app; keep exported files as backups.

3. Use Import Scala to choose a .scl file. The importer accepts compatible octave scales with up to 12 notes, preserving the chromatic-key mapping and supported -100 to +100-cent offsets. Unsupported intervals or ambiguous mappings are rejected rather than silently changed.

4. Recall a saved scale to apply its tuning. Export Scala writes the current scale when its pitches are ascending and its root has zero offset. Select the musical root separately.

5. On the FM-1, choose a preset scale with SEL, then hold SEL for three seconds to fine-tune it in CUSTOM TUNE. Editing copies the preset into the custom tuning slot; KNOB 2 adjusts the selected key in 10-cent steps. The factory scale remains available.

HYBRID editing and Vital import

1. Open Shape → Native / import. Select HYBRID / Vital, then choose Read part to edit the current native patch, or Load Vital to open a .vital JSON file.

2. Review the translation report. Custom wavetables, spectral processing, unison, effects and unsupported routes are not reproduced. If no native waveform matches, sending is disabled until you explicitly allow native-wave approximation.

3. Expand oscillator, filter, envelope or LFO sections to edit exact values. Under Modulation, select M1-M4 as a source, choose a destination and amount to assign a macro. Multiple routes may share a macro.

4. Choose Apply to selected part. Draft controls do not change the device before this step. Return to the original part if you changed the selection while editing.

5. Save to the editor library for a single sound, or save a program to preserve all four parts. Macro assignments are native patch data and are saved with the sound/program.

This first 3.5.1 build is **USB-only**. Bluetooth is a separate experimental adaptation and is not enabled by this release.

Before you install

You need **Chrome or Edge**, a **data USB cable** plugged straight into the computer (not a charge-only cable or hub if you can avoid it), and the FM-1 powered and connected.

• Switching to another custom firmware? Choose **Back up and install TransitFM1** in the installer first. Once the FM-1 shows **TRANSIT**, use the next firmware's own installer. Use TransitFM1 **0.1.3**; 0.1 has startup and MIDI stalls, and 0.1.2 was withdrawn. **For Baud Girl, install official V15 first, then use Baud Girl's installer**: its image can exceed TransitFM1's write range.

• **Every SynthMaven firmware installation resets user settings, programs and other musical storage to factory state.** Export a complete backup before installing, even when updating from this version. Keep older backups with their matching firmware; they are not a migration format. The 3.5.1 shared sound bank has 384 slots (U001-U384), separate from the eight four-part program slots.

• Projects, templates and presets from Melodee 1.0 or later load. Anything saved before 1.0 shows **OLD FORMAT** and stays untouched until you choose to replace it. If you still have those, open and save them with 1.1.0 before updating. A backup the device cannot read is refused before anything is written.

• Melodee is a field-testing release with **no warranty**. Use at your own risk.

Installing

Open the web installer in Chrome or Edge, connect the FM-1, and press Install. Leave the cable connected until writing finishes. When it succeeds, the FM-1 reports Melodee's product identity (for example `FM-1_9xxxx`).

Current builds use **both cores**. Voice limits are shared and depend on the chosen engines, effects and four-part workload; per-engine maxima are not guaranteed simultaneously. Single-core builds have been retired. If an update leaves the unit in the loader, install a working package again using the recovery steps below.

From a terminal you can also run `tools/fm1_install.py` (see the project README).

Getting around

One navigation language everywhere:

• **SELECT** turns the pages of the open section. On Stage it selects one of the four program parts.

• **KNOB 1** moves across (or edits a value). On list pages, **KNOB 2** moves the row and **KNOB 1** sets its value.

• **OCT+** enters, confirms, or opens a sheet. **OCT-** goes back or cancels, closing the innermost thing first (a question, a popup, a sub-screen) and only then returning to Stage.

• On HOME, short OCT taps shift octave and a single OCT hold for three seconds changes program; both together reset octave. In menus they are Enter / Esc.

• Named values often open a picker. Hold **HOME** and set **PICKER POPUPS** to **OFF** to edit those names on the knob instead. The choice is remembered. Sheets, and lists you open yourself, stay as they are. Short pages (CHORD, ENV DEST, LFO DEST, LFO 2, ARP 2) put their settings straight on the four knobs.

Holds and locks

SynthMaven shortcuts have fixed thresholds: GLO for two seconds toggles layering; EDIT for two seconds retains the four-part mixer; ENV for three seconds opens chords; SEL/SCL for three seconds opens custom tuning. A short press must not flash a long-press screen. FX retains the Melodee performance controls. Hold HOME for settings, SEQ for the pattern grid and REC to capture. See the program/tuning section above for the changed knob functions.

Sheets and undo

Hold **OCT+** on pages that have a sheet (Stage, sound pages, NOTES, PATTERNS, SONG, and others) to open actions such as Init sound, Copy to…, or Sound designer.

Undo / redo:

• Hold **EDIT** and press **OCT-** to undo. On NOTES, that undoes the last step edit first. Otherwise a finished recording take is undone first (**UNDO RECORDING**), and if there is none, the last sound or pattern load (or Capture).

• While **EDIT** is held, **OCT-** undoes further and **OCT+** redoes. On NOTES those keys stay on the step edits. Elsewhere **OCT+** redoes the take (**REDO RECORDING**) before the load history.

• A take keeps one undo: recorded notes, manual steps, length, division, swing, gate and that track's motion, put back in one step. Playback can keep going. It is refused while that track is still recording (**FINISH RECORDING**). A newer take replaces it. Switching pattern or engine, or loading a project, drops it. Other tracks stay as they are. If the snapshot cannot be kept, the screen says **NO RECORD UNDO MEMORY** or **UNDO MEMORY FULL**.

• Sound and pattern loads keep eight levels. A new edit clears redo. Recording, changing track or bank, or changing loop length starts a fresh step-edit history.

• If a song or chain is busy, the device may ask you to stop first (**STOP TO UNDO**).

The panel

Step by step

1. Start with MASTER low, connect headphones or speakers, then raise MASTER to a comfortable level. MASTER also affects USB stereo recordings.

2. Press HOME and select the part with SELECT. Turn the knob whose on-screen label matches the parameter you want to change.

3. Use short button presses for normal pages. Hold EDIT/GLO for two seconds or ENV/SEL for three seconds only when you want the documented secondary function. Release before repeating a shortcut.

Control	Job
Piano Keys	Play notes on the selected track. In layers and grids they also select patterns, effects, roots and drum lanes.
KNOB 1-4	Edit the four controls on the open page, or Stage's four sound knobs.
MASTER KNOB	Overall output volume.
ALGORITHM KNOB	On HOME, change the selected part's engine. On LAYER MIX, set its lower split point. On other pages, select a track.
PRESETS KNOB	Browse sounds; on engine EDIT pages, jump between sections (or pick the operator / line); on NOTES, zoom.
SELECT KNOB	Turn the pages within the open section, both ways. On Stage, select a program part. On PROGRAM, select one of eight slots. On MASTER TUNE, choose the scale.
FX SCL ENV LFO EDIT GLO ARP SEQ	Open that family's pages; press again for the next page.
HOME	Return to HOME. Tap again for independent drums. Hold for the settings menu.
SAVE	Tap for PROGRAM; SELECT chooses a slot, LOAD recalls it and SAVE opens name entry. Other sound/project pages are retained for the Melodee sequencer.

Control	Job
PLAY/STOP	Start and stop the tracks in use.
REC	Arm or disarm the selected track. While playing, recording starts at the next pattern start and continues across loops until you disarm or stop. Hold to capture (or take a jam on SONG).
OCT- OCT+	On Stage: shift octave (both together reset). In menus and everywhere else: go back / enter or confirm.

Key and button lights share one grammar: **dark** nothing there, **dim** something there, **bright** happening now, **slow pulse** waiting (for example REC armed but stopped, or a pattern queued for the loop end). The menu's **LIGHTS** sets how bright idle indicators are.

Stage

Step by step

1. Press HOME until the four-part overview is shown; turn SELECT to highlight the part you want to play or edit.

2. Turn ALGORITHM to choose its engine, then PRESETS to choose a sound within that engine. Audition with the keys.

3. Hold GLO for two seconds to compare the selected part alone with the layered program. Press HOME again for independent drums; return HOME to the four-part overview.

HOME opens the four-part program overview. All four parts remain visible with their sound names, levels and keyboard ranges. SELECT chooses a part, ALGORITHM changes its engine and PRESETS browses sounds within that engine. The four assignable sound knobs edit the selected part; turning one displays its current value. Tap HOME again for the independent drum machine, then again to return. Hold OCT- or OCT+ for three seconds on HOME to load the previous or next saved program.



Tracks in use: sound, pattern, step dots.



Notes you play sit in front of the live waveform.



Turning a knob shows what it shapes.

The selected track's four **sound knobs** stay in the bottom strip. Turning a knob updates its value there; the Stage display remains visible. What they do depends on the engine:

Engine	KNOB 1	KNOB 2	KNOB 3	KNOB 4
Prophet	CUTOFF	RES	filter env amount	amp release
FM6	MLVL	MRAT	MEG	DTUN
CZ-1	WAVE	DCW	DETUNE	VIBRATO
SID	WAVE	PW	CUT	RES
Drum	TUNE	TONE	DECY	SNAP

On a Drum track, lanes light as they hit. Hold **OCT+** for the Stage sheet: Init sound, Save as user…, Favourite, Undo last load, Sound designer, **Add track** and **Remove track**.

The second HOME screen shows the independent eight-lane drum pattern. The drum part is separate from the four melodic parts and shares tempo and the processor's audio capacity. Use Studio → Drums for the full pattern grid, lane levels and velocity controls. The menu's legacy HOME SELECT preference does not override SELECT's part-selection function in the SynthMaven program overview. GLO SELECT retains its assigned secondary function.

Tracks

Step by step

1. Hold EDIT for two seconds, then release it: LAYER MIX remains visible.

2. Set part 1-4 volume with knobs 1-4. Turn SELECT to the part to pan, then PRESETS to position it left or right.

3. Turn ALGORITHM to set the selected part's lower split point. For an exact upper limit use Perform → Layers in the editor. Play notes on both sides of the boundary to check the split.

4. Save the four-part program using SAVE → PROGRAM. A sound preset alone is not a backup of the complete mix.

One to four tracks, one instrument and sound each. A factory program starts with four. **SELECT** on HOME selects among the four program parts; the screen and its knobs take that track's color. In the SynthMaven theme the four tracks are red, pink, light blue and gray. Knob 1 is the track color; the other three knobs are shades of it.

• The eight factory programs already contain four parts. On HOME, hold OCT- or OCT+ for three seconds to move between programs. Use Perform → Layers in the editor to mix or mute their parts.

• **Remove track** removes the selected track after a question, and only while stopped (**STOP FIRST** otherwise). The tracks after it move down, taking their sounds, patterns, song sections, motion and recordings with them. The last remaining track cannot be removed.

• Stage, the mixer, PATTERNS, SONG and GLO show the tracks in use. Projects and the template keep their own track count. A project saved before this version keeps all four.

• Each track carries its own engine, sound, patterns, transpose, effect sends, modulation and voice settings. ROOT, SCL, QNT and MPC degree are shared by the project.

• Hold **EDIT** for two seconds for the four-volume mixer. **GLO** opens device-wide settings; its LAYERING page controls layered playback.

• Sounding voices share one **BUDGET** (**GLO** → TUNE/CLOCK, **KNOB 2**): 8-32 points, default 18, remembered with the device. An idle track reserves nothing. FM6 uses 1 point per voice (up to 16). Prophet uses 2 per voice (up to eight voices a track). Most other engines use 2; the ensemble Strings engine uses 4 to leave processing headroom. These are shared limits, not four independent allocations. Melodee runs at the boot loader's 360 MHz and does not change the clock. A unit updated while it was still at 320 MHz keeps that clock until the next power-off, and the allowance is smaller until then.

Sounds

Step by step

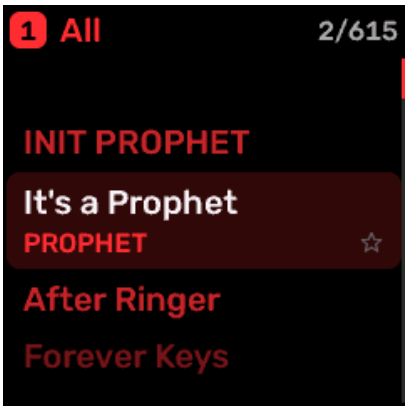
1. Select the part on HOME and choose its engine and starting preset.

2. Press EDIT for the engine's sound pages; tap again or turn SELECT to reach the required page. Match knobs 1-4 to the four on-screen labels.

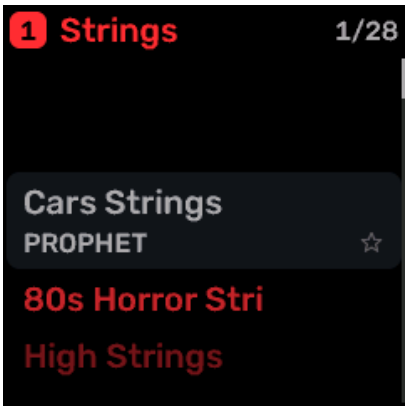
3. Change one parameter, play a note, and compare before changing the next. The tables below explain each page's parameters and engine-specific behavior.

4. Use SAVE → USER for a single sound, or SAVE → PROGRAM for the complete four-part setup.

Turn **PRESETS** (or open SAVE → PRESETS) to browse. **KNOB 1** chooses the list and **KNOB 2** (or keep turning PRESETS) chooses a sound. Each engine's factory presets come first, then the shared user slots that use that engine.



Browser: list on KNOB 1, sound on KNOB 2.



Category lists: BASS, LEAD, PAD, FAV, RECENT.

Control	What it does
KNOB 1 LIST	Choose the list - categories (BASS, LEAD, PAD, …), FAV, or RECENT.
KNOB 2 / **PRESETS**	Scroll sounds. While the list moves fast, the sound loads when the knob rests.
KNOB 3 FAV	Mark or unmark the current sound as a favourite (device favourites survive power-off).
KNOB 4 ENGINE	Change the engine and start from its **init** sound (dry, nothing changed): INIT PROPHET, FM6 INIT VOICE, INIT TONE, INIT SID, or the 808 kit.
OCT+ / **OCT-**	Keep the browsed sound / revert to the one you had.

• Knobs accelerate on a spin; MENU → **KNOB ACCEL** OFF keeps every detent one step.

• Use **EDIT** + **OCT-** to undo the last load. Loading a sound never changes your patterns.

• Hold **OCT+** on a sound page for the sheet: Init sound, Save as user…, Favourite, Undo last load, Sound designer. On Stage that sheet also has **Add track** and **Remove track**.

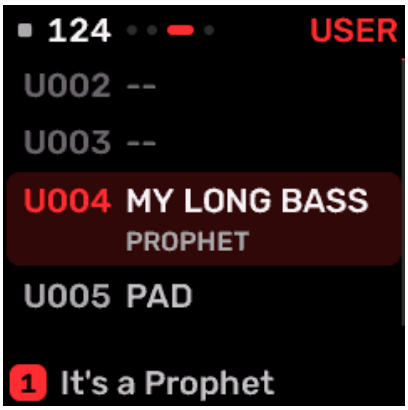
Saving sounds

Step by step

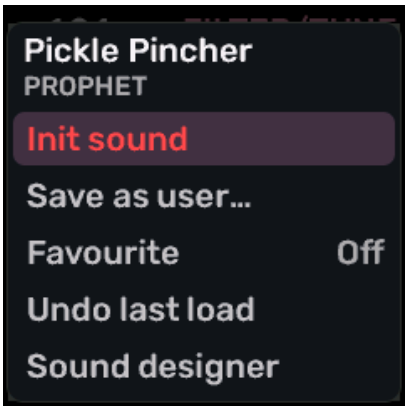
1. Stop playback, then tap SAVE repeatedly to reach USER. Choose the destination slot with knob 1 or knob 2.

2. Press OCT+ for the action sheet, select Save here, and confirm with OCT+. Enter the name with the keyboard or cursor/character knobs shown below.

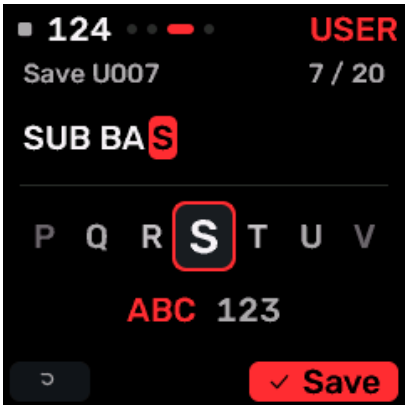
3. Confirm the name with OCT+, or cancel with OCT-. To recall it later, select the slot and choose Load. Saving here stores one sound, not the whole program.



USER slots; OCT+ opens the slot's sheet.



Sound sheet (hold OCT+ on a sound page).



NAME: type with the keys.

Stop, tap **SAVE** repeatedly until **USER** appears, pick a slot with **KNOB 1**/**KNOB 2**, press **OCT+** for the sheet (Load, Save here, Rename, Erase), choose **Save here**, then **OCT+** again. Name it on the NAME screen.

• Every engine shares one bank, **U001-U384**. A slot can hold a Prophet program, an FM6 voice, a CZ-1 tone, a SID sound, a drum kit, or any mix of those. **SAVE** → **USER**, **FM6** → **STORE** and **P5 STORE** are that same bank. PRESETS lists a user sound under its own engine's factory tones.

• Saving keeps the native voice and the sound's envelopes, modulation, voice controls and spread. Loading leaves the track's mix, effects, scale, arpeggiator and notes alone.

• **Import here:** a DX7, CZ-1, Prophet or Melodee preset sent over USB opens the USER list. Pick the destination with **KNOB 1**, press **OCT+** and choose **Import here**. A DX7 cartridge needs 32 consecutive slots from there. Occupied slots ask before they are replaced. **Cancel import** leaves the slots unchanged. Playback must be stopped. An import does not load a track.

• **Bulk presets:** hold **OCT+** and turn **SELECT**, **KNOB 1** or **KNOB 2** to select a range of USER or STORE slots. Release, then tap **OCT+** for **Send selected** (USB SysEx), **Erase selected** (one confirmation) or **Clear selection**. Empty slots are skipped. A single USER slot can also **Send**.

NAME screen

Control	What it does
White keys	Type letters, multi-tap like a phone keypad (press again within the window to step A → B → C).
Black keys	Switch mode: ← , SPACE , → , DEL (and 123 for numbers).
KNOB 1	Move the cursor.
KNOB 2	Change the character under the cursor.
OCT+ / **OCT-**	Save the name / cancel. Name length depends on the engine: Prophet 20 characters, FM6 10, CZ-1 16, others 12.

Projects

Step by step

1. Stop playback and tap SAVE repeatedly until PROJECT is visible. Choose A-D with knob 1.

2. Use the Save action for a new project; name and confirm it when requested. Save projects to retain pattern banks and arrangement as well as sounds.

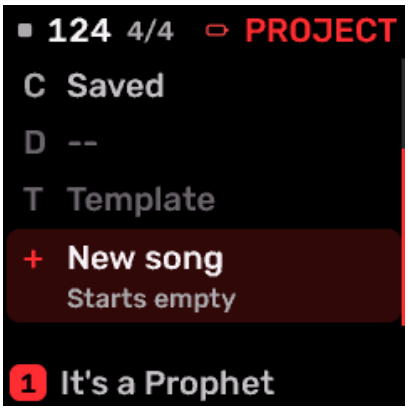
3. To recall, select the saved project and use Load. Save current edits first: loading replaces the working project.

4. To create a template, select TMPL and save; to start an arrangement, select NEW and complete the key/tempo and track-role pages. Back up before replacing an occupied slot.

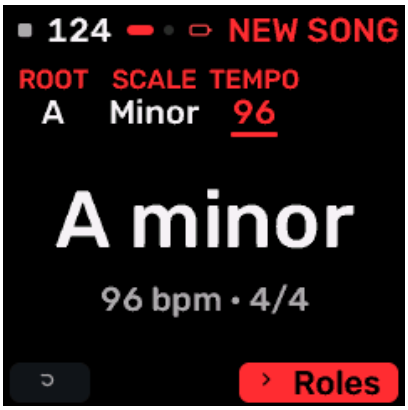
Four project slots **A-D**, plus a template and a new-song wizard. Tap SAVE repeatedly until **PROJECT** appears. PROGRAM and PROJECT are different storage formats. A project holds the tracks in use, their pattern banks (eight each, up to 32), timing, arrangement, automation and settings (FBKG format).



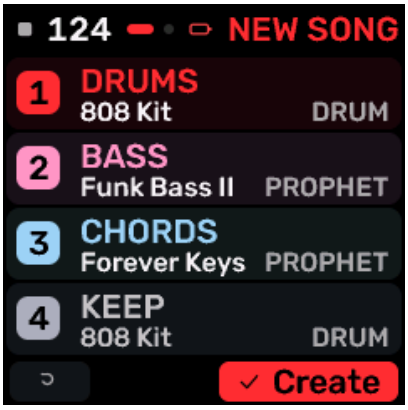
PROJECT: slots A-D, template, New song.



NEW song past the template.



NEW: key, scale and tempo.



Roles on the four knobs. -- skips that place and creates no track for that slot.

Control	What it does
KNOB 1 SLOT	A, B, C, D, TMPL (template), or **NEW** past the template.
KNOB 2 BOOT	OFF or A-D - which project loads at power-on. BOOT OFF with a saved template uses the template.
KNOB 3 LOAD	GO to load the chosen slot.
KNOB 4 SAVE	GO to save into the chosen slot.
OCT+	The slot's sheet: Load, Save here, Rename, Boot, Erase (Erase asks first).

• **NEW** opens the wizard, stopped only. Set key, scale and tempo (**KNOB 1** ROOT, **KNOB 2** SCALE, **KNOB 3** TEMPO), then a role on each of the four knobs: **--** (no track), **KEEP**, **DRUMS**, **BASS**, **CHORDS**, **LEAD** or **PAD**. The song keeps the places that have a role, in order, with empty patterns. With no template, track 1 starts as KEEP and

the rest as --, so the song has one track. A saved template starts KEEP on as many tracks as the template has. A role's first factory sound is used when the template's sound does not already fit that role.

• **TMPL**: save your current sounds, settings and number of tracks as the template for new projects (empty patterns). In SynthMaven 3.5.1 the factory-program startup loads P1 with four parts; this differs from the original Melodee one-track startup.

• Holding **SAVE**, or **SAVE**+**REC**, writes the project back to its slot (SAVED B, and so on). If it is playing, the screen says **STOPPING TO SAVE** and the transport stops first. A project with no slot yet opens PROJECT on a free slot (**NEW PROJECT: PICK SLOT**).

• **Autosave:** stopped and untouched for five seconds (no key, button, knob or MIDI), a project that already has a slot saves itself when it changed. Hold **HOME** and set **AUTOSAVE** to **OFF** to save only by hand. It defaults **ON**, and the choice survives power-off. Manual **SAVE** stays available either way.

Engines

Selectable instruments: **PROPHET, FM6, CZ-1, SID, DRUM, HYBRID, ANALOG, DIGITAL, VOICE, STRINGS, JUNO, ACID, VPS and WSEQ**, plus the MIDI-only track type. Press **EDIT** for that engine's pages; press again or use **SELECT** to move through them. On engines with several sections, **PRESETS** jumps between sections (and on FM6 / CZ-1 picks the operator / line). On an EDIT page, tap **EDIT** again for the section map. In that map, **KNOB 2** moves by category, and **KNOB 1** or **SELECT** picks a page within it, one step per detent. Each category remembers its page. **OCT+** opens the page; **OCT-** or **EDIT** closes the map. Use **EDIT** + **OCT-** to undo; holding EDIT for two seconds opens the layer mixer.

A fresh installation starts with P1 DX7 STAGE, including all four captured parts. FMX, VECTOR, TERRAIN and MACRO are excluded from this build; the selectable instruments listed above are supported. Older projects containing unavailable engines require a replacement sound. Ranges below are the device's own; the same values appear in the web editor.

Additional SynthMaven instruments

Step by step

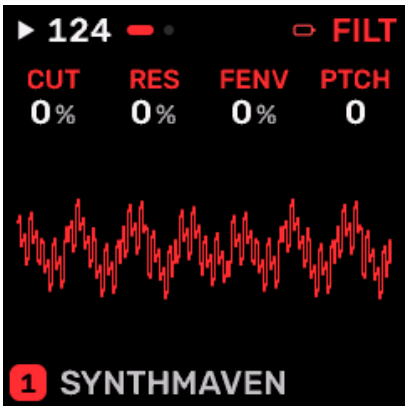
1. On HOME choose the part, then select the required engine with ALGORITHM. Choose a starting preset with PRESETS.

2. Press EDIT and move through that engine's pages. Begin with oscillator shape/tuning, then filter and amplifier response using the controls in the following tables.

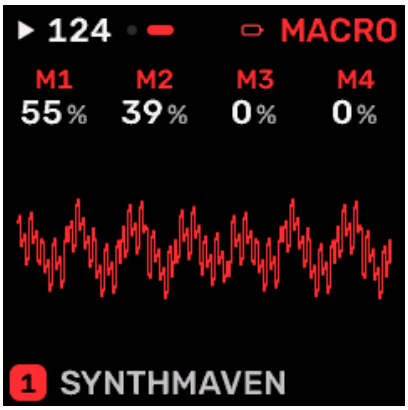
3. Audition low and high notes, short notes and held notes. Adjust the sound level in the mixer before saving it to a USER slot or PROGRAM.

Select the part on HOME, turn ALGORITHM to the instrument, then PRESETS to a starting sound. Tap EDIT and use SELECT to move between the two engine pages. Knobs 1-4 follow the left-to-right labels. In the editor, choose Shape → Sound and use the paginated Tone controls. The shared ENV, LFO and effects controls remain available where supported; the native HYBRID patch editor exposes its additional oscillator, filter and matrix settings.

HYBRID



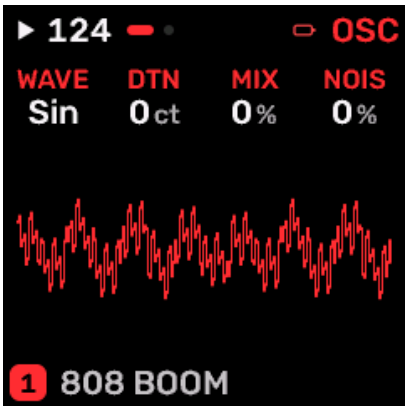
HYBRID: engine page 1



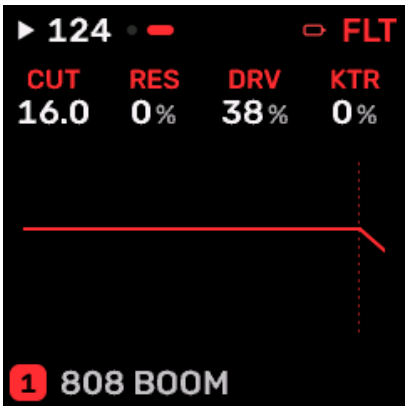
HYBRID: engine page 2

Page / knob	Setting	Meaning
1 / 1	CUT	Filter cutoff
1 / 2	RES	Filter resonance
1 / 3	FENV	Filter-envelope depth
1 / 4	PTCH	Patch pitch
2 / 1	M1	Assignable macro 1
2 / 2	M2	Assignable macro 2
2 / 3	M3	Assignable macro 3
2 / 4	M4	Assignable macro 4

ANALOG



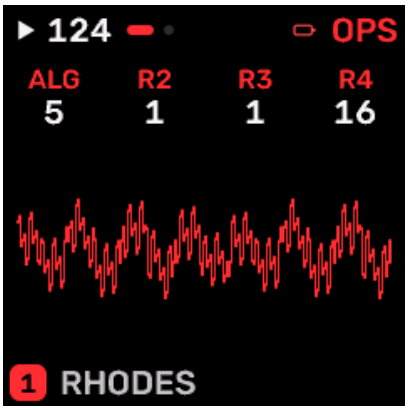
ANALOG: engine page 1



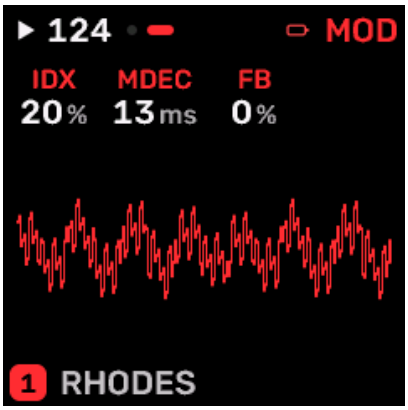
ANALOG: engine page 2

Page / knob	Setting	Meaning
1 / 1	WAVE	Oscillator waveform Options: SAW, SQR, TRI, SIN, PWM.
1 / 2	DTN	Oscillator detuning
1 / 3	MIX	Oscillator blend
1 / 4	NOIS	Noise level
2 / 1	CUT	Filter cutoff
2 / 2	RES	Filter resonance
2 / 3	DRV	Drive
2 / 4	KTR	Keyboard tracking

DIGITAL



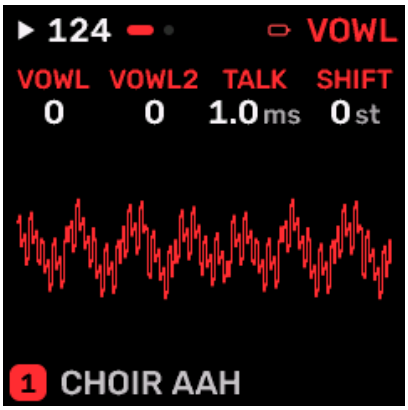
DIGITAL: engine page 1



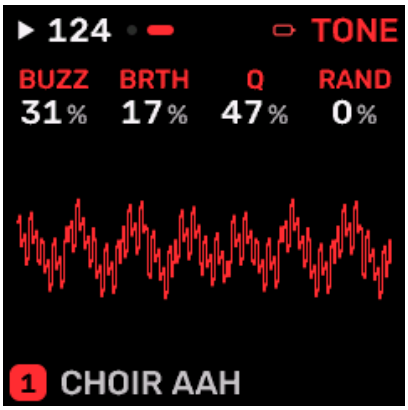
DIGITAL: engine page 2

Page / knob	Setting	Meaning
1 / 1	ALG	FM algorithm Options: 1, 2, 3, 4, 5, 6, 7, 8.
1 / 2	R2	Operator 2 ratio Options: .5, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 14, 16.
1 / 3	R3	Operator 3 ratio Options: .5, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 14, 16.
1 / 4	R4	Operator 4 ratio Options: .5, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 14, 16.
2 / 1	IDX	Modulation index
2 / 2	MDEC	Modulator decay
2 / 3	FB	Feedback
2 / 4	-	Unused

VOICE



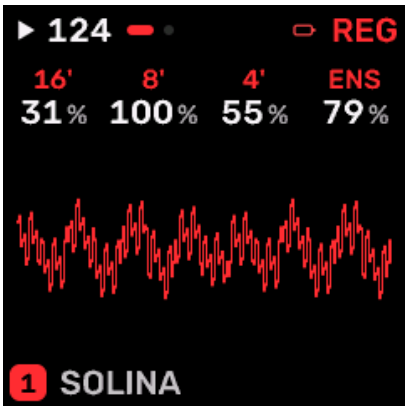
VOICE: engine page 1



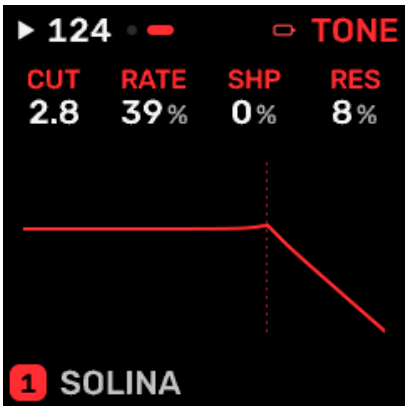
VOICE: engine page 2

Page / knob	Setting	Meaning
1 / 1	VOWL	First vowel
1 / 2	VOWL2	Second vowel
1 / 3	TALK	Vowel blend/movement
1 / 4	SHIFT	Formant shift
2 / 1	BUZZ	Excitation buzz
2 / 2	BRTH	Breath level
2 / 3	Q	Formant resonance
2 / 4	RAND	Random variation

STRINGS



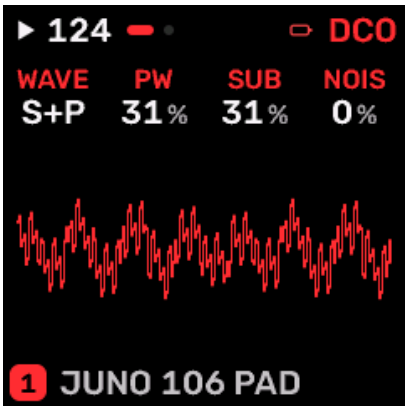
STRINGS: engine page 1



STRINGS: engine page 2

Page / knob	Setting	Meaning
1 / 1	16'	16-foot register level
1 / 2	8'	8-foot register level
1 / 3	4'	4-foot register level
1 / 4	ENS	Ensemble depth
2 / 1	CUT	Filter cutoff
2 / 2	RATE	Ensemble rate
2 / 3	SHP	Wave shape
2 / 4	RES	Resonance

JUNO



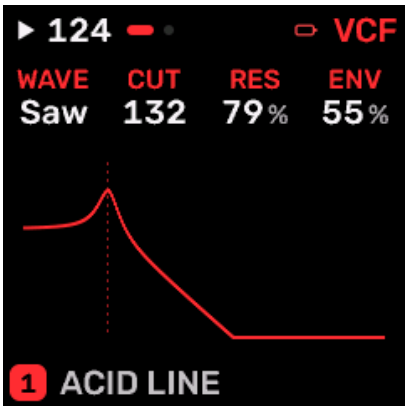
JUNO: engine page 1



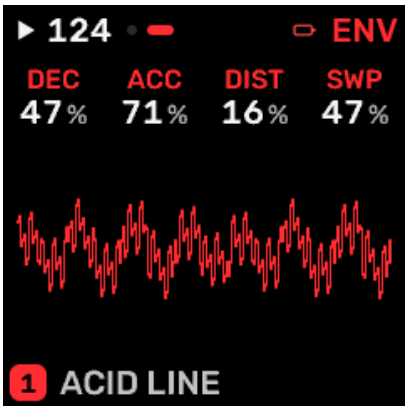
JUNO: engine page 2

Page / knob	Setting	Meaning
1 / 1	WAVE	Saw/pulse mix selection Options: SAW, PLS, S+P.
1 / 2	PW	Pulse width
1 / 3	SUB	Sub-oscillator level
1 / 4	NOIS	Noise level
2 / 1	HPF	High-pass filter step Options: 0, 1, 2, 3.
2 / 2	CUT	Low-pass cutoff
2 / 3	RES	Resonance
2 / 4	CHOR	Chorus mode Options: OFF, I, II, I+II.

ACID



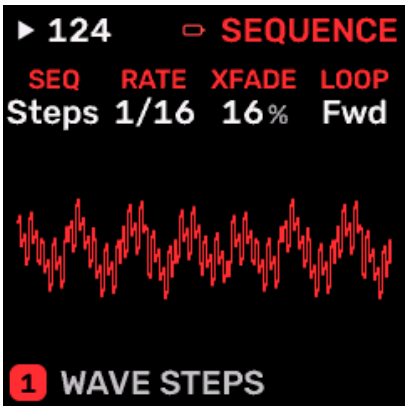
ACID: engine page 1



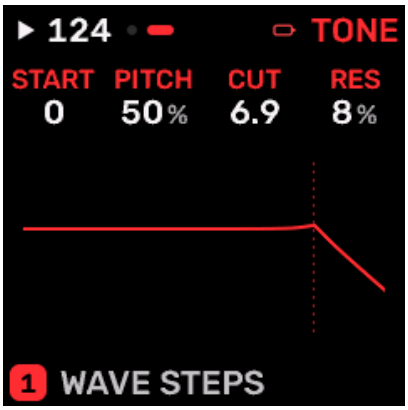
ACID: engine page 2

Page / knob	Setting	Meaning
1 / 1	WAVE	Saw/square waveform Options: SAW, SQR.
1 / 2	CUT	Filter cutoff
1 / 3	RES	Resonance
1 / 4	ENV	Filter-envelope depth
2 / 1	DEC	Envelope decay
2 / 2	ACC	Accent depth
2 / 3	DIST	Distortion
2 / 4	SWP	Sweep character

WSEQ



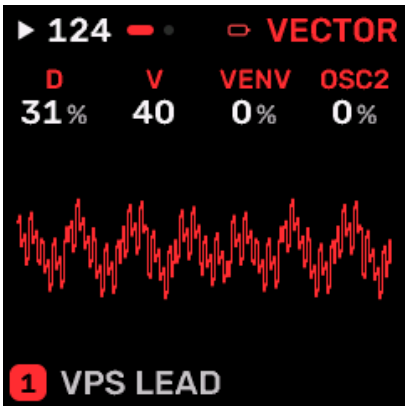
WSEQ: engine page 1



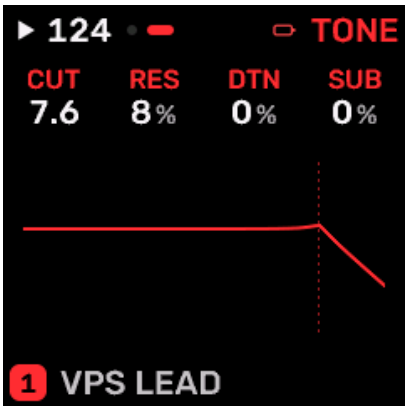
WSEQ: engine page 2

Page / knob	Setting	Meaning
1 / 1	SEQ	Wave-sequence selection Options: STEPS, RHYTHM, SWEEP, FIFTHS, MINOR, MAJOR, OCTAVE, GATE, BELLS, 8BIT, MORPH, VOX, NOISY, DUB, DORIAN, PING.
1 / 2	RATE	Sequence rate
1 / 3	XFADE	Crossfade amount
1 / 4	LOOP	Loop direction/mode Options: FWD, BACK, PING, ONCE.
2 / 1	START	Starting step
2 / 2	PITCH	Sequence pitch depth
2 / 3	CUT	Filter cutoff
2 / 4	RES	Resonance

VPS



VPS: engine page 1



VPS: engine page 2

Page / knob	Setting	Meaning
1 / 1	D	Phase-shape breakpoint
1 / 2	V	Phase-shape slope
1 / 3	VENV	Envelope to phase slope
1 / 4	OSC2	Second-oscillator level
2 / 1	CUT	Filter cutoff
2 / 2	RES	Resonance
2 / 3	DTN	Detuning
2 / 4	SUB	Sub level

WSEQ uses compact oscillator waveforms with timed crossfades; it does not play the older FMX wavetable engine. VPS is phase-shaping synthesis, not the removed VECTOR engine. These instruments share the four-part voice budget.

Prophet

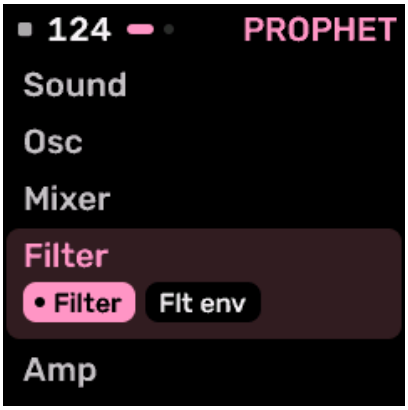
Melodee's own Prophet-5-style engine. Factory programs from Sequential's Prophet-5/10 SysEx.

Sequential Prophet-5 Rev 4-style programs: two oscillators (saw, pulse; B also triangle), hard sync, mixer, SSI or Curtis four-pole filter, separate filter and amp envelopes, Poly-Mod, LFO, wheel, unison, glide and Vintage. PRESETS browses INIT, Sequential's 200 factory programs, then the shared user slots that hold Prophet programs (**U001-U384**). Navigate between sections/pages with PRESETS knob, Navigate between individual settings per page

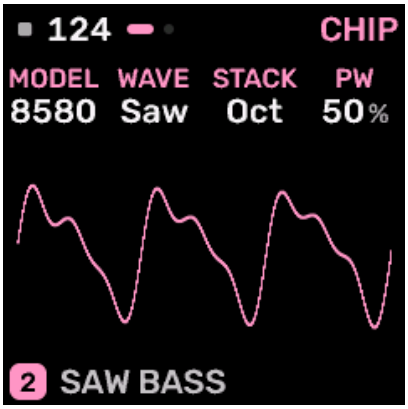
(e.g. OSC A vs OSC B), with SELECT knob. In the SECTION map, use KNOB 2 for the category, KNOB 1 or SELECT for its page, and OCT+ to open it.



Prophet Stage knobs: cutoff, res, env amount, release.



The section map (tap EDIT on an EDIT page).



OSC A page.

Oscillators

Page	Controls
OSC A	FREQ A (pitch, semitones), SAW A (on/off), PULSE A (on/off), WIDTH A (pulse width).
OSC B	FREQ B (semitones), FINE B (fine detune), WIDTH B, KEY B (B tracks the keyboard on/off).
B WAVES	SAW B, TRI B, PULSE B (on/off), LOW B (B becomes an LFO: low frequency oscillator).

Mixer and filter

Page	Controls
MIXER	OSC A level, OSC B level, NOISE level, SYNC (hard-sync B to A).
FILTER	CUTOFF, RES (resonance), KEY (keyboard tracking OFF / HALF / FULL), MODEL (SSI or CURTIS four-pole).

Envelopes

Page	Controls
FLT ENV	ATK (attack), DEC (decay), SUS (sustain), REL (release) for the filter envelope.
AMP ENV	ATK, DEC, SUS, REL for the amplifier.
ENV MOD	ENV AMT (the amount to open the filter, following the envelope's shape), VEL FLT (velocity → filter), VEL AMP (velocity → level), RELEASE (release stage of both amplifier and filter envelopes toggle on/off).

Poly-Mod, LFO and wheel

Two separate modulation paths. **Poly-Mod** takes the filter envelope and/or Oscillator B as sources and routes them into Osc A's pitch, pulse width, and the filter. Osc B can do this even when its mixer level is zero, so it works as a pure modulator (audio-rate or low-rate with LOW B). The **LFO / wheel** path is independent: the LFO (and optional noise) is the source; its depth is INITIAL plus the MIDI mod wheel (CC 1), then you choose destinations on the WHEEL pages.

Page	Controls
POLYMOD	Sources (amounts): ENV AMT (filter envelope), OSC B. Destinations (on/off): FREQ A (Osc A pitch), PW A (Osc A pulse width).
MOD FLT	FILTER (Poly-Mod destination: filter cutoff, on/off). Also VINTAGE (component drift), AT FLT (aftertouch → filter), AT LFO (aftertouch → LFO depth).
LFO	Shapes the wheel-path source: RATE, and waveform SAW / TRI / SQUARE (on/off).
WHEEL	Depth: INITIAL (always-on amount; mod wheel CC 1 adds on top), N MIX (blend noise into the LFO). Destinations (on/off): FREQ A, FREQ B (→ oscillator pitch).
WHEEL 2	More destinations (on/off): PW A, PW B (→ pulse width), FILTER (→ cutoff). GLIDE is portamento time (not a modulation destination).

Voice and store

The Voice section has Prophet's own **UNISON** and **BEND** pages, plus Melodee's shared **VOICE** list for the track (poly/mono, glide, priority, pan, mute). UNISON is not the VOICE page - VOICE is the list of global engine voice settings. Loading or editing UNISON copies into Melodee's VCE / PRIO / DTUNE / ALLOC; you can override those afterwards on VOICE. Program glide stays on WHEEL 2; VOICE's GLD is Melodee's track portamento.

Page	Controls
UNISON	UNISON (on/off), VOICES (1-5), DETUNE (0-7), note PRIORITY (LOW / LOW RET (retrigger) / LAST / LAST RET).
VOICE	List (**KNOB 2** row, **KNOB 1** value): VCE (POLY / MONO / LEG / UNI), GLD (glide time), GLMOD (RATE / TIME), PRIO (LAST / LOW / HIGH), ALLOC (ROT / REUSE), DTUNE, PAN (-100…+100%), MUTE, SPRD (stereo spread). Same shared page as other engines - see Voice.

Page	Controls
BEND	BEND (pitch-bend range, 1-12 semitones).
STORE	SLOT (U001-U384, the shared bank), STORE, SEND, INIT. Controls whose required waveform or switch is off are drawn dim; they stay editable.

FM6

Step by step

1. Select FM6 on the required part, then choose a factory voice or import a DX7 voice using Shape → 6-OP FM in the editor.

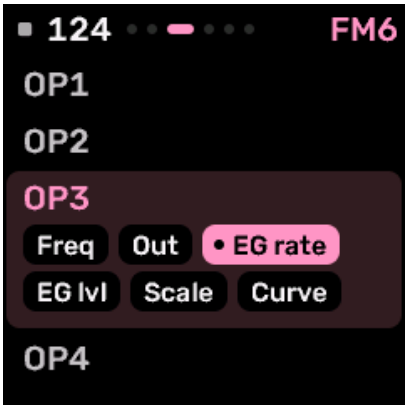
2. Open its operator pages and select the operator to edit. Set frequency ratio or fixed frequency and output level; use the algorithm diagram to distinguish carriers from modulators.

3. Adjust operator envelope rates/levels and velocity response. Carrier level changes loudness; modulator level usually changes brightness and harmonics.

4. Save the native FM6 voice to a USER slot, or save the program. In the editor send a draft to the part before saving; merely importing a file does not save the device.

Uses msfa from Dexed (Google Inc., Pascal Gauthier).

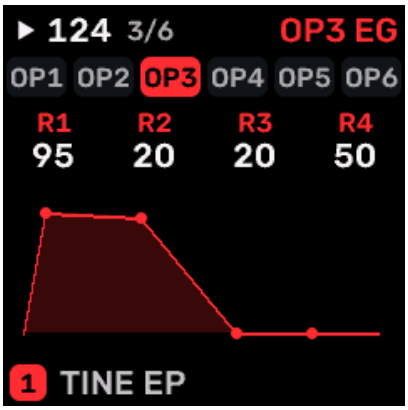
Six-operator FM (Dexed / DX7-compatible): 32 algorithms with an on-screen chart, per-operator frequency, levels, envelopes and scaling, pitch envelope, LFO, and controller / portamento settings. On operator pages, **PRESETS** picks the operator. FM6 voices live in the shared slots **U001-U384**. DX7 librarians can send a 32-voice bank over USB into any 32 consecutive slots.



Section map. Tap EDIT on an EDIT page.



FREQ for the selected operator.



EG RATE: the operator envelope.



ALGO: the algorithm chart.



STORE: save into the shared user bank.

Voice

Page	Controls
ALGO	ALG (algorithm 1-32), FB (feedback 0-7), SYNC (oscillator key sync), TRNS (transpose, ±24 semitones).
PITCH EG	R1-R4: pitch envelope rates.
PITCH LV	L1-L4: pitch envelope levels.
FM LFO	WAVE (TRI / SAWDN / SAWUP / SQR / SIN / S&H), SPEED, DELAY, KSYNC (key sync).
FM LFO 2	PMD (pitch mod depth), AMD (how much the LFO wobbles amplitude - global amp-mod depth), PMS (pitch mod sensitivity 0-7).

Per operator

Page	Controls
FREQ	CRS (coarse ratio, or Hz in FIXED), FINE, DTN (detune ±7), MODE (RATIO / FIXED).
OUT	OUT (output level 0-99), VEL (velocity sensitivity 0-7), AMS (how much this operator follows amp mod, 0-3; see below), ON (operator on/off).
EG RATE	R1-R4: envelope rates (0-99).
EG LVL	L1-L4: envelope levels (0-99).
SCALE	Keyboard level and rate scaling. BRK (break point - at this note, level is unchanged), LDEP / RDEP (how strongly level changes left / right of BRK, 0-99; 0 = off), RSCL (rate scaling 0-7: higher keys run the envelope faster).
CURVE	LCRV / RCRV: shape and direction of level scaling on each side of BRK (-LIN / -EXP lower the operator's level as you move away; +EXP / +LIN raise it). Depth (LDEP / RDEP) sets how much; the curve's sign sets which way.

AMS is not the operator envelope (that's EG RATE / EG LVL). It's DX7-style *amplitude modulation sensitivity*: how strongly this operator's level follows amp mod from the LFO's AMD, or from controllers routed to amplitude (wheel / breath / aftertouch with destination A). AMD sets how much amp wobble exists; AMS (0 = ignore, 3 = strongest) sets whether *this* operator hears it. Use it for tremolo on a carrier, or put AMS on a modulator so the LFO breathes the brightness instead of the overall volume.

SCALE and CURVE are DX7-style keyboard level scaling: they raise or lower that operator's output as you play above or below BRK. Classic use: -LIN or -EXP on the right so high notes get quieter, or so a modulator thins out and the sound gets less bright up the keyboard.

Controllers (per track, saved with the project)

Page	Controls
FM BEND	BEND+ / BEND- (bend range up / down, 0-12 st), STEP (bend step), DXVEL (DX velocity curve on/off).
FM PORTA	PORTA (PEDAL / ON), TIME, GLISS (glissando on/off), ENGINE (MODERN / MARK I / OPL - algorithms 4 and 6 use longer feedback loops in MARK I).
FM WH/FT	WHEEL range, W.DST (wheel destination: P / A / E combinations), FOOT range, F.DST.
FM BR/AT	BRTH (breath) range and B.DST, AFTER (aftertouch) range and A.DST.
STORE	SLOT (U001-U384, the same bank as SAVE → USER), STORE, SEND, INIT.

CZ-1

Step by step

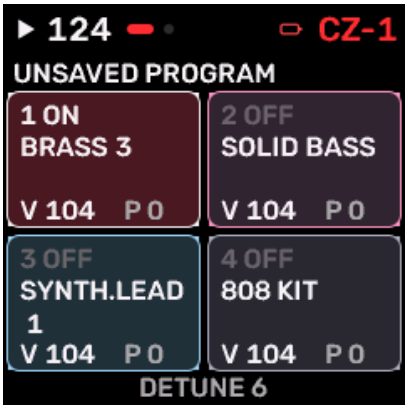
1. Choose CZ and a starting tone on HOME. Open its EDIT pages.

2. Choose line/wave settings first, then the DCO, DCW and DCA envelope pages in the reference below. Set pitch, timbre and volume separately.

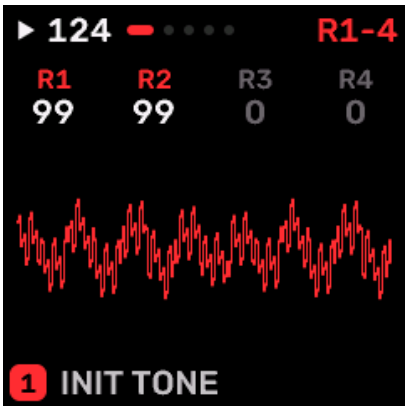
3. Audition and save to USER or PROGRAM. Use the editor native import workflow for compatible CZ data and explicitly apply imported changes.

Phase-distortion core after MAME's uPD933 model by Devin Acker. Factory tones from Casio.

Native Casio CZ-1 tones: two lines with waveforms and windows, ring and noise, detune, vibrato, and six eight-step envelopes (pitch, wave, amp × two lines). Factory: Casio's 64 presets (A-1 BRASS 1 … H-8 TYPHOON SOUND). User tones use the shared slots **U001-U384**, the same bank as the other engines; a fresh device starts with them empty. Values that do nothing on the current tone are drawn dim but stay editable.



CZ-1 Stage knobs: WAVE, DCW, DETUNE, VIBRATO.



An eight-step envelope page (rates R1-R8, levels L1-L8).

Common

Every tone is built from one or two **lines**. A line is a DCO → DCW → DCA chain: pitch, then phase-distortion (timbre), then amplitude - each with its own eight-step envelope. Common pages set how the lines combine, plus shared detune, vibrato and windows. **PRESETS** picks which line's pages you are editing when you are in the per-line section.

Page	Controls
CZ TOOLS	NAME (16-character tone name; **OCT+** to edit), 1 > 2 / 2 > 1 (copy one line's wave, window, key follow, level, velocity and all three envelopes onto the other - MOD stays with line 1 and is not overwritten), COMP (swap the edited tone with the snapshot taken before your first edit on this track; "NO EDITS" if you have not changed anything yet).
CZ LINE	LINE chooses how many oscillators sound: LINE1 or LINE2 alone (the unused line's params are drawn dim), 1+2 (both lines with their own settings), or 1+1 (two voices that both use line 1's params - line 2's own wave/envelopes are unused, but detune and MOD can still affect the second oscillator). MOD mixes the two lines when more than one is active: OFF (sum), RING (multiply line 1 × line 2), or NOISE (randomly stretch line 2's phase - a noise-modulated second line). OCT is a global pitch offset (0 = -1 octave, 1 = unison, 2 = +1 octave).
CZ DETUNE	Pitch offset for *line 2 only*: SIGN (±), OCT (0-3 octaves), NOTE (0-11 semitones), FINE (0-60). Inactive (dim) in LINE1 mode; SIGN only matters when some detune is set. Use with 1+2 or 1+1 for chorusing, intervals, or beating.
CZ VIBRATO	A pitch LFO shared by the tone: WAVE (TRI / SAW UP / SAW DN / SQR), RATE (speed), DEPTH (amount - set this above 0 or the vibrato is silent and WAVE / RATE / DELAY go dim), DELAY (wait after note-on before the vibrato fades in).
CZ WINDOW	A repeating amplitude mask inside *every* oscillator cycle, for as long as the note plays. WAVE picks the phase-distortion carrier shape; WINDOW then fades that carrier up/down within each period (like a tiny amp envelope that retriggers on every cycle). That changes the waveform's silhouette and spectrum; OFF leaves the cycle flat (no extra shaping). SAW DN, TRIANGLE, TRAPEZ, HALF SAW, 2 SAW UP, 2 SAW 6 and 2 SAW 7 are different masks. LINE1 and LINE2 each have their own. This is separate from the WAV envelope (which moves DCW / timbre over the note) and the AMP envelope (overall loudness over the note).

Per line (LINE 1, LINE 2)

Each line has a waveform page, a touch (level / velocity) page, and three envelopes: **PIT** (pitch / DCO), **WAV** (wave depth / DCW - the phase-distortion amount that brightens or darkens the tone), and **AMP** (loudness / DCA). These are Casio-style rate/level multi-stage envelopes, not ADSR.

Page	Controls
WAVE	WAVE is the primary carrier (SAW, SQUARE, PULSE, NULL = silent, DBL SINE, SAW-PULS, MULTISIN for a resonance-like spectrum, PULSE2). WAVE2 is an optional second carrier (same set, plus OFF): when on, the line alternates WAVE ↔ WAVE2 each phase wrap. W.KEY (0-9) is DCW key follow - higher notes get less wave/timbre depth (bright → darker up the keyboard). A.KEY (0-9) is DCA key follow - higher notes run the amp envelope faster (shorter notes up the keyboard).
TOUCH	LEVEL (1-15) is the line's output level. V.PIT / V.WAV / V.AMP (0-15) set how much MIDI velocity scales pitch, wave depth and amp - soft notes pull those paths down. (The FM-1's own keys send fixed velocity; velocity matters over MIDI or from the sequencer.)
PITCH / WAVE / AMP envelopes	The envelope shape of the CZ-1 is controlled by eight rates R1-R8 (how fast the envelope moves toward each step's level - higher = faster) and eight levels L1-L8 (0-99 targets). END (1-8) is the last active step: the envelope only runs through END, and the END step always targets 0 (release to silence; its level value is ignored). SUS selects which level is the sustain step while the key is held; on release the envelope jumps past SUS and finishes through END → 0. SUS must sit before END (otherwise it shows "-" = no sustain). With no SUS, the whole path keeps playing after note-off - useful for long rings. Rates after END and levels at/after END are inactive (dim). Each line has its own Pitch, Wave and Amp envelopes (six envelopes total).

SID

Step by step

1. Choose SID on the selected part and load a starting preset.

2. Open EDIT; select oscillator waveform, tuning and modulation using the labeled controls. Set envelope and filter afterward.

3. Compare single notes and chords, then balance the part in the mixer and save. SID emulation character is intentional; lower drive or resonance if an aggressive timbre is unwanted.

Uses reSID by Dag Lem, including measured combined-waveform tables from a real 6581 and 8580.

Probably the simplest synth engine (in terms of numbers of parameters to program) on the device; use this if you're looking to dial in a quick sound. It models a Commodore MOS 6581 / 8580 chip per note: waveforms, pulse width, ring, sync, combined waves, the chip's ADSR, filter and drive. Thirty-two factory presets. Stage knobs: WAVE, PW, CUT, RES. With a C64 MODEL, MONO / LEG plays one note at a time; POLY turns a held chord into a fast chip arpeggio, one note per PAL frame, with one gate and envelope for the whole chord. This is separate from the ARP pages. New notes hard-restart the chip envelope.

Page	Control	What it means
CHIP	MODEL	6581 or 8580 (one chip per note), or C64 6581 / C64 8580 (one voice for the track, registers updated once per PAL frame at 50 Hz).
	WAVE	TRI, SAW, PULSE, NOISE, or a combined wave (TRI+SAW, TRI+PLS, SAW+PLS, T+S+P).
	STACK	Second-oscillator stacking: ONE, DETUNE, OCT, FIFTH, MAJOR, SYNC, RING, SUB.
	PW	Pulse width.
FILTER	CUT	Filter cutoff.
	RES	Resonance (0-15).

Page	Control	What it means
	MODE	LP, BP, HP, NOTCH, LP+BP, BP+HP, ALL, or OFF.
	DRIVE	Input drive into the filter.

Drum

Step by step

1. For independent drums press HOME to reach drum Home. Select a lane, select a step, and turn knob 2 right to add the hit or left to remove it.

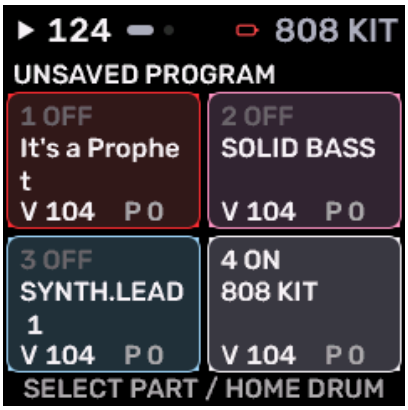
2. Set lane level with knob 3 and tempo with knob 4. PLAY starts/stops the drum pattern independently of the melodic sequencer.

3. For the separate DRUM sound engine, choose DRUM as a program part instead; this consumes that part. Use the kit and hit controls in the tables below.

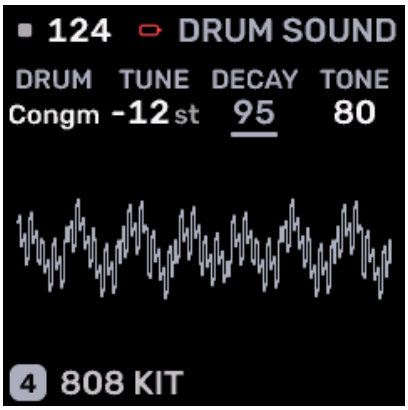
4. In the editor use Studio → Drums → Save pattern to retain the independent drum pattern. Save a program for a DRUM-engine sound part.

808 kit: Hügelton's TR-808 circuit models. 909 kit: the schwung module 9W9 by AThousandDetails (after ER-99), ported via Charles Vestal's x0x.

Synthesized TR-808, TR-909 and SID kits on the General MIDI key map. SID KIT uses a single 6581 chip whose three voices share the hits; SID DIGI also plays its synthesized kick and snare through the chip's four-bit volume register. In SID mode, DIGI replaces KICK on the HIT page and RES replaces DRV. The DRUM EDIT pages have **Sound** (KIT / HIT macros), **Drum** (per-instrument Sound and Mix), **Lanes** and **Voice** tabs. On DRUM SOUND and DRUM MIX, black keys pick the drum lane and white keys pick a drum within it; the knobs then shape that drum.



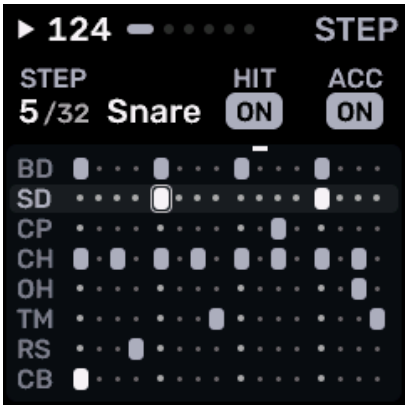
Drum Stage: lanes light as they hit.



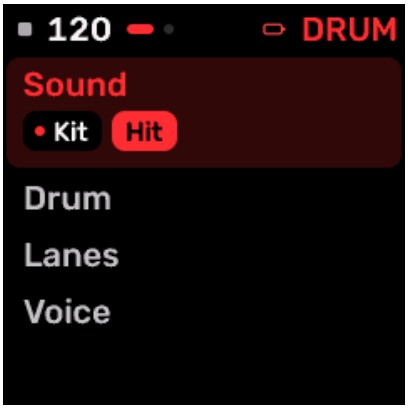
DRUM SOUND: TUNE, DECAY, TONE for the picked drum.



DRUM MIX: LEVEL per drum.



The drum grid: white keys step, black keys pick lanes.



EDIT map: KNOB 2 chooses a category, KNOB 1 its page.

Kit macros (Sound section)

These four-knob pages shape the whole kit at once (also the Stage knobs for TUNE, TONE, DECY and SNAP).

Page	Controls
KIT	KIT (808, 909 or SID), TUNE (overall pitch), TONE (overall character / brightness), DECY (overall decay).
HIT	SNAP (snare snappy / noise amount), ACC (how strongly velocity changes level - the circuits alone only move about 2 dB from soft to hard), KICK (PUNCH keeps the bass-drum attack click; ROUND softens it), DRV (drive into a soft clip, level kept).

Per drum (Drum section)

Pick the lane on the black keys and its drum on the white keys, then offset that instrument from the kit macros. DRUM HIT uses the same lane-first ordering and shows the selected drum's name.

Page	Controls
DRUM SOUND	DRUM (which instrument), TUNE, DECAY, and TONE (or SNAPPY on the snare; ATTACK on 909 kick / toms).
DRUM MIX	LEVEL for the picked drum.
LANES	Levels for KICK, SNARE, CLAP, HATCL (closed hat).
LANES 2	Levels for HATOP (open hat), TOM, RIM, BELL.

Stage knobs: TUNE, TONE, DECY, SNAP. The SEQ sequencer drum grid uses white keys for steps and black keys to select which lane; recorded takes use the same NOTES piano roll as other engines after recording stops. A recorded hit can keep its own pitch and length (see Notes and DRUM HIT).

Voice

Step by step

1. Select the sound part before editing voice controls. Choose POLY for overlapping notes or a mono mode for a single lead/bass line.

2. Adjust glide and priority while playing overlapping notes. Then set detune, pan and spread on the following voice pages.

3. Audition the full layered program; four parts share the voice budget. Save the program after balancing levels.

All synthesis engines have a VOICE settings page, with the same options regardless of engine. This includes shared voice allocation and tuning for the track, under **EDIT** → VOICE (a list; **KNOB 2** moves the row, **KNOB 1** sets its value). Prophet and FM6 also have their own voicing pages; VOICE still sets poly/mono mode, glide, priority, pan, mute and spread for them.



VOICE list.

Row	Values	What it means
Voice	POLY / MONO / LEG / UNI	Play the instrument polyphonically (multiple tones at the same time), monophonic (only one tone at a time), legato (single voice but don't retrigger envelopes when you change note), or unison (monophonic playback using multiple voices thickened together).
Glide	time	Glide (portamento) time.
Glide Mode	RATE / TIME	Glide follows a fixed rate or a fixed time.
Priority	LAST / LOW / HIGH	Note priority when voices run out.
Allocation	ROT / REUSE	Rotate through voices, or reuse the same one.
Detune	0-127	Detune between stacked / unison voices.
Pan	-100…+100%	Track pan.
Mute	OFF / ON	Mute the track.
Spread	0-100%	Stereo spread of stacked / unison voices.

Sound designer

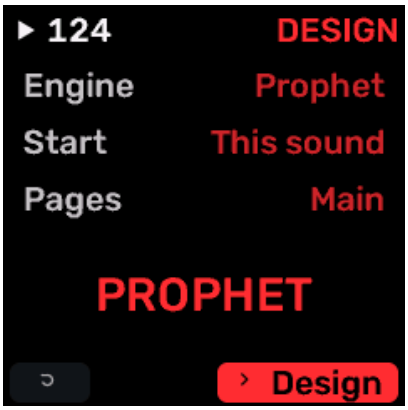
Step by step

1. Open SOUND DESIGNER from the HOME-hold settings menu. Start from a sound you have already backed up.

2. Choose the displayed sound-design operation and adjust its amount using the control table below. Audition the result on the selected part.

3. Keep useful results by saving to a new USER slot or program; cancel or reload the original if the result is unsuitable.

Build a sound on the device from a goal or from the engine's own pages. Open it from the HOME hold menu (**SOUND DESIGNER**), or from a sound page's sheet (**OCT+** held → Sound designer).



START: engine, starting point, and which pages.

1. START - engine; start from CURRENT, INIT, or INIT with a recipe (BASS, PAD, LEAD, KEYS, PLUCK, FX); MAIN pages only or ALL.

2. Stages: **Features** → **Source** → **Tone** → **Env** → **Motion** → **Play**. Empty stages are skipped. Audition on the keys; hold **SAVE** to compare with the original.

3. Finish: Keep the sound, Save as user…, or Discard (patterns and mix stay).

Motion on the designed track is paused while you design so it cannot overwrite edits. Knob turns in the designer do not record motion.

Envelopes and modulation

Step by step

1. Select a part, press LFO for periodic modulation or open the modulation routes under EDIT/editor Shape → Sound → Modulation.

2. Choose a source, destination and amount. Start with a small amount and operate that source: velocity needs new notes; pressure needs a pressure-capable controller.

3. Set LFO rate and shape, then increase the destination depth to taste. Use sync when the movement should follow tempo.

4. For a template, open the editor Modulation card, choose a template and apply it to a free route. Apply native HYBRID drafts to the part, then save.

Shared Melodee modulation, separate from each engine's own envelopes where those exist. Press **ENV** for ENV and ENV DEST, **LFO** for the LFO and MOD pages.



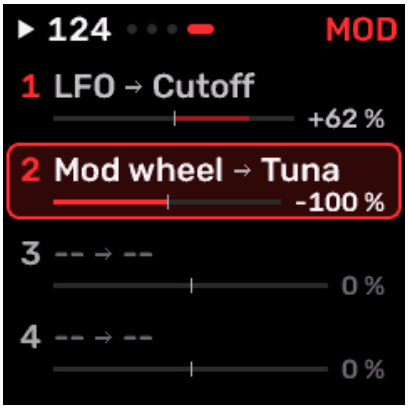
ENV: the shared amp envelope.



LFO.



ENV DEST: envelope to filter, pitch, shape.



MOD: four routes.

Page	Controls
ENV	ATK (Attack - how long the sound takes to reach its peak), DEC (Decay - how the sound went down after its initial peak), SUS (Sustain - how loud to play the note while the key is pressed down), REL (Release - how long to fade the sound out after a note is released) - shared amp envelope.
ENV DEST	FLT, PIT, SHP - envelope amount to filter, pitch and wave shape (±).
LFO	RATE (speed), WAVE (SIN / TRI / SAW / SQR / S&H) (type of wave), PHS (phase), FADE (fade-in).
LFO 2	SYNC (OFF, 4BAR…1/32 tempo divisions), TRIG (NOTE / FREE), Polarity (Bidirectional up and down in amount, or Unidirectional only).
LFO DEST	pitch, filter, wave shape (±), amp amount .

MOD matrix

Four routes. **KNOB 2** picks the route (1-4); **KNOB 1** sets its source, **KNOB 3** its destination, **KNOB 4** the amount (±).

Field	Choices
SRC	OFF, LFO, ENV, VEL, KEY, RAND, MODW (mod wheel), AT (aftertouch), EXPR (expression), S&H, SLEW.
DST	OFF, PITCH, CUT, SHP, AMP, PAN, RATE, VIB, E1-E8 (engine macros, shown with the engine's labels), DEPTH.
AMT	-100…+100%.

Edit Prophet, FM6 and CZ-1 envelopes on those engines' EDIT pages (and in the web editor). Hold **LFO** while turning sound or effect knobs for temporary changes that restore when you release (or keep with **OCT+**).

Effects

Step by step

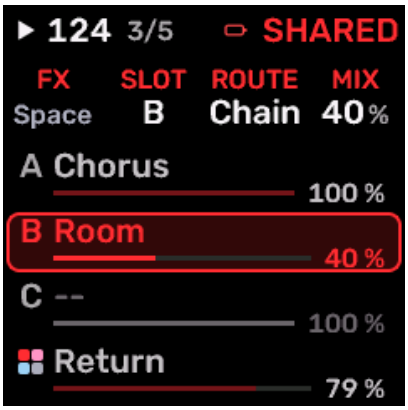
1. Select a part and open FX. Select an effect type before adjusting its mix and type-specific controls.

2. For shared sends, set Send FX 1, 2 and 3, choose chain/separate routing, then raise the selected part's send and the shared return. A zero send or return is silent.

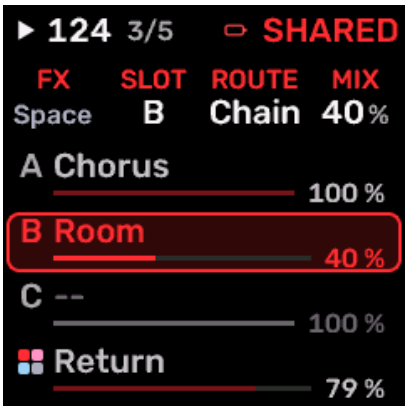
3. Audition with other layers playing. Reduce send/mix if reverb or delay obscures the lead. Shared effects are not separate copies for each layer.

4. Save the program to preserve the full shared routing and mix; a single sound save has a narrower scope.

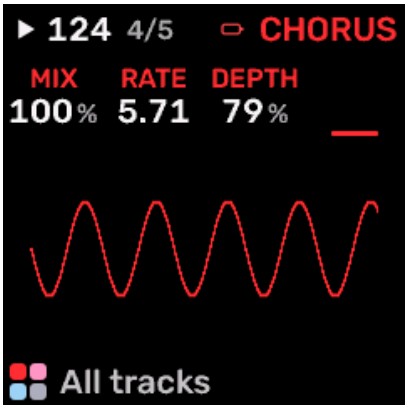
Press **FX** to open the selected track's rack: three note-effect slots for NOTE TAIL and NOTE ECHO, followed by a Shared row. Audio processors run in SHARED; each track sends audio to that shared rack. Effects save with the project and template. Loading a sound, changing the engine, Init sound and sound undo leave them in place.



FX: note-effect slots and the Shared row.



SHARED: slots A-C, ROUTE and RETURN.



TONE: the lit band reaches the effect.

On the rack, **KNOB 2** picks a row, **KNOB 1** chooses its effect, and **KNOB 4** sets its amount (NOTE TAIL's REL, NOTE ECHO's LEVEL, or the track's SEND on the Shared row in CHAIN mode). Empty slots show **--**. **OCT+** opens the selected slot, its effect picker if empty, or the SHARED rack. **SELECT** visits the racks and occupied slots; empty slot pages are skipped. Pressing **FX** from a slot returns to its rack.

The shared audio-effect picker groups choices into Space, Motion, Dirt and Colour. Track slots offer only note effects. On an empty shared slot, **OCT+** opens the browser: **KNOB 1** picks a family, **KNOB 2** an effect, and **OCT+** confirms. On a shared audio slot's page, **KNOB 1** is MIX, **KNOB 2** is RATE / TIME / SIZE / DRIVE / BITS, and **KNOB 3** shapes its character. **KNOB 4** is TONE for ROOM and REVERB, SHAPE for DRIVE, or RESO for the swept pedals:

Effect	KNOB 2	KNOB 3
PHASER	RATE	DEPTH - swept notches.

Effect	KNOB 2	KNOB 3
WIDTH	RATE	DEPTH - all-pass on the left, dry on the right. Mono changes the tone.
FORMANT	RATE	FOCUS - vowel shapes.
TAPE	RATE	JITTER - drifting filters.
SPIT	RATE	BITE - a resonant band that jumps and saturates.
ROOM	SIZE	DECAY - a small space.
REVERB	SIZE	DAMP - SLOOP's stereo reverb; TONE shapes its input.
DRIVE	DRIVE	TONE, and SHAPE on KNOB 4: Soft, Hard, Fold or Fuzz - SLOOP drive processors, unity output gain.
CRUSH	BITS	ALIAS - bit reduction and sample hold (the held rate in kHz).
FLANGER / PEDAL PHASER / PEDAL CHORUS	RATE	DEPTH and signed RESO - feedback pedals; shared returns exclude their dry half.
CHORUS	RATE	DEPTH - stereo pitch movement. Start MIX near 50%.

TONE goes from DARK through FULL to THIN. Below FULL it cuts highs; above FULL it cuts lows from what enters the wet effect. The dry sound stays intact. ROOM also keeps a low cut at FULL to stop bass building up in their tails.

MIX at zero, or OFF, fades that slot dry. If a time effect cannot get its buffer, the picture says **MEMORY FULL**; turn the slot off and on to try again. Hold **OCT+** for the sound sheet. Tap **OCT-** to return from a slot to its rack, from SHARED to FX, and from FX to Stage.

SLOOP REVERB replaces Echoverb at effect ID 8, with SIZE, DAMP and TONE. The native-rate stereo tank uses 19,396 bytes per slot, down from Echoverb's 60,340 bytes; bypass releases it. Existing ID 8 settings select the replacement reverb. Old audio Echo slots load OFF. ROOM and CHORUS retain their 2 KiB buffers. NOTE TAIL and NOTE ECHO remain note inserts. SLOOP audio processors run in the shared rack; the existing PHASER and CHORUS remain distinct choices.

Master punch effects

Hold **FX** and press a key to apply one of SLOOP's 27 audio effects to the whole stereo master, including shared returns and USB stereo capture. Release the effect key for dry audio. Double-tap FX to lock the layer and tap again to close. KNOB 1 browses the key reference. Keys held before FX keep playing; effect keys produce no notes or MIDI.

The effects include loops, reverse, tape stop, half speed, filter sweeps, phone, crush, alias, gate, echo, wobble, band kills, rewind, octave up, flanger, auto-pan, freeze, mono and fade. FREEZE uses SLOOP Reverb to hold 186 ms of captured master audio. The optional punch history uses 32 KiB while the layer is in use (372 ms of mono history; replay spans up to 348 ms); hold FX briefly before reverse/rewind. If memory is unavailable the header shows FX MEMORY; close FX, reduce effect memory use and reopen to retry.

Shared return

The Shared row feeds one shared three-slot return after this track's level and pan. In the **SHARED** rack, **KNOB 2** selects slots A-C or RETURN, **KNOB 1** chooses a slot's effect, **KNOB 3** sets **ROUTE**, and **KNOB 4** sets the selected amount. **OCT+** opens a slot. Hold **OCT+**, or tap it on RETURN, for Routing and Clear shared FX (asks first).

• **CHAIN** (default) - raise the selected track's SEND on FX's Shared row. Slots A-C run in series. On SHARED, **KNOB 4** sets each slot's MIX, initially fully wet, or the overall RETURN.

• **SPLIT** - each shared slot has its own send. On SHARED, **KNOB 4** sets this track's SEND to the selected slot, or RETURN on the last row. Each slot's return joins the dry output. SEND 2 and SEND 3 are remembered when you go back to CHAIN. Shared slot pages still edit their MIX.

Only one shared REVERB can be active: choosing it in another shared slot moves it there; imported duplicate REVERB slots keep the first. Historical per-track audio processors load OFF and are not moved automatically. Recreate those processors in SHARED and set the track sends. NOTE TAIL and NOTE ECHO remain per-track. Existing shared settings keep their supported compatibility paths; older firmware may not reproduce the new routing.

Scales and chords

Step by step

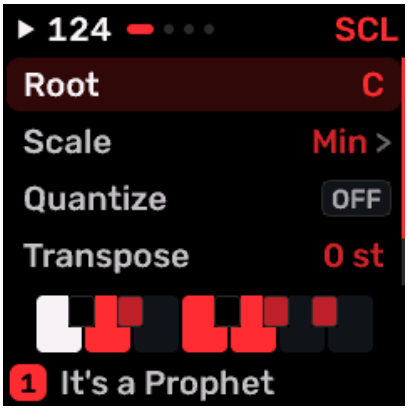
1. Tap SEL, choose the scale with SELECT/knob 1, set the first affected octave with knob 2 and root with knob 3, then enable master tuning with knob 4.

2. Hold SEL for three seconds to edit a copy: knob 1 selects a chromatic key, knob 2 changes cents in 10-cent steps, knob 3 quantizes all, and knob 4 applies tuning to the selected part.

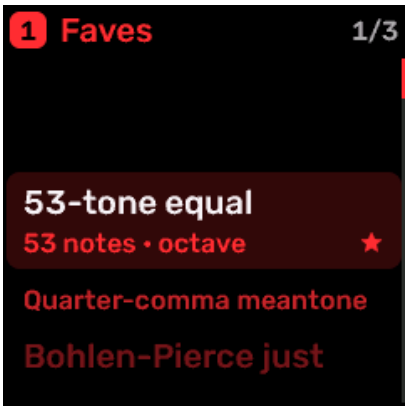
3. Hold ENV for three seconds to open chords. Choose chord type, voicing and strum, then audition each scale degree. Microtonal-compatible harmony is filtered by the active master scale.

4. Use the editor custom-scale library to name/export the result. Enable Save tuning override only if a program should deliberately recall its own tuning instead of retaining the global tuning.

Tap SEL (labelled SCL in the inherited reference screens) for MASTER TUNE. Hold SEL for three seconds for CUSTOM TUNE. Hold ENV for three seconds for CHORD. The native scale controls below are also exposed in the editor; master tuning takes precedence when enabled for the selected part. Use the master-tuning instructions above for the 3.5.1 keyboard, root, octave boundary and cents workflow.



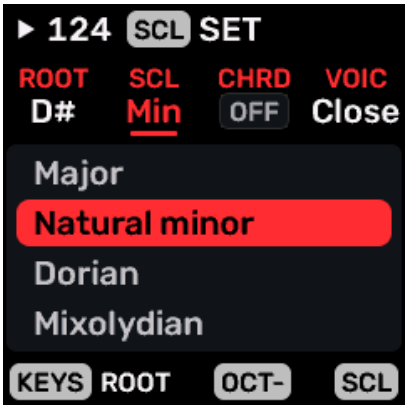
SCL settings.



SCALES picker.



CHORD.



Inherited native scale overlay reference; the 3.5.1 SEL hold opens CUSTOM TUNE instead.

Page / control	What it does
SCALES	**KNOB 1** family (or ALL / FAV), **KNOB 2** scale, ROOT, QNT. PRESETS also browses scales. Seventy scales including microtonal equal divisions, just intonation, temperaments, maqam models and Bohlen-Pierce.
SCL	ROOT (shared by the project), scale favourite, QNT, TRN (per-track transpose ±24 semitones).
CHORD	CHRD (OFF, DIA3, DIA7, MAJ, MIN, DOM7, MAJ7, MIN7, SUS4, POW, AUTO, AUTO2) and VOIC (CLOSE, OPEN, INV1, INV2, +OCT). STRUM: OFF, 1/8, 1/16, 1/32, 1/64, 1/128, 16T or 32T for one complete sweep per press. STROKE: DOWN (low to high), UP or ALT (alternates each press). The first note sounds immediately; the last lands one selected division later. Release cancels unplayed notes. ARP receives the complete chord and uses its own timing.

AUTO chords: Select AUTO with CHRD for automatic harmony in any of the 16 Western scales. Each key supplies the root; AUTO chooses a three- or four-note chord entirely inside that scale. It prefers familiar triads, considers recent chords and melody on other unmuted melodic tracks with CHRD OFF, and can choose suspended or seventh chords. With VOIC CLOSE it chooses nearby inversions for smooth transitions; OPEN, INV1, INV2 and +OCT override that voicing. The chord name appears on the CHORD page. A rare combination without a supported chord name is labelled “set”; its pitches still belong to the scale.

AUTO remembers melody onsets for two beats and also considers held live notes and sounding step notes. Generated chords, kits, slices and microtonal tracks do not guide its choices. An out-of-scale root takes the scale note below it (the next available note at the bottom of the MIDI range). TRN shifts the complete harmonic scale. At the ends of the MIDI range, the complete chord moves by octaves. Repeated presses with the same context keep the same chord; held chords stay unchanged through strumming, sustain and release. After a two-beat pause, STOP, panic or a scale/root/transpose/layout/voicing change, the harmonic history starts fresh. Recordings retain the notes you heard. CHR allows all twelve notes and defaults to major triads. AUTO plays only the root in mono modes, keeps kit mappings, and plays the input note alone for microtonal selections; use the existing chord modes for those tunings.

AUTO2: melody with harmony underneath. This second automatic option keeps the mapped key as the highest note. It chooses a familiar three-note chord containing that melody, using only notes in the selected Western scale. The same key chooses the same harmony regardless of the backing tracks, previous presses or timing. Original AUTO retains its existing behaviour.

With AUTO2, CLOSE places accompaniment within the octave below your melody. OPEN lowers the bottom voice; INV1/INV2 choose the third/fifth as bass, adding a lower copy when the melody is that chord tone; +OCT adds a lower root. Your melody stays on top in every voicing. During a strum it sounds immediately, followed by accompaniment in DOWN/UP/ALT order. At the bottom of the MIDI range unavailable lower notes are omitted. Out-of-scale inputs map downward as in AUTO; QNT ALL makes successive keys walk the scale. Mono modes play the mapped melody alone. Kits and microtonal selections keep AUTO's bypass behaviour. The CHORD page highlights the melody and names the resulting harmony.

QNT: OFF (chromatic), SNAP (every key, rounded down to the scale), WHITE (white keys only; black silent), ALL (every key is the next scale degree), MPC (for an Akai MPC pad controller). With QNT MPC, MPC Sample banks **E, F, G and H play Melodee tracks 1, 2, 3 and 4** respectively, on any MIDI channel, overriding ROUT and the DRUM channel for notes. Playing a valid MPC pad silently selects its Melodee track, so the controls and octave indicator follow the most recently played bank. MPC mode suppresses the “MIDI IN → T…” notices; normal MIDI routing also remains silent and keeps the manually selected track. Each melodic bank’s 16 pads play successive scale degrees. The **DEG** control sets the degree pad 02 plays (1 = ROOT); pad 01 is one degree below, pad 03 one above. On Stage, the octave buttons shift only the selected track’s bank (-3 to +3); pressing both resets that bank. The header and button lights show its octave. Each bank’s setting survives track/sound changes and is saved with projects and templates. Outside MPC mode, the shared keyboard octave stays independent. This uses the default program pad map: E = MIDI 100-115, F = 116-127 then 0-3, G = 4-19, H = 20-35. Banks A-D and absent tracks are silent. Enable the MPC’s Pad MIDI Out (Always, or Empty for empty pads). Controllers retain their MIDI channel routing; sustain and panic also follow that channel’s held notes across banks. Incoming MIDI follows WHITE and ALL the same way the keys do, without the octave buttons. ROOT, SCL, QNT and DEG are shared by all tracks; TRN (transpose) remains per track. Older projects with differing roots use the saved selected track’s root when loaded. Chord keys work on the keys, MIDI in, recording and the arpeggiator. Native drum keys keep their chromatic key map and octave; other kits and slices keep their own maps.

MPC drum pads: A DRUM track automatically uses this finger-drumming layout in its assigned bank. Pad 01 is bottom left. Drum pads ignore bank octave, scale, ROOT, TRN and DEG, so each pad keeps its sound. Normal MIDI outside MPC mode and recorded patterns retain their GM drum mapping.

Left	Middle left	Middle right	Right
13 Conga low	14 Conga mid	15 Conga high	16 Clave
09 Tom low	10 Tom mid	11 Tom high	12 Maraca

Left	Middle left	Middle right	Right
05 Closed hat	06 Open hat	07 Cowbell	08 Cymbal
01 Kick	02 Snare	03 Clap	04 Rim

The 909 keeps these main positions, with 07 Ride and 08 Crash. Its extra pads repeat useful hits: 12 Ride, 13 Kick, 14 Snare, 15 Closed hat and 16 Open hat. Each pad retriggers the hit even while its duplicate is held.

Arpeggiator

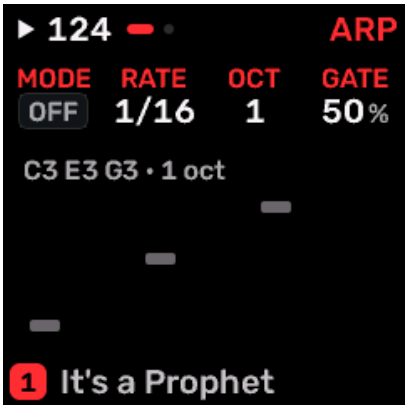
Step by step

1. Select the part and open ARP through the LFO/Motion controls. Choose a mode other than OFF.

2. Hold a chord and set rate, octave range and gate. Use ARP 2 for swing, probability, hold and note ordering.

3. Turn HOLD off or stop/panic to end latched notes. Save the program after choosing the pattern and timing.

The beat light follows the arp when it runs.



ARP.

Page	Controls
ARP	MODE (Off, Up, Down, Up then Down, Random, Order as played, Repeat), Rate (note division), Octave (1-4), Gate (note length %).
ARP 2	SWG (swing), PROB (probability each step plays), HOLD (keep arpeggiating after release), ORD (ordered by NOTE pitch or PLAY order).

Sequencer

Step by step

1. Select a part, press SEQ and choose PATTERN. Set length and division, then set project BPM on TEMPO.

2. Open NOTES and enter steps, or arm REC and play a take. Use PLAY to start/stop; an empty pattern will remain silent.

3. Adjust swing, gate and recording quantize using the separate controls below. Recording quantize affects the next take, not already recorded notes.

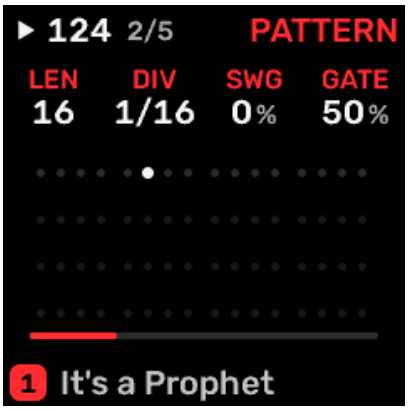
4. Save a PROJECT to retain pattern banks and arrangement. Program changes preserve the current sequence rather than replacing it.

Press **SEQ** for the sequence family; press again or turn **SELECT** through the pages. Patterns are up to **64 steps**; divisions run from 1/32 to 4 bars. **PLAY** starts and stops; **REC** arms the selected track without starting the transport; **REC**+**PLAY** arms and starts recording together.

Page	What it does
NOTES	Edit entered steps and recorded takes (piano roll; drum grid on Drum).
PATTERN	How long the selected track's loop is, how fast its steps run, and its own swing and note length.
TEMPO	Project speed (BPM) and song swing; the project-wide quantize grid for the next recorded take on any track.
PATTERNS	One row per track in use, eight patterns each.
SONG	Arrangement of sections.
MOTION	Knob automation for the selected pattern (shown once that pattern has some automation saved).
DRUM HIT	On a Drum track: the macros of a selected recorded hit.

PATTERN and TEMPO

Think of a **pattern** as one looping phrase on one track - a drum groove, a bass line, a chord stab. **PATTERN** sets how that loop is built for the selected track. **TEMPO** sets how fast the whole song runs, plus a song-wide groove feel. Each track can still have its own pattern length, step size, swing and note length, so the parts need not line up the same way.



PATTERN: this track's loop shape.



TEMPO: song speed and feel.

Control	What it does
LEN	How many steps before the pattern repeats (1-64). Sixteen steps at 1/16 is one bar of common 4/4; eight steps is a half-bar loop; longer lengths stretch a phrase across more bars.
DIV	How long each step lasts relative to the beat: 1/4, 1/8, 1/16, 1/32, triplets (8T, 16T), or slower spans (1/2, 1/1, 2BAR, 4BAR). Finer divisions pack more events into the same time; coarser ones make each step last longer.
SWG	Swing for this track's pattern: delays the even-numbered steps so the rhythm leans instead of sitting dead straight. 0% is even; raise it for a shuffled or "swung" feel. Independent of song swing on TEMPO.
GATE	How long manually entered steps stay on, as a percentage of the step. Low values are short and clipped; high values are sustained (ties and recorded note lengths still follow what you played or edited).

Control	What it does
BPM	Song speed in beats per minute (40-240). Saved with the project. Set this on SEQ → TEMPO; the 3.5.1 GLO hold toggles layering.
SWG	Song swing: a project-wide lean applied on top of timing. Use this when every track should share the same groove; use PATTERN SWG when only one part should shuffle.
QNT	One project-wide grid for the *next* recorded take on any audio or MIDI track, saved with the project and template. **OFF** (default) keeps the timing you played. A division snaps that take's note onsets to the swung grid when recording ends; lengths and velocities stay. The piano roll then shows the committed timing. Changing QNT later chooses the next take's grid - it does not redo a take already recorded. If a manual step is in the way, the take is left alone and the screen says **QNT STEP OCCUPIED**. This is separate from scale QNT on SCL.

Capture into an empty pattern can set LEN for you (1, 2 or 4 bars) - see Recording.

Notes

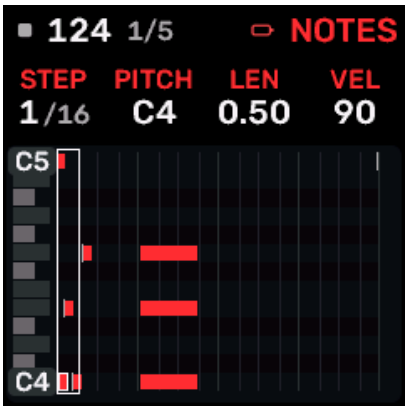
Step by step

1. Stop playback and open SEQ → NOTES for the selected part and pattern. Move to the desired step using the controls shown below.

2. Enter a note or chord, then adjust its length, velocity and probability. Use accent/slide only where the selected sound supports them.

3. Enable NOTE PREVIEW in GLO → RECORDING if you want to hear edits. Play the loop and save the project.

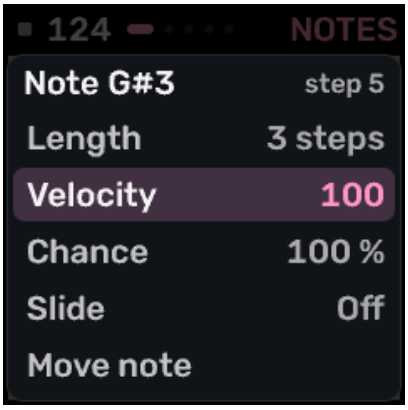
SEQ → NOTES is the editor for manual steps and recorded takes. The highlighted note has a bright outline; the playhead is a separate line. Recorded notes keep their original timing beside ordinary steps and tie tails.



Piano roll: steps, recorded notes, ties.



A single step highlighted.



Note sheet (OCT+ on a note).

Control	What it does
KNOB 1 STEP	Walk every interval, including empty ones and each pitch of a chord.
KNOB 2 PITCH	Change pitch; on an empty manual step, insert a note. Keys and MIDI enter notes or chords when stopped (then advance when released).
KNOB 3 LENGTH	Resize at quarter-step boundaries (0.25, 0.50, 0.75, 1.00, …). Hold **ENV** for fine 1/16-step adjustments on recorded notes.
KNOB 4 VEL	Velocity 1-127. Hold **ENV** and turn KNOB 4 for SLIDE on a manual note.
PRESETS	Zoom (16 steps down to one). Hold **SCL** and turn SELECT or KNOB 1 to move; hold **ENV** and turn them to resize.

Control	What it does
EDIT	Tap to delete the highlighted pitch. Hold **EDIT** while recording to erase under the playhead (header: ERASING). Otherwise a hold undoes.
OCT+	On empty space places a note; on a note opens its sheet: Length, Velocity, Chance, Slide, Move note, Copy note, Delete note. Recorded notes offer Length, Velocity, Move, Copy and Delete.

Stop live recording before editing recorded properties. Hold **EDIT** to undo step edits; **EDIT**+**OCT-** / **OCT+** undo / redo up to eight edits. An erase pass starts a fresh history. Tap **EDIT** deletes the highlighted pitch, including a recorded hit.

Hold **OCT+** and turn **SELECT**, **KNOB 1** or **KNOB 2** to select a range, then turn **KNOB 1** to move those notes at quarter-step boundaries. Tap or hold **OCT+** for **Copy: KNOB 1**, **Quantize**, **Delete selected** or **Clear selection**. Delete selected removes manual and recorded notes in the range. A single note's **Move note** or **Copy note** works the same way: choose it, then turn **KNOB 1**. Copies and moves wrap around the loop. Hold **OCT+** without a range for the pattern tools: **Quantize** (the whole pattern, or the selection if there is one), **Clear pattern** and **Clear motion**. On Quantize, **KNOB 1** chooses **1**, **0.5** or **0.25** step and **OCT+** applies it. Durations and velocities stay. Each operation is one undo. Moving, copying and quantizing notes need the transport stopped. Clear pattern and Clear motion requested during playback wait for the track's loop end.

Patterns

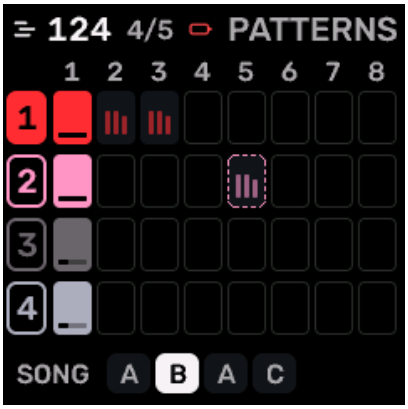
Step by step

1. Open SEQ → PATTERNS, or hold SEQ to reach the pattern grid. Select the track and its pattern slot.

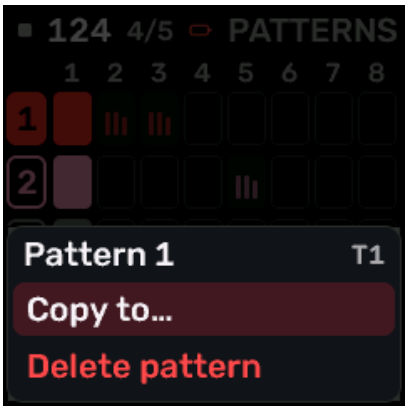
2. Use the displayed pattern actions to select, copy or clear a pattern; confirm destructive actions only after saving the project.

3. Play the pattern to check transitions, then save the project to retain the banks.

Eight patterns per track. The PATTERNS grid shows one row for each track in use; **KNOB 1**-**4** queue patterns for tracks 1-4 when that track exists. **SELECT** goes on to SONG.



PATTERNS: one row per track, eight patterns.



Pattern sheet: Copy to…, Delete.

• Hold **SEQ** and press one of the first eight white keys to select pattern 1-8 (or use the PATTERNS grid / **SEQ**+**SELECT**). The playing pattern is lit; ones with notes glow; one waiting for the loop end breathes. Selecting an empty pattern arms one recording pass. Double-tap a used pattern key to arm one pass there. Those shortcuts disarm after the pass; an already explicit REC arming stays on. On PATTERNS while unarmed, or with SEQ held on any page, the first eight black keys select tracks 1-8 directly. Unused track keys stay dark. On PATTERNS, pressing one creates the missing tracks through that slot and selects it without leaving the page; tracks 5-8 are MIDI-only.

• Turn **PRESET** on PATTERNS to change the active audio track's sound. A temporary card shows its name and engine immediately. Keys play the sound while the card is open; it closes two seconds after your last turn or released audition note, then keys select tracks and patterns again. Hold **SEQ** to select during preview, or press **OCT-** / **OCT+** to close the card. PRESET leaves MIDI tracks 5-8 unchanged.

• Hold a pattern key alone for a second to open its action dialog: press that key again to delete the pattern, another pattern key to copy it there, or **OCT-** to cancel. In the dialog the fourth and fifth black keys (C■, D■) blink: C■ halves the pattern (the steps past the new length stay hidden), D■ doubles it (its steps, recorded notes and motion copied after themselves, up to 64 steps). Keep the original pattern key or **SEQ** held to leave the dialog open and halve or double repeatedly. After a resize, releasing both closes it. On the playing pattern, hold **EDIT** to undo.

• On PATTERNS, **OCT+** opens the sheet: **Copy to…** (confirm before replacing a used slot) or **Delete pattern** (confirm). Delete removes the pattern's manual notes, recorded notes and motion together. **OCT-** clears what is queued. Use **EDIT** + **OCT-** to undo; holding EDIT for two seconds opens the layer mixer.

• While playing, a pattern change waits until that track's loop ends. Selecting the active pattern cancels a queued change. Destructive pattern operations (delete, clear notes or motion, or copy over a used or playing pattern) also wait for that track's loop end. The header shows the pending action with **AT LOOP END**; the pattern key breathes. Live note, hit, erase and motion-lane edits stay immediate. STOP applies pending choices.

Songs

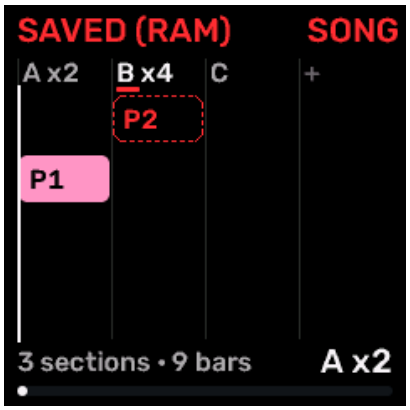
Step by step

1. Open SEQ → SONG or Studio → Projects in the editor. Add a row to begin the arrangement.

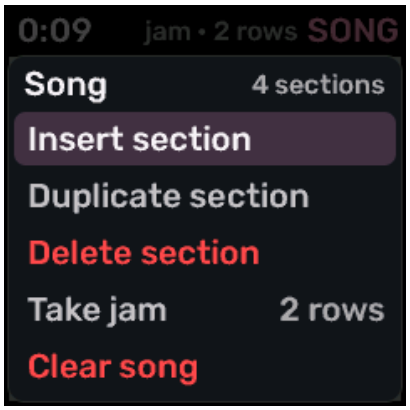
2. Choose the patterns and repeat count for each row. Add subsequent rows in playback order.

3. Play from start to audition the arrangement, then stop and save the project. A setlist of programs is separate from this pattern arrangement.

On **SEQ** → **SONG**, the knobs select section, track, that track's pattern, and how many times the section repeats (up to 16 rows, repeats 1-16).



SONG: sections of patterns.



Song sheet.

• While you jam with no song playing, every loop of track 1 logs the tracks' patterns as a section (same patterns again: a repeat).

• Hold **REC** on SONG (or use Take jam on the sheet) to turn that log into the arrangement. Overwrite asks for confirmation.

• PLAY runs the arrangement; sections change every track at track 1's loop boundary; the final section stops. STOP returns to the patterns selected before SONG.

• Sheet actions: Insert / Duplicate / Delete section, Take jam, Clear song.

Motion

Step by step

1. Select a part and pattern containing notes. Start playback and arm the motion-recording operation described below.

2. Move the desired control during the pattern, then stop recording. Open MOTION to inspect or edit captured automation.

3. Clear only the unwanted automation lane using its action, then replay to verify and save the project.

Knob moves recorded into the pattern: Prophet and CZ-1 Stage knobs and sound pages, FM6 operator and voice pages, and other motion-capable sound and effect controls. MOTION lists one lane per automated control on the pattern that is playing.



MOTION: KNOB 3 picks the step, KNOB 4 sets its value.

Control	What it does
KNOB 1	Motion playback on or off for this track.
KNOB 2	Select a lane. Hold **SEQ** and turn it to clear that lane.
KNOB 3	Select a step in the lane.
KNOB 4	Set that step's value. A step with no point gains one at the value the sound is holding. Hold **SEQ** and turn it to remove the point.
OCT+	The lane's sheet: Play, **Clear** that parameter, Clear all motion.

• Arm **REC**, start playback, then turn a sound knob. Armed but stopped: the screen shows **PLAY TO RECORD MOTION**.

• Edits and clears need the song stopped. If a song is playing or armed, the screen says **STOP TO EDIT**. Hold **EDIT** to undo one edit or clear. Other lanes, tracks and pattern banks stay as they were.

• Stopping or bypassing motion restores the original patch; project snapshots keep that original.

• Hold **LFO** while turning knobs for temporary tweaks that do not record as motion; release to restore, or press **OCT+** while LFO is held to keep them.

• Shared pool of **64 events** across the project. MOTION is hidden until the selected pattern has events. **MOTION FULL** when the pool is exhausted.

Recording

Step by step

1. Select the target track and an empty pattern. In GLO → RECORDING set click, click level and count-in.

2. Set recording quantize on SEQ → TEMPO, then arm REC. Press PLAY (or REC+PLAY) to begin and perform the notes.

3. Stop recording and inspect NOTES. Correct individual notes or repeat the take; save the project when satisfied.

• Tap **REC** to arm or disarm the selected track. Only one track can be armed at a time: arming another finishes the previous track's take and disarms it. Transport stays put.

• Arm before **PLAY** to record from step zero. Arm during playback to record from that track's next pattern start. Explicit **REC** arming stays on across loops until you tap REC again or stop. STOP finishes the take and leaves the track armed for the next start. **REC**+**PLAY** arms and starts in one gesture. Timing, velocity and held length are kept (including repeats and overlapping chord releases).

• **Undo a take:** when you stop or disarm, or a one-pass shortcut finishes, hold **EDIT** to put that pattern back as it was (**UNDO RECORDING**). Motion recorded while armed is included, even when no notes were played. There is one such undo. Details are under Sheets and undo.

• While stopped and armed, the first key or MIDI note starts playback and records immediately, skipping count-in. A count-in you start with **PLAY** still runs (internal clock only; external clock skips it).

• **Capture (Take Taker):** sometimes referred to as "retroactive recording" - while the transport runs, Melodee keeps only what you play on the selected track. Switching tracks clears the previous capture buffer; notes on other tracks are ignored. Hold **REC** to write the selected track's last bars into its pattern. An empty pattern takes 1, 2 or 4 bars and sets LEN (the pattern length on SEQ → Pattern); a pattern with notes takes its last LEN steps. Use **EDIT** + **OCT-** to undo; holding EDIT for two seconds opens the layer mixer. Capture pauses while a track is actively recording.

• **SEQ** → **TEMPO** → **QNT** sets one recording grid for all audio and MIDI tracks, saved with the project or template. It quantizes the new take's onsets when recording ends. **OFF** (default) keeps the timing you played. This is separate from scale QNT on SCL. Track GATE affects manually entered steps.

• A project shares **1,024 timed notes** across all 32 banks. **RECORDING FULL** leaves existing takes intact; clear or replace recorded steps to free space.

The header shows **WAIT LOOP** while recording waits for the selected track's loop boundary, and **ERASING** during an erase pass. These stay visible while the state applies. An armed track has a REC border around its Stage row.

There are some Menu (long hold **HOME**) preferences related to recording: AUDIO CLICK, CLICK LEVEL, COUNT-IN.

Global: ten named pages

Tap GLO to open the global settings. Tap it again or turn SELECT to cycle pages. Each screen shows its name and GLO page number (1/10-10/10). Knobs 1-4 control the corresponding labeled positions; blank positions do nothing. HOME returns to the program. Hold EDIT for two seconds for the four-volume mixer. MIDI Learn stays in the editor; sound, tuning, effects, envelope, LFO and sequencer editing remain under their own buttons.

1. Tap GLO and turn SELECT to the required named page.

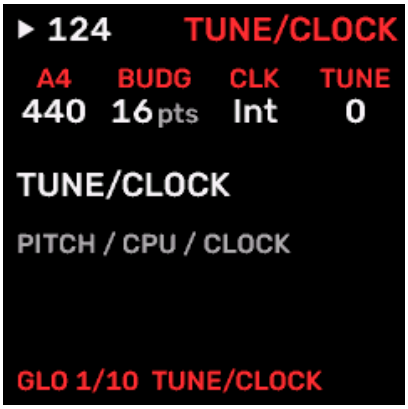
2. Turn the knob matching the setting shown on screen.

3. For a color change, choose DISPLAY (6/10), then turn knob 1 through the theme presets.

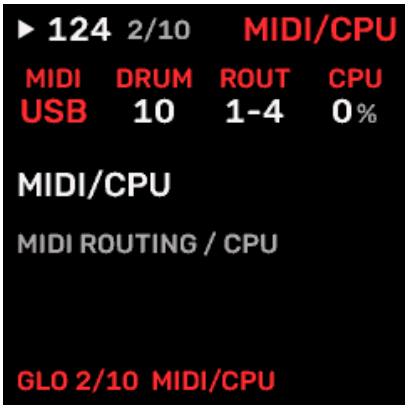
4. Return HOME. Device preferences persist separately from saved programs; a firmware flash resets them.

Page	Knobs and purpose
1/10 TUNE/CLOCK	A4 reference (400-480 Hz), shared voice BUDGET (8-32 points), clock source (INT/USB/TRS), fine TUNE (±50 cents).
2/10 MIDI/CPU	Input USB/TRS, drum MIDI channel, channel routing, CPU information. This is routing, not MIDI action mapping.
3/10 VOICE COST	Per-voice costs for Prophet, CZ, FM6 and SID. AUTO uses the engine defaults. Higher costs reduce available polyphony to protect processing headroom.
4/10 LAYERING	Switch all four parts between layered/split playing and selected-part playing. Hold GLO for two seconds for the same toggle.
5/10 CPU GUARD	Drum voice cost, shared voice budget and overload SAFETY. Raising the budget or disabling safety can cause overload; the four parts and drums share processing capacity.
6/10 DISPLAY	COLOR theme preset, FONT, button/key LIGHTS and automatic SCREEN OFF delay. Turn knob 1 to cycle the theme presets; the selected palette is saved with device settings.

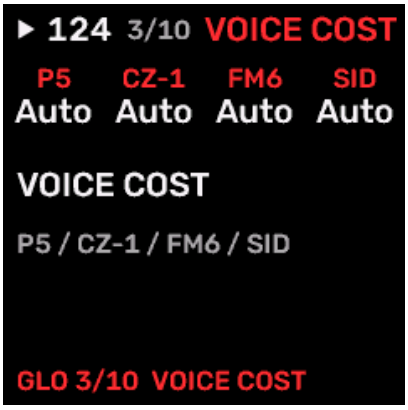
Page	Knobs and purpose
7/10 USB / SAVE	USB AUDIO direction, USB INPUT stereo/four-track capture, USB ROLE COMPUTER/PEDAL, and AUTOSAVE. COMPUTER is required for the editor and installer. USB role changes re-enumerate the connection.
8/10 CONTROLS	Parameter PICKER, knob ACCEL and FX HOLD duration. The fixed SynthMaven two-/three-second shortcuts remain unchanged.
9/10 RECORDING	Audio CLICK mode, click LEVEL, COUNT-IN and note PREVIEW.
10/10 DEVICE	Chord-entry ADD/HOLD preference, panel CALIBRATION and ABOUT. Turn the corresponding knob right to open calibration or device information.



TUNE/CLOCK



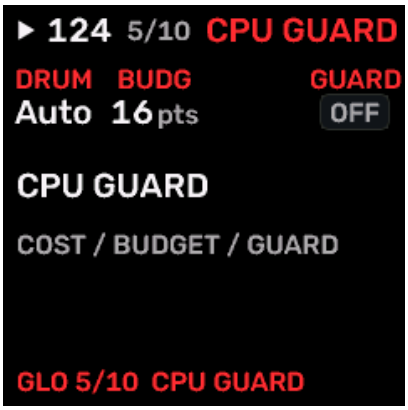
MIDI/CPU



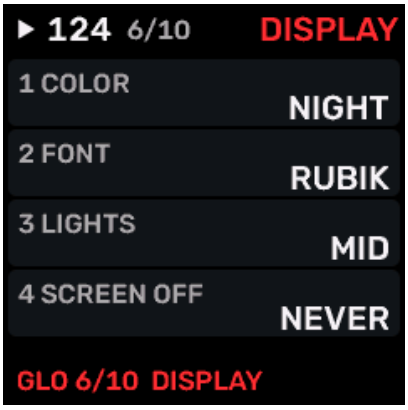
VOICE COST



LAYERING



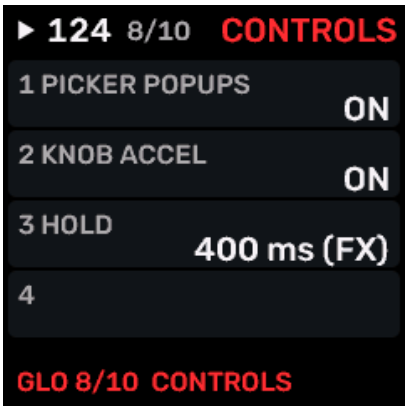
CPU GUARD



DISPLAY



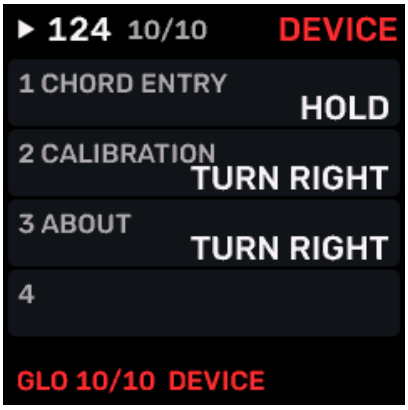
USB / SAVE



CONTROLS



RECORDING



DEVICE

Menu

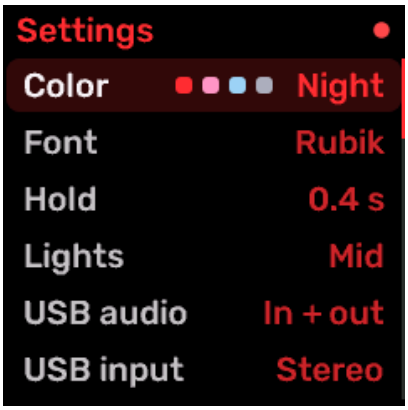
Step by step

1. Hold HOME to open the settings list. Turn knob 2 to the row, then knob 1 or OCT+ to edit its value.

2. For ordinary global settings use the named GLO pages instead: the table below identifies the same preferences and legacy options.

3. Return HOME after changing a preference. Fixed SynthMaven shortcuts and HOME part selection take precedence over inherited menu preferences.

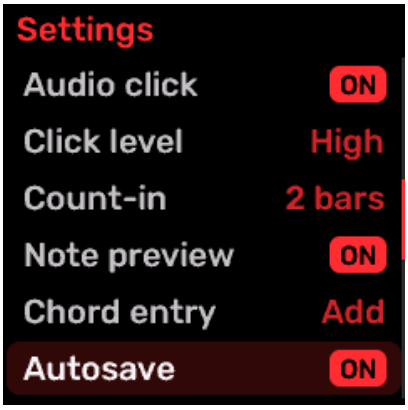
Hold **HOME** for the settings menu (a list; **KNOB 2** moves the row, **KNOB 1** or **OCT+** changes the value).



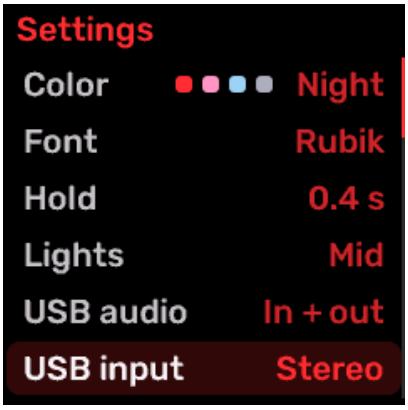
MENU.



HOLD time.



AUTOSAVE.



USB INPUT: stereo master.

Item	What it does
COLOR	**NIGHT** (default), **DAY**, **CONTRAST**, **SUNSET**, **OCEAN**, **NEON**, **CANDY** or **LO-FI**. Knob 1 uses the selected track's color; the other three knobs are shades of it. Remembered.
FONT	**RUBIK** (default) or **PIXEL** (Melodee Pixel, a block face). Pages keep the same layout in either. Remembered.
HOLD	Inherited performance-hold timing (0.3-0.6 s), including FX. It does not change the fixed EDIT/GLO two-second or ENV/SEL/OCT three-second shortcuts.
LIGHTS	Idle indicator brightness, OFF (only pressed / engaged / selectable) through FULL.
USB AUDIO	In+Out / In / Out / Off.
USB INPUT	**STEREO** (default) records the master, including track and shared effects. **4 TRACKS** records an isolated mono stem per track place, after that track's effects and level and before the shared effects and the master. A track that is not in the song is silent on its stem.
MIDI KNOBS	**OFF** (default), **GENERAL** or **MINILAB**. Operate the eight panel controls from a MIDI controller. Remembered; see MIDI knobs.
MIDI MODE	**ABSOLUTE** (default), **REL 1**, **REL 2** or **REL 3**. Match the controller's encoder output. Remembered.
AUDIO CLICK	**OFF** / **REC** (while recording) / **ON**.
CLICK LEVEL	Metronome volume.
COUNT-IN	Bars of count-in before an armed track records when you start with **PLAY** (internal clock only).
NOTE PREVIEW	Hear notes as you edit steps.
CHORD ENTRY	How keys and MIDI build a chord on a stopped step: HOLD = press pitches together, then release to advance; ADD = tap pitches one by one to add or remove them at the same cursor (up to four; a full step shows CHORD FULL).
AUTOSAVE	**ON** (default) writes a changed project back to its slot after five seconds stopped and untouched. **OFF** leaves saving to you. The choice survives power-off.
PICKER POPUPS	**ON** (default) opens a picker for a long list of names. **OFF** edits those names on the knob. Remembered across power-off.
KNOB ACCEL	Knob acceleration on a spin; OFF keeps every detent one step.
HOME SELECT	Legacy Stage preference. SynthMaven HOME reserves SELECT for part selection; this setting does not replace that four-part navigation.
GLO SELECT	What **GLO**+**SELECT** turns on any page: the same choices as HOME SELECT, default BPM. Remembered.
SCREEN OFF	Blank the screen after a time.
CALIBRATION	Hardware calibration (also: hold **OCT-**+**OCT+** at power-on).
ABOUT	Version / credits.
SOUND DESIGNER	Opens the sound designer.

MIDI

Step by step

1. Choose COMPUTER for a host computer/iPad or PEDAL for a directly attached class-compliant controller. Follow the powered-hub wiring instructions above.

2. Set MIDI input and channel routing on GLO → MIDI/CPU. Match the controller transmit channel to the intended part/drum channel.

3. For action mapping open editor Settings → MIDI Learn, capture the message first, assign its action, and save it. Test the received-message/match indication.

4. Return to COMPUTER before reconnecting the editor or installer. MIDI mappings do not substitute for saving edited sounds/programs.

USB and TRS MIDI in and out.

• ROUT chooses which MIDI channels play. **CH1-4**, **CH5-8**, **CH9-12** and **CH13-16** map those four channels onto tracks 1-4, and only onto tracks in use. Notes and controllers on a channel outside the group are ignored. **SEL** sends every channel to the selected track.

• The **DRUM** channel (default 10) plays the first Drum track only when that channel is inside the ROUT group, or when ROUT is **SEL**. With **CH1-4**, channel 10 does nothing: set ROUT to **CH9-12** or **SEL**, or set DRUM to a channel in the group (DRUM 2 plays the kit instead of track 2). With no Drum track, the channel follows ordinary routing. Set OFF to disable.

• A note that already started still releases on the track that played it if you change ROUT so its channel is no longer included.

• The keys send on the selected track's channel.

• Pitch bend, sustain, panic and clock (INT / USB / TRS) are supported. Common CCs control level, pan, cutoff/brightness, envelope times and glide. CC91 does nothing. CC93 operates KNOB1 with the MINILAB profile; otherwise it does nothing.

• Local **PLAY** keeps the internal tempo until the selected MIDI input sends Start, Continue or Stop. Clock pulses alone do not take over.

• Microtonal MIDI out sends degree addresses; an external synth needs the same tuning and mapping to match.

MIDI knobs

Hold **HOME** and choose **MIDI KNOBS** → **MINILAB** for an Arturia MiniLab 3 with its factory Arturia or User CC assignments. Use its MIDI port and Arturia/User program, outside arpeggiator editing; DAW/MCU mode sends a different mapping. Leave **MIDI MODE** → **ABSOLUTE** for factory settings. No controller reprogramming is needed. The profile and mode survive power-off.

There is no universal eight-knob panel layout. **GENERAL** uses MIDI's eight general-purpose CCs for controllers you can assign. Both profiles use this order:

Controller knob	Melodee control	GENERAL CC	MINILAB CC
1	Volume	16	74
2	Select	17	71
3	Presets	18	76
4	Algorithm (track selection)	19	77

Controller knob	Melodee control	GENERAL CC	MINILAB CC
5	KNOB1	80	93
6	KNOB2	81	18
7	KNOB3	82	19
8	KNOB4	83	16

KNOB1-4 follow the current page, selected track and held panel-button layers. In ABSOLUTE mode, the first value establishes the knob's position; subsequent movement sets ordinary parameter values across their full range (0 = minimum, 127 = maximum). Navigation and special controls jog by the change in position, so connecting a controller does not jump to a different page, track or preset. For unlimited navigation with endless encoders, select a matching relative mode: REL 1 uses 63/65 for down/up (64 neutral), REL 2 uses 127/1 (0 neutral), REL 3 uses 15/17 (16 neutral). Relative movement retains each transmitted step without extra panel acceleration.

Volume responds immediately and attenuates the physical master knob: 0 mutes, 127 keeps the physical knob's level. It also applies to computer playback. Turning MIDI KNOBS OFF restores the physical master level. MIDI volume resets to full on power-on.

USB and TRS work identically. A channel admitted by ROUT controls the selected panel track, even when that channel's notes play another track. Mapped CCs are consumed by the panel; they do not also change a synth parameter or pass through a MIDI Track. Other CCs retain their normal behavior. Set MIDI KNOBS OFF to keep all conventional CC behavior, including CC74 brightness and CC71 resonance. The MiniLab assignments are documented in Arturia's MIDI implementation chart.

USB audio

Step by step

1. Connect to a computer/iPad in COMPUTER role. On GLO → USB / SAVE choose the USB AUDIO direction you need.

2. Choose STEREO USB INPUT for the full mix or 4 TRACKS for isolated mono track stems. Select the FM-1 as the recording interface in the host app.

3. Start with host monitoring low; raise FM-1 MASTER and check recording meters. Record a short test and play it back before a longer session.

Class-compliant, no driver. 16- or 24-bit at 44.1 kHz. The computer sees one device named **Melodee**, with playback and recording together:

• **Playback** - play the computer through the FM-1. The computer's volume and mute apply to that playback, on top of the **MASTER KNOB**. Computer playback and the monitor click stay out of recordings, and the computer's volume does not change what is recorded.

• **Recording** - **STEREO** by default: a copy of the master, including track effects, shared effects, pan and the output protection. Hold **HOME** and set **USB INPUT** to **4 TRACKS** for four isolated mono stems, after each track's effects and level and before the shared effects and the master. A track that is not in the song is silent on its stem.

Switch either direction off from the HOME menu's **USB AUDIO**. Changing a USB option reconnects the device; reselect Melodee in your DAW if it disappears. A computer that still lists the older separate playback and recording devices needs that same reselection.

Web editor

Step by step

1. Use a USB data cable and COMPUTER mode. Open the web editor in a Web MIDI-capable browser or SM FM1 on iPad landscape.

2. Connect matching FM-1 MIDI input/output in the connection wizard. Check that the status identifies real hardware rather than Mock device.

3. Select a part, edit controls and audition on the FM-1. Use page buttons for additional controls; save to the appropriate program, user sound, library or project store.

Edit the FM-1 from the computer in Chrome or Edge: Open the patch editor.

Connect over USB for sound and mixer controls, a 6-operator FM editor, Prophet and CZ-1 panels, sequencer and motion, the shared library of 384 user slots, SysEx import/export, projects, settings (including BUDGET), and complete backup / restore.

Backup and switching firmware

Step by step

1. Before flashing, connect in COMPUTER mode and stop playback. Open Studio → Projects in the editor and choose Save backup. Keep the downloaded file outside the app/browser.

2. Also export any editor-local library entries and named custom scales; these are not all stored in the FM-1 backup.

3. Install firmware only after the backup finishes. A firmware flash resets previous device settings and programs to the fresh factory state.

4. To restore, reconnect to the editor, choose Restore backup and select a compatible archive. Wait for completion, then verify all eight programs and important settings. Restore only data you intend to keep over the new factory setup.

• Before leaving Melodee for another custom firmware, choose **Back up and install TransitFM1** in the installer, then use the destination firmware's own installer once the FM-1 shows **TRANSIT**. Use **0.1.3**. **Baud Girl requires official V15 first**, then its own installer; TransitFM1 refuses images beyond its write range.

• TransitFM1 makes no sound. It guards the next firmware's write against interruptions except possibly its last sector. Close DAWs and other apps sending MIDI. Keep power and USB connected; if interrupted, resume the same firmware.

• The installer's **Return to official V15** saves a complete Melodee backup first when the firmware can export one.

• **Skip backup** is available when export is impossible; confirm that Melodee music, sounds and settings may be lost.

• After reinstalling Melodee, restore from the web editor's backup tools.

• Older Melodee builds cannot read newer data (for example native motion events need 1.0.2 or later; the new effects and recorded takes need 1.3.0 or later; the shared 384-slot bank needs this version). Export user sounds and keep the installer's backup before downgrading. A backup made before the preset migration is what an older version can restore.

• Startup maintenance stays on screen while it works: the job's name, then **Keep power connected** and **Do not switch off.** A short **CHECKING SETTINGS** pass can appear when saved settings need a small repair. The first start that converts older preset banks is **MIGRATING PRESETS**. If it cannot finish, the screen says **Could not finish** and **Restart to retry**. **OCT+** (**KEEP**) keeps the presets and continues (preset writes stay protected until conversion succeeds). **OCT-** (**WIPE**) asks **Wipe all user presets?**, then **OCT+** confirms. Projects, settings and factory sounds stay. A wipe that was confirmed resumes if power is lost during it.

• **RESETTING PRESETS** runs when a bank or a slot cannot be read. An unreadable bank becomes empty. Unreadable slots in a bank that can be read are cleared; presets this version understands stay, including a recognized older conversion. A newer preset record this version cannot read is cleared. If the reset cannot finish, the same **KEEP** / **WIPE** choice applies.

• **RESETTING SETTINGS** replaces device settings this version cannot read with the defaults (color, calibration, and the globals kept on the device). Projects stay. A template stored with those settings is not kept. Settings saved by a newer Melodee are left alone. If the write cannot finish, the screen stays until you restart; **OCT+** continues, and there is no wipe.

• The updater does not scan or move legacy Prophet banks from the abandoned flash area. Sounds that lived only in that old area are not recovered on upgrade; keep an export or a full backup if you still need them.

• A slot saved before Melodee 1.0 shows **OLD FORMAT** and is left as it is until you choose to replace it. The device asks first. A backup it cannot read is refused before anything is written.

If something goes wrong

• Install stops or fails: unplug and replug the cable, use a data cable and a direct port, open Chrome or Edge, and press Install again.

• Three short key-light flashes mean the MIDI update loader is running; one long flash means ROM UBOOT. A unit that shows **Melodee Update** is in the MIDI update loader: press Install again, or run `tools/fm1_install.py` with a working package. An update that loses power while the application is being written comes back here on its own. Hold **OCT-** alone as you power on to force that rescue. Holding **OCT-** and **OCT+** together is calibration, not rescue. The first install of this recovery, and an update that rewrites the boot stub, can still need the ROM recovery below if power is lost during that write.

• A unit that no longer starts and appears as USB **4c4a:8057** (inquiry **WL82 / UBOOT1.00**) is already in ROM UBOOT. Release downloads include `fm1-recover` for Windows x64, Linux x64, and macOS Intel and Apple Silicon. Extract the archive, open a terminal in that folder, and run `fm1-recover --info` (`.\fm1-recover.exe --info` in Windows PowerShell). Passing a `.fwsc` file makes a full backup and leaves flash alone; add `--write` to restore application flash. The bootloader and the current saved-data regions are kept. Read `RECOVERY.md` in the archive for drivers and the full procedure. The tool does not force a running device into UBOOT. Melodee 1.4 does not change the system clock. A black screen left by a 1.3.0 or 1.3.1 clock change is this same UBOOT state: recover, then install.

• Report Melodee bugs on GitHub Issues, not to Felucca or M-VAVE.

Factory programs

These eight four-part programs preserve the owner's final instrument edits. Start on P1; hold OCT+ for three seconds for the next program or OCT- for the previous program. Use SELECT on HOME to choose a part, ALGORITHM to change its engine and PRESETS to change its sound. Save after editing.

Program	Lead engine	Supporting engines
P1 DX7 STAGE	FM6 / DX7	Analog, Hybrid, FM6
P2 E. Organ	FM6 / DX7	Juno, Hybrid, FM6
P3 RHODES EP	Digital	Juno, Hybrid, FM6
P4 DISCO LEAD	Hybrid	Analog, CZ, Prophet
P5 VINTAGE LEAD	Analog	Hybrid, CZ, Analog
P6 LEAD GUITAR	FM6 / DX7	FM6, Digital, VPS

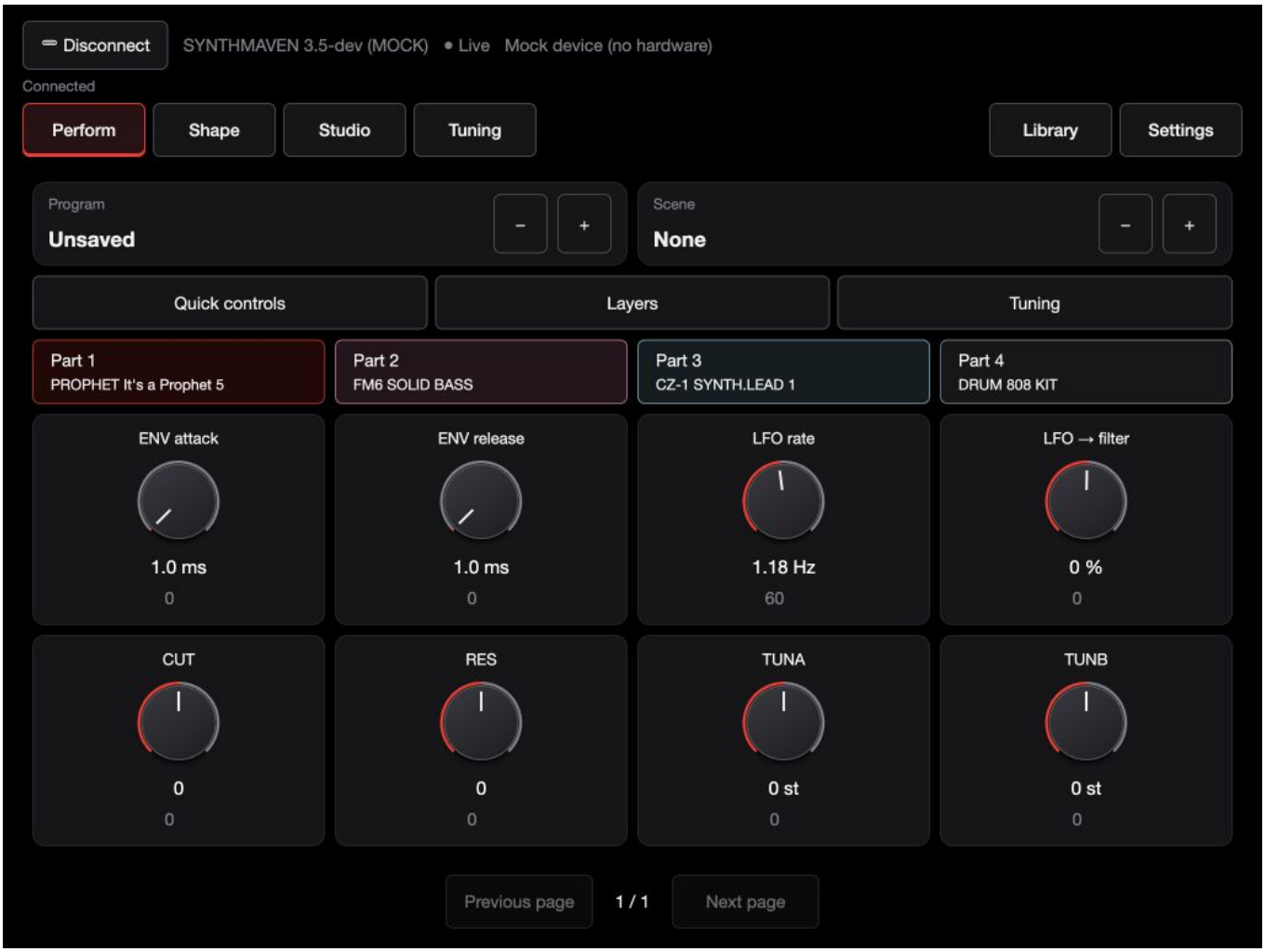
Program	Lead engine	Supporting engines
P7 ARAB STRINGS	Strings	Strings, Prophet, Strings
P8 RNB BASS	CZ	Analog, Digital, Analog

Effects and scene data are retained from the captured programs. Global tuning stays in effect unless a saved tuning override is enabled. Polyphony is shared across layers and drums; the heavier ensemble string engine reserves more capacity per voice. This may shorten overlapping notes under load, without changing the stored sound settings.

Illustrated editor walkthrough

These screenshots show the shared web/iPad interface at a 1024 × 768 landscape viewport using mock demonstration data. Factory-program names and sounds on the connected instrument differ from this mock example. The same workflows operate over USB in the browser and in SM FM1. All edits described below require a real connection to affect the instrument.

Perform: quick edits



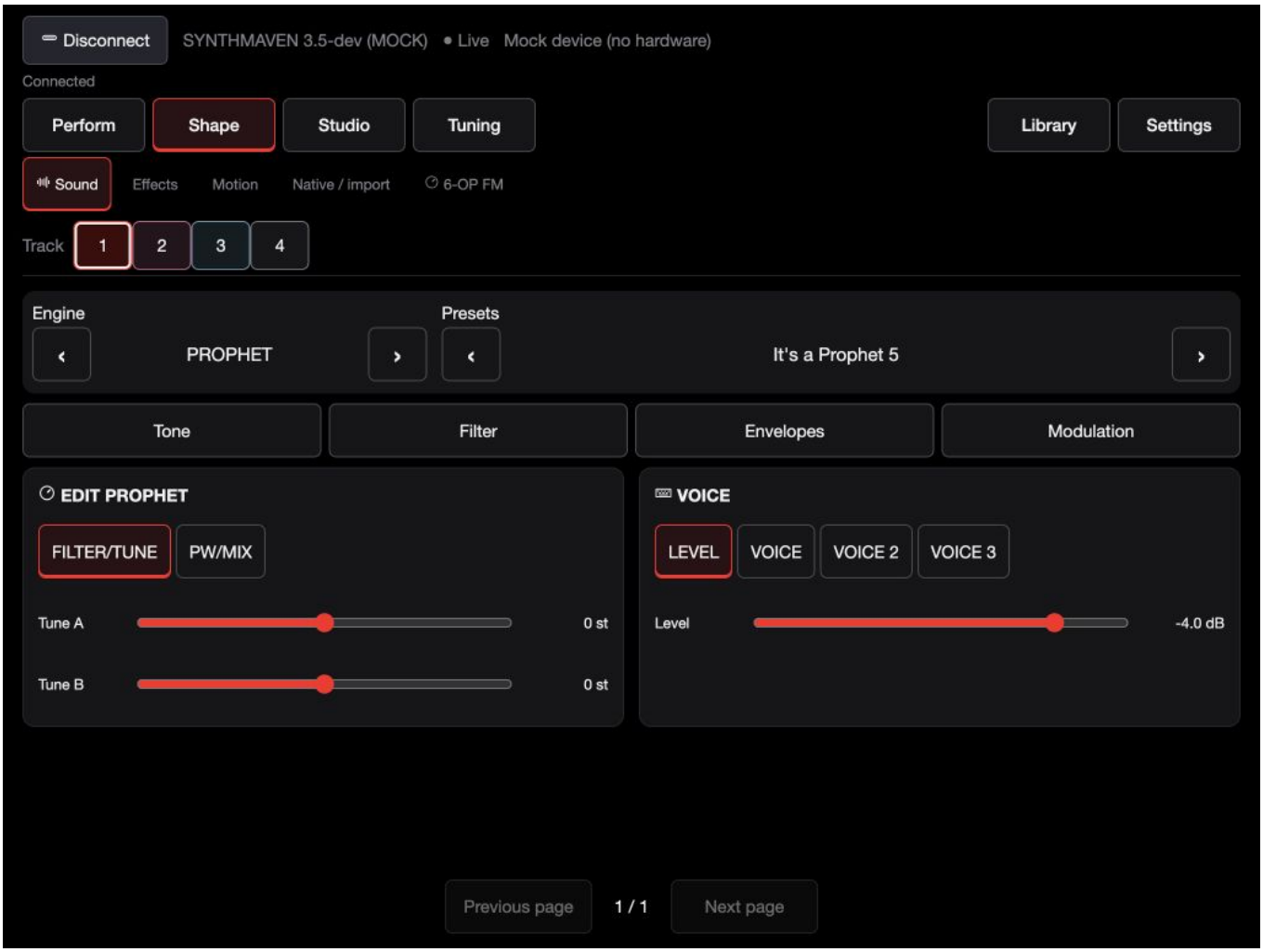
Perform: quick edits - shared web and iPad layout.

1. Select a part, adjust ENV or LFO and the engine controls, then use Program Up/Down or Scene Up/Down.

2. The names show the active selection.

3. Save edits in Programs before loading a different program.

Tone



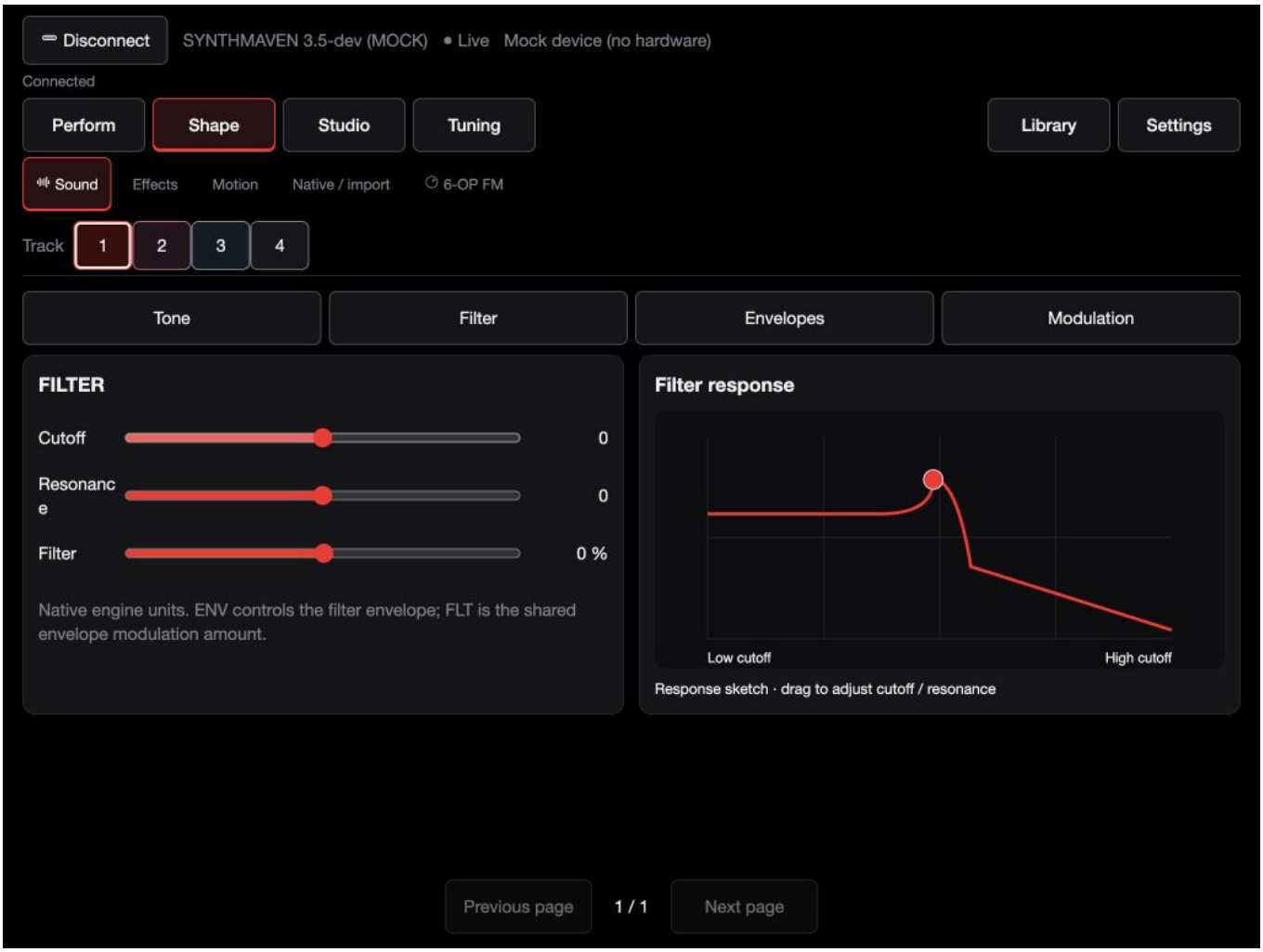
Tone - shared web and iPad layout.

1. Choose an engine and preset with the side arrows.

2. Select a card page to reach additional voice settings.

3. Each page changes the same selected part.

Filter

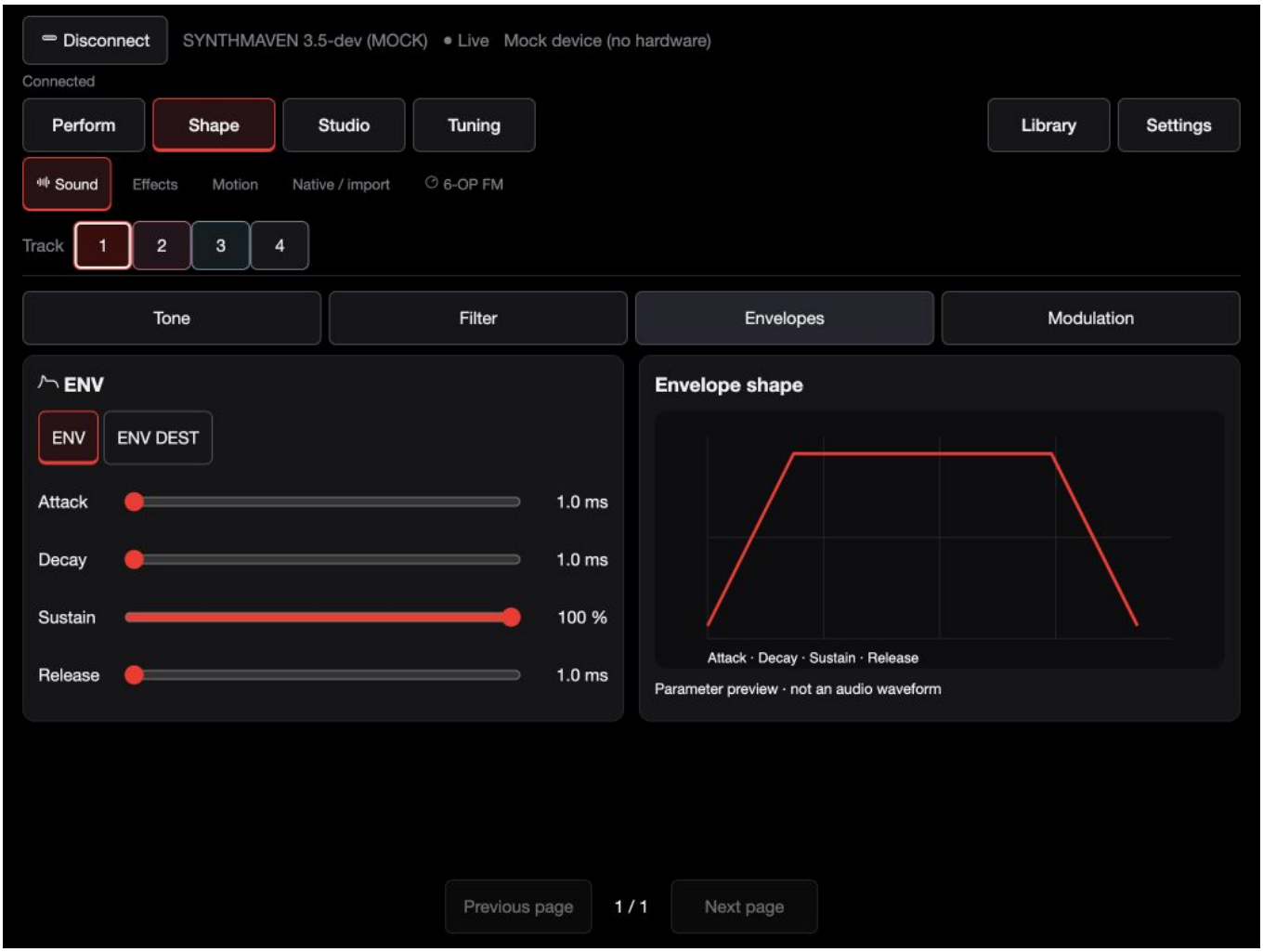


Filter - shared web and iPad layout.

1. Select Filter, choose the supported model and adjust cutoff, resonance and envelope amount.

2. Drag the response sketch to change cutoff and resonance together.

Envelopes



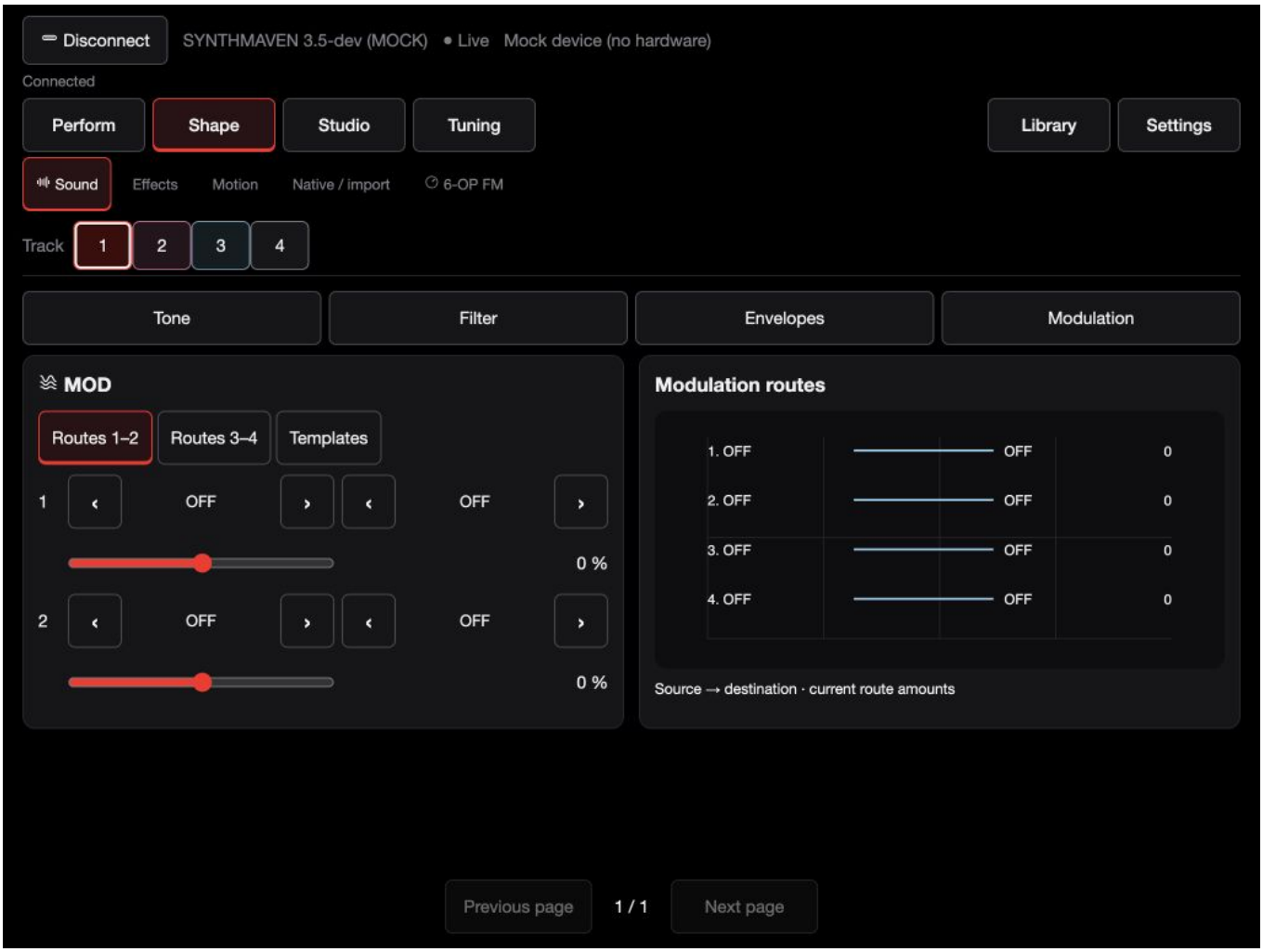
Envelopes - shared web and iPad layout.

1. Choose Envelopes.

2. Set attack, decay, sustain and release; use the card pages for further controls.

3. The adjacent graph previews those settings.

Modulation



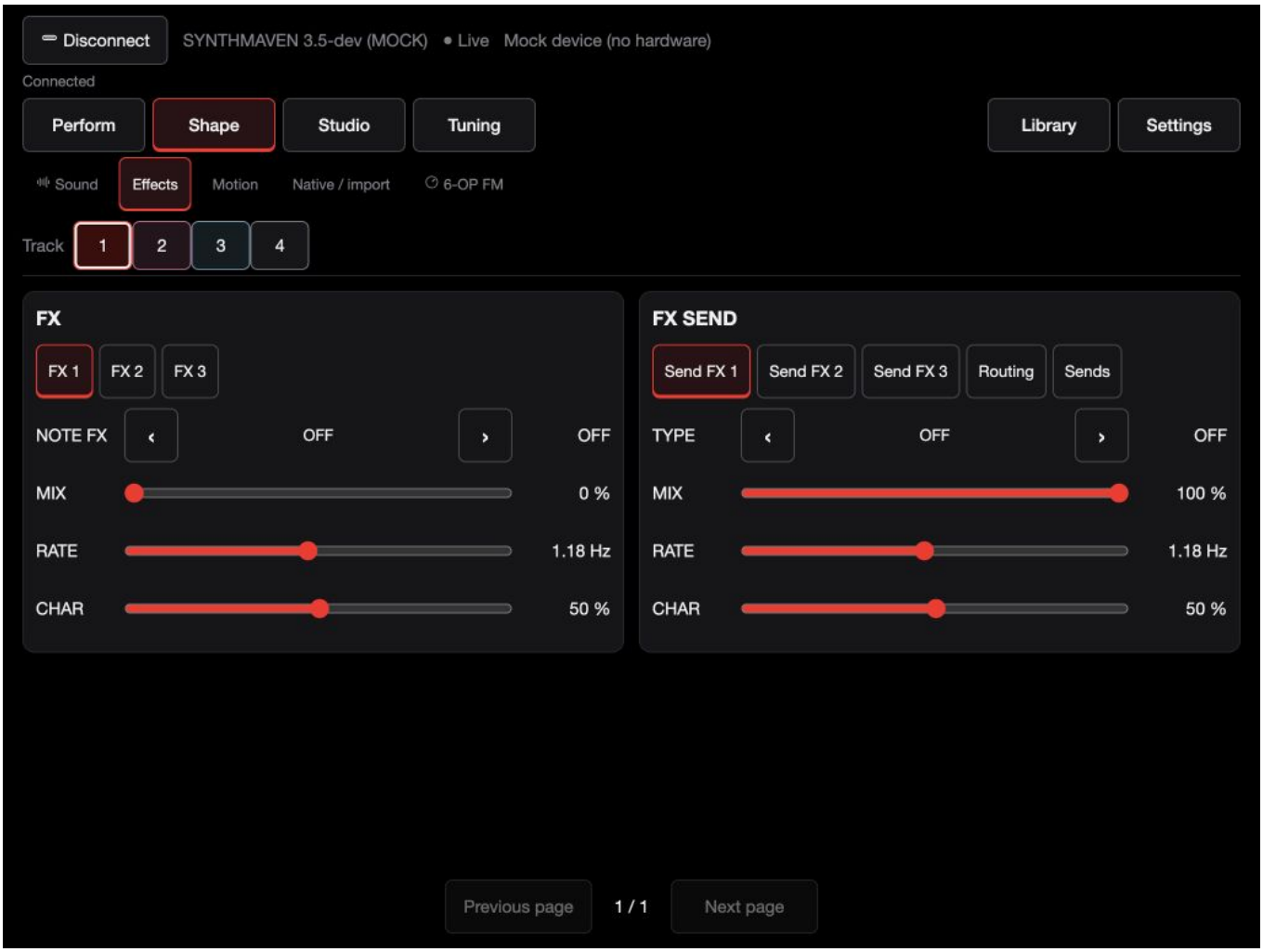
Modulation - shared web and iPad layout.

1. Choose Routes 1-2 or Routes 3-4, then source, destination and amount.

2. Templates fill a free route.

3. Save the sound/program after applying a template.

Effects



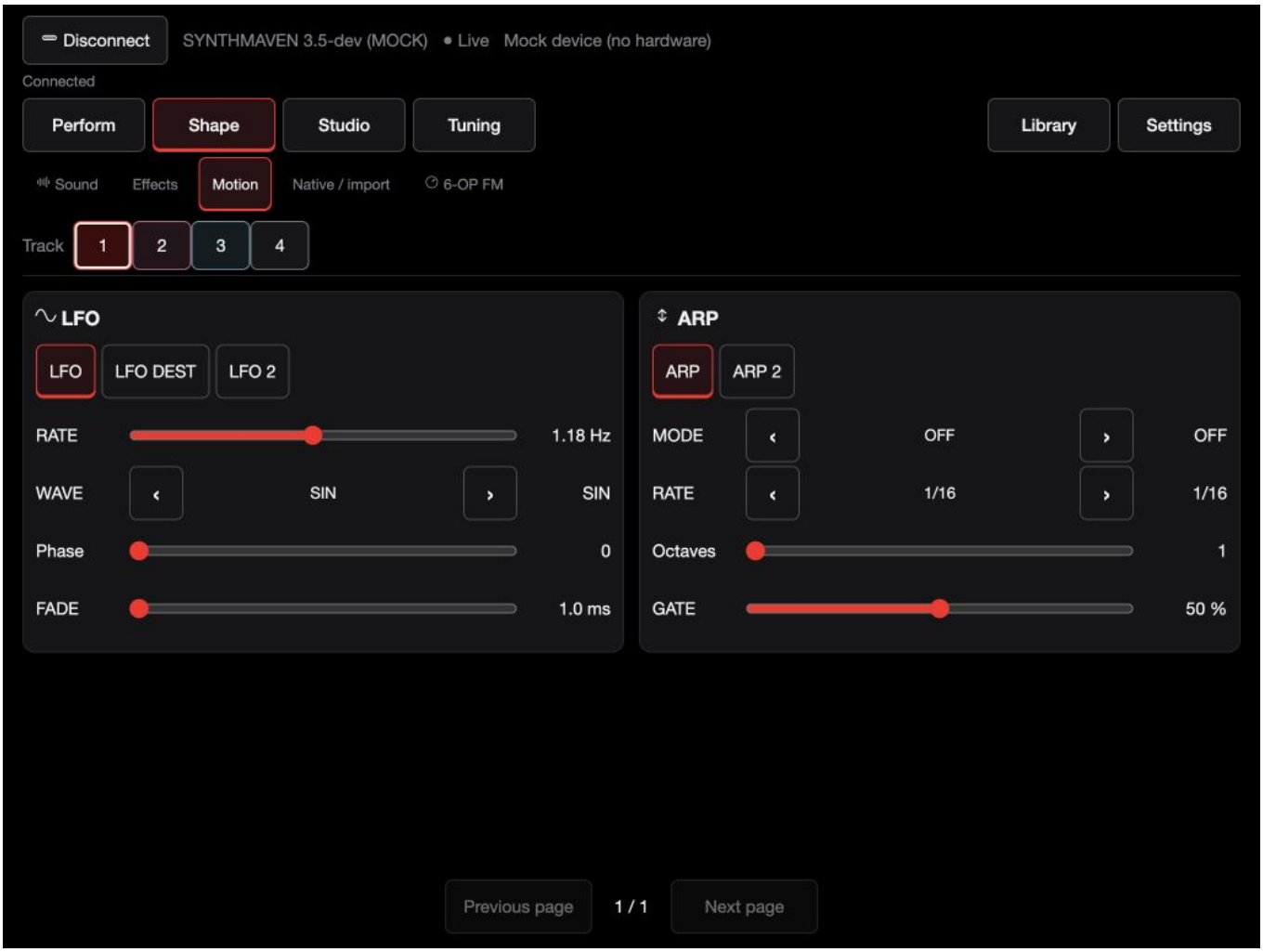
Effects - shared web and iPad layout.

1. Select the effect slot and adjust its parameters.

2. For shared audio effects, select Send FX 1, 2 or 3, then Routing and Sends.

3. Raise the selected part's send and shared return to hear the effect.

LFO and arpeggiator

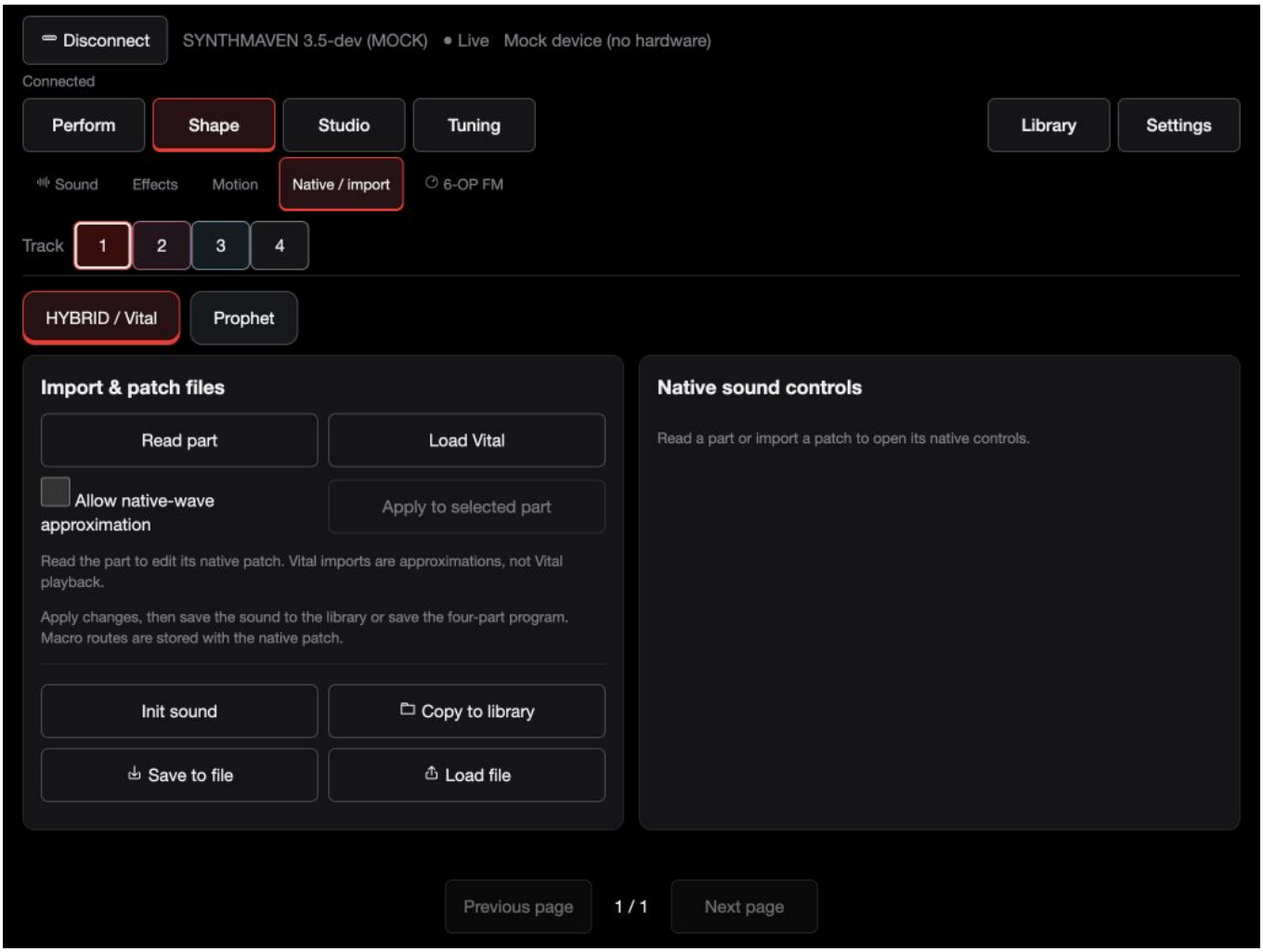


LFO and arpeggiator - shared web and iPad layout.

1. Use LFO, LFO DEST and LFO 2 to set rate, waveform, destinations and synchronization.

2. Use ARP and ARP 2 for pattern, range, gate, swing, probability and hold.

Native patch import



Native patch import - shared web and iPad layout.

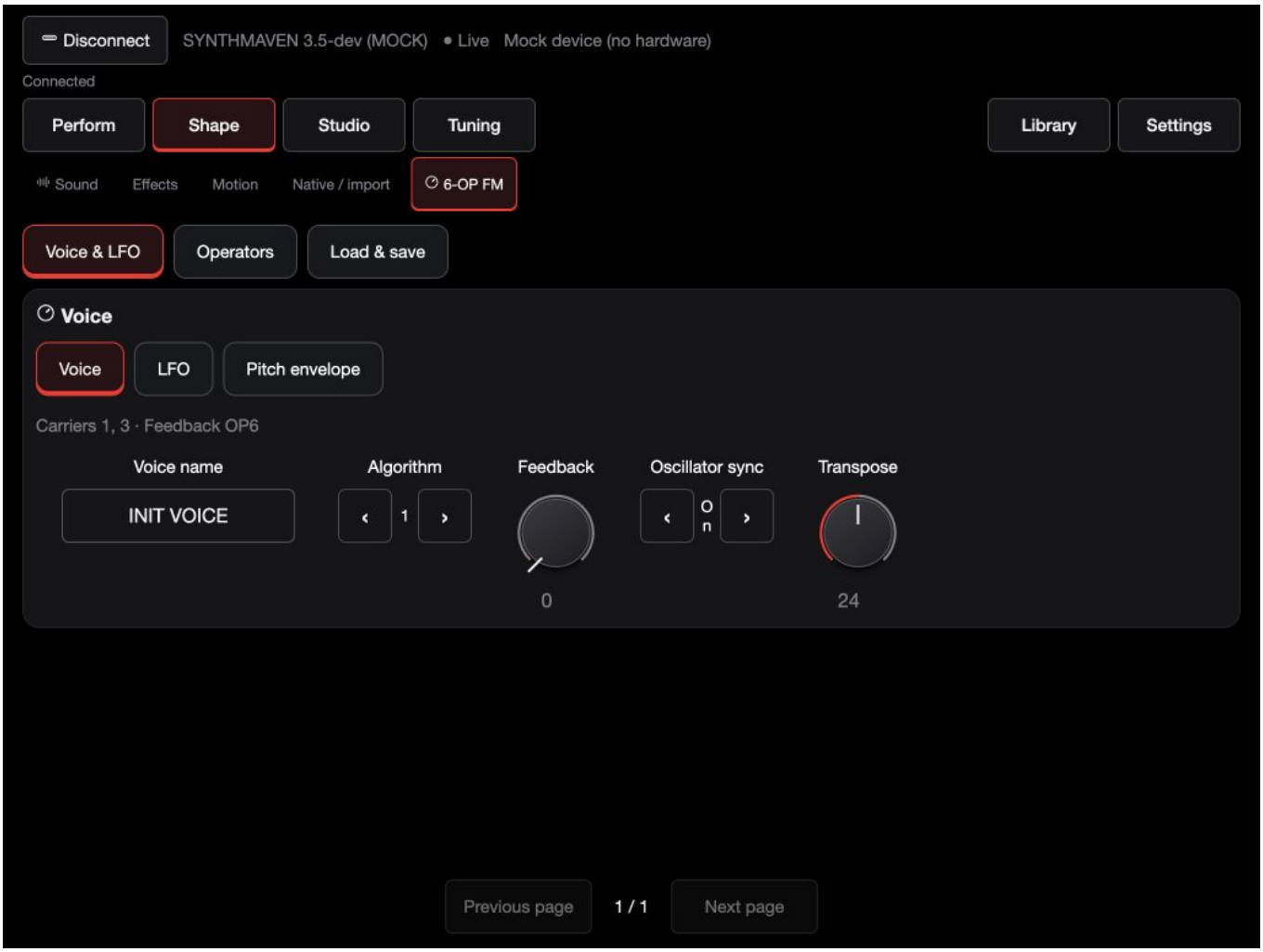
1. Select HYBRID / Vital or Prophet.

2. Read the selected part before editing, or import a compatible patch.

3. Review approximations, apply to the part and save.

4. Vital import is translation, not the Vital synthesizer running on the FM-1.

DX7 / FM6



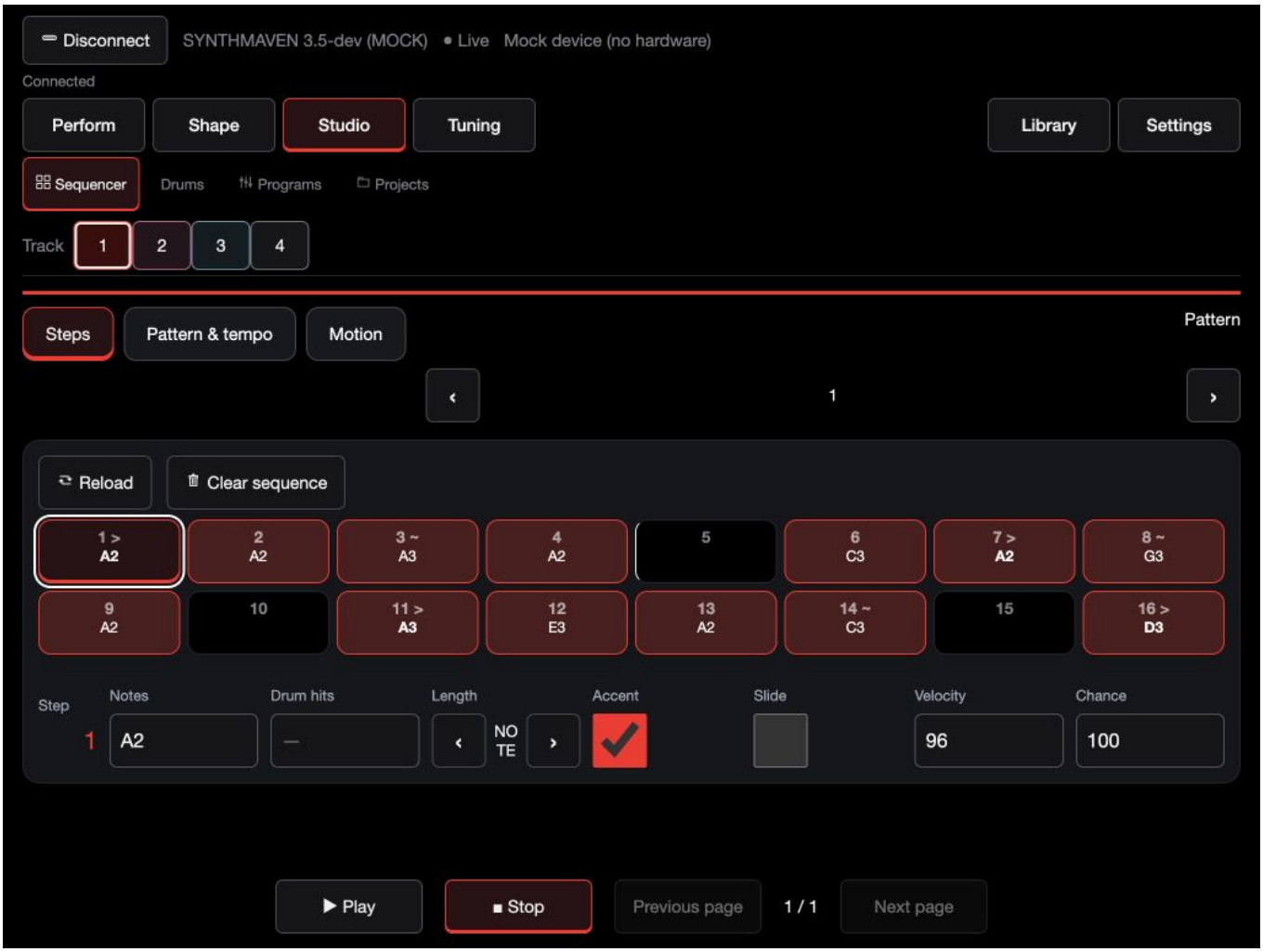
DX7 / FM6 - shared web and iPad layout.

1. Use Voice & LFO, Operators and Load & save.

2. Select an operator to edit its frequency and envelope.

3. Import a DX7 SysEx file, choose a voice, send it to the selected part and save the program.

Sequencer



Sequencer - shared web and iPad layout.

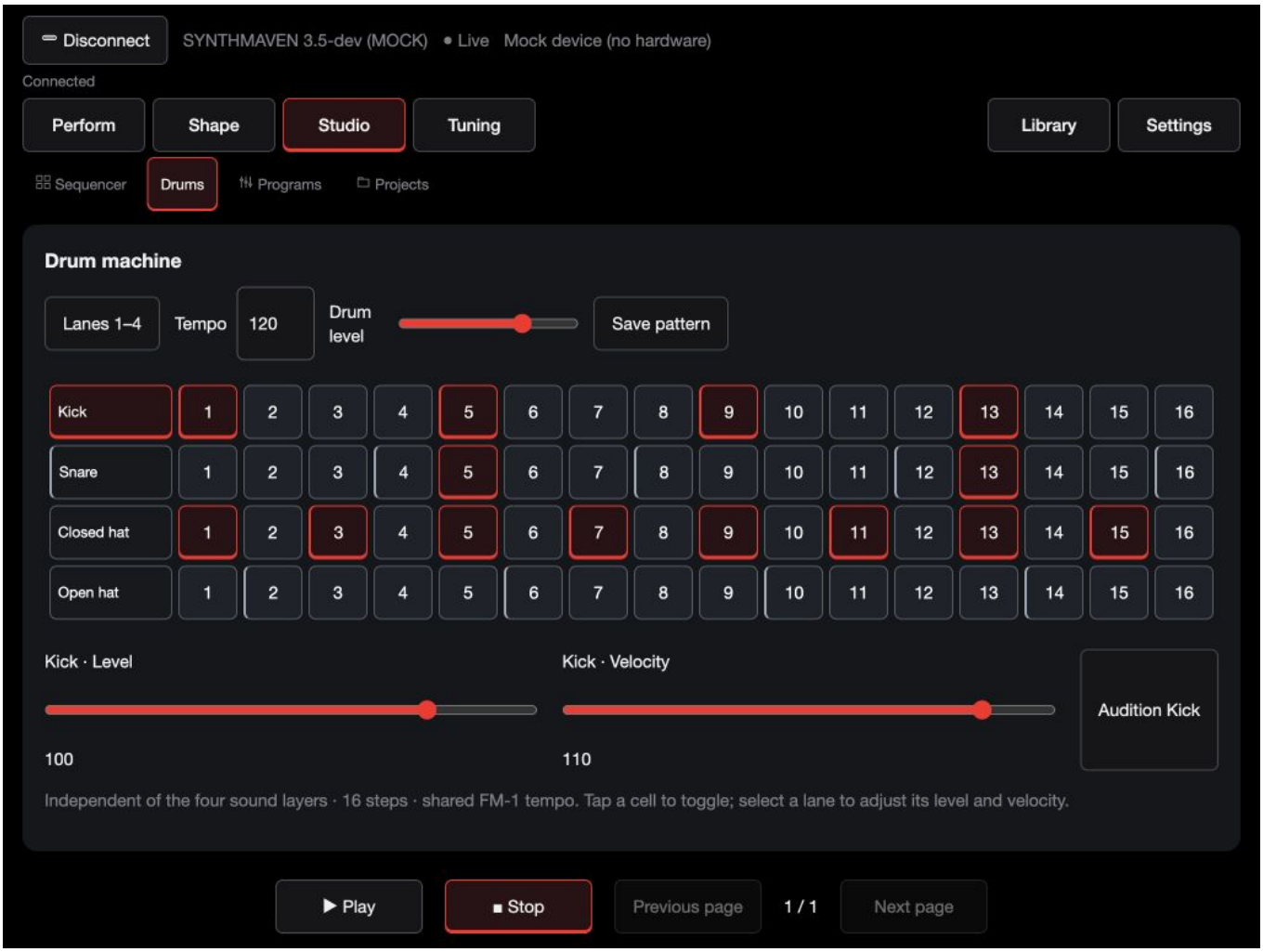
1. Select a track and pattern.

2. Tap a step, enter its note(s), length, velocity and chance, and toggle accent or slide.

3. Pattern & tempo sets loop behavior; Motion edits automation.

4. Play and Stop operate the connected FM-1.

Independent drum machine



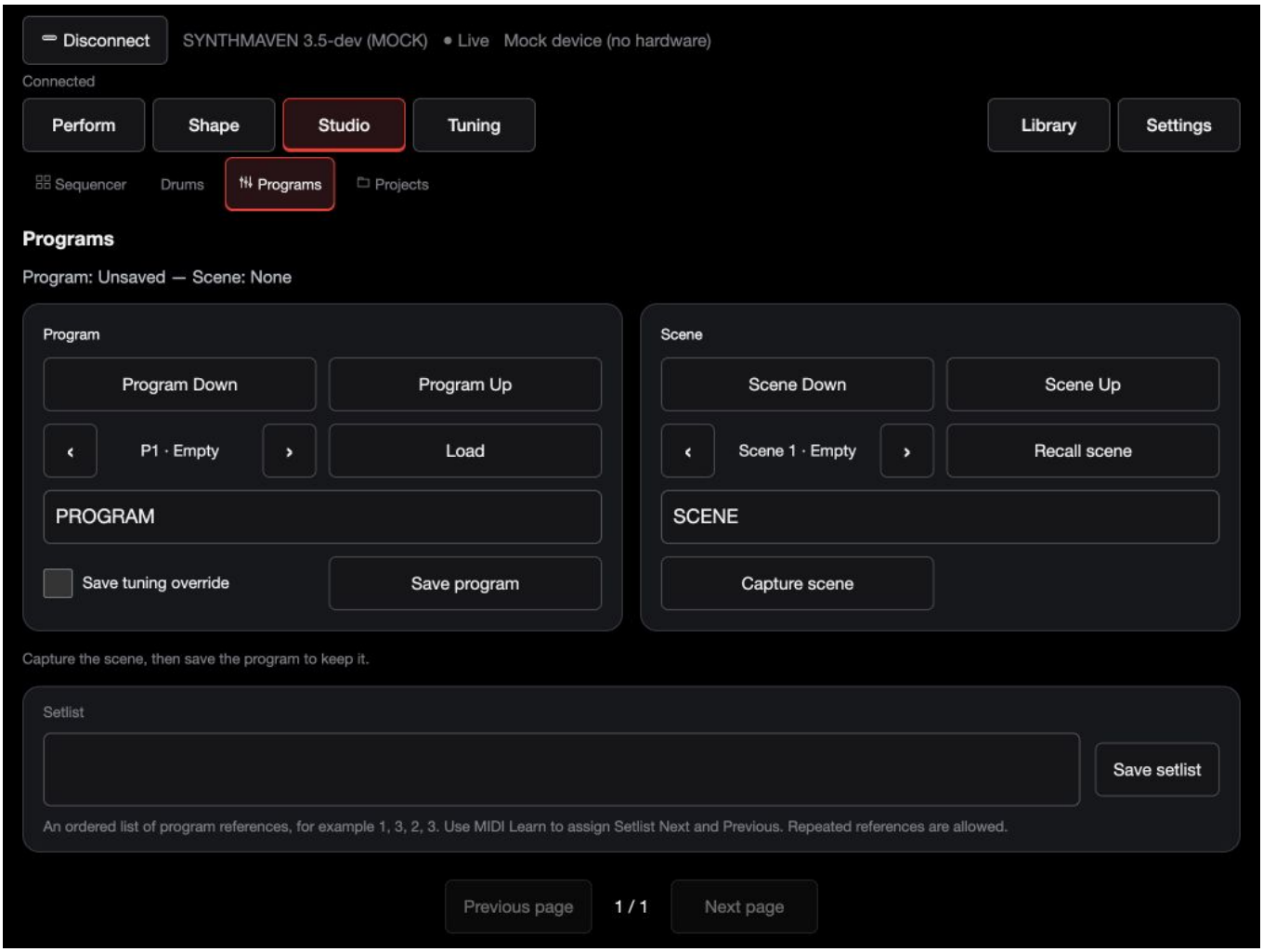
Independent drum machine - shared web and iPad layout.

1. Tap steps to enable hits.

2. Select a lane for level, velocity and audition; switch lane banks for the other four sounds.

3. Set tempo and overall drum level, use Play/Stop, and Save pattern to retain edits.

Programs, scenes and setlists



Programs, scenes and setlists - shared web and iPad layout.

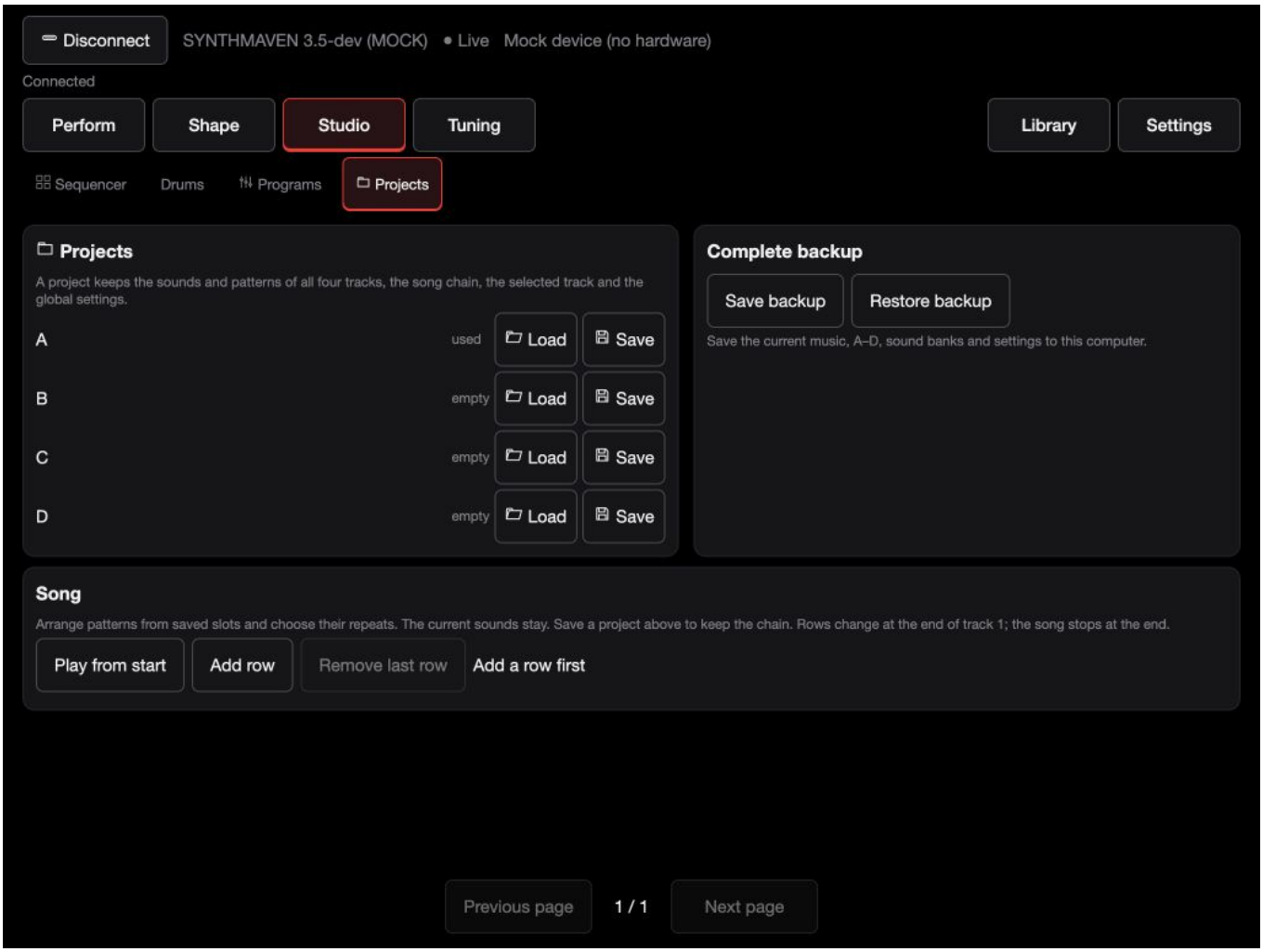
1. Select P1-P8, name the program and Save program.

2. Enable Save tuning override only when this program should replace global tuning.

3. Capture each scene, then save the program.

4. Enter an ordered setlist such as 1, 3, 2, 3 and Save setlist.

Projects and full backup



Projects and full backup - shared web and iPad layout.

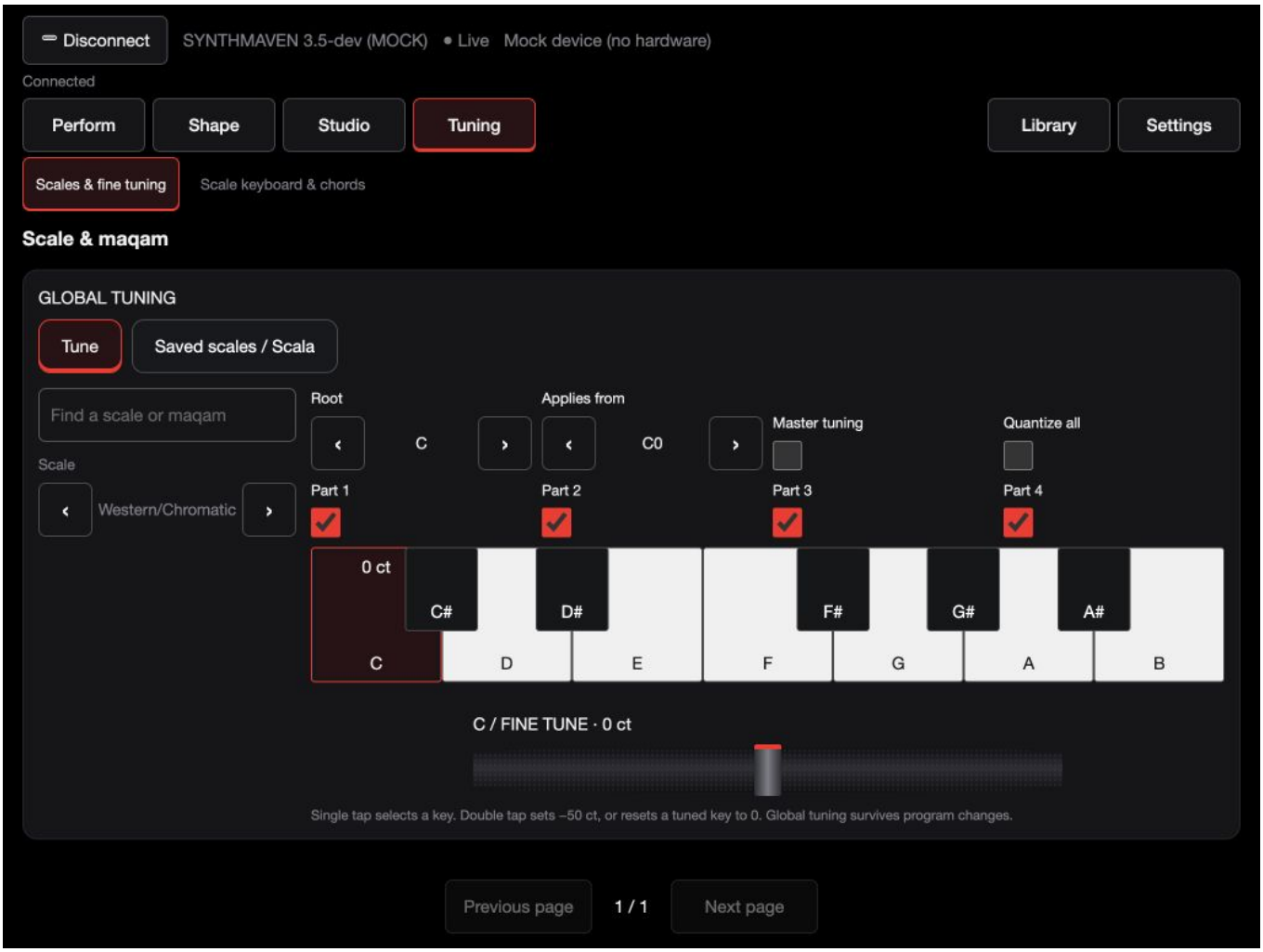
1. Save an A-D project to preserve sounds, patterns and arrangement.

2. Load recalls that project.

3. Save backup downloads a full archive; Restore backup restores one.

4. Add song rows to arrange patterns and repeats, then save the project.

Global microtuning



Global microtuning - shared web and iPad layout.

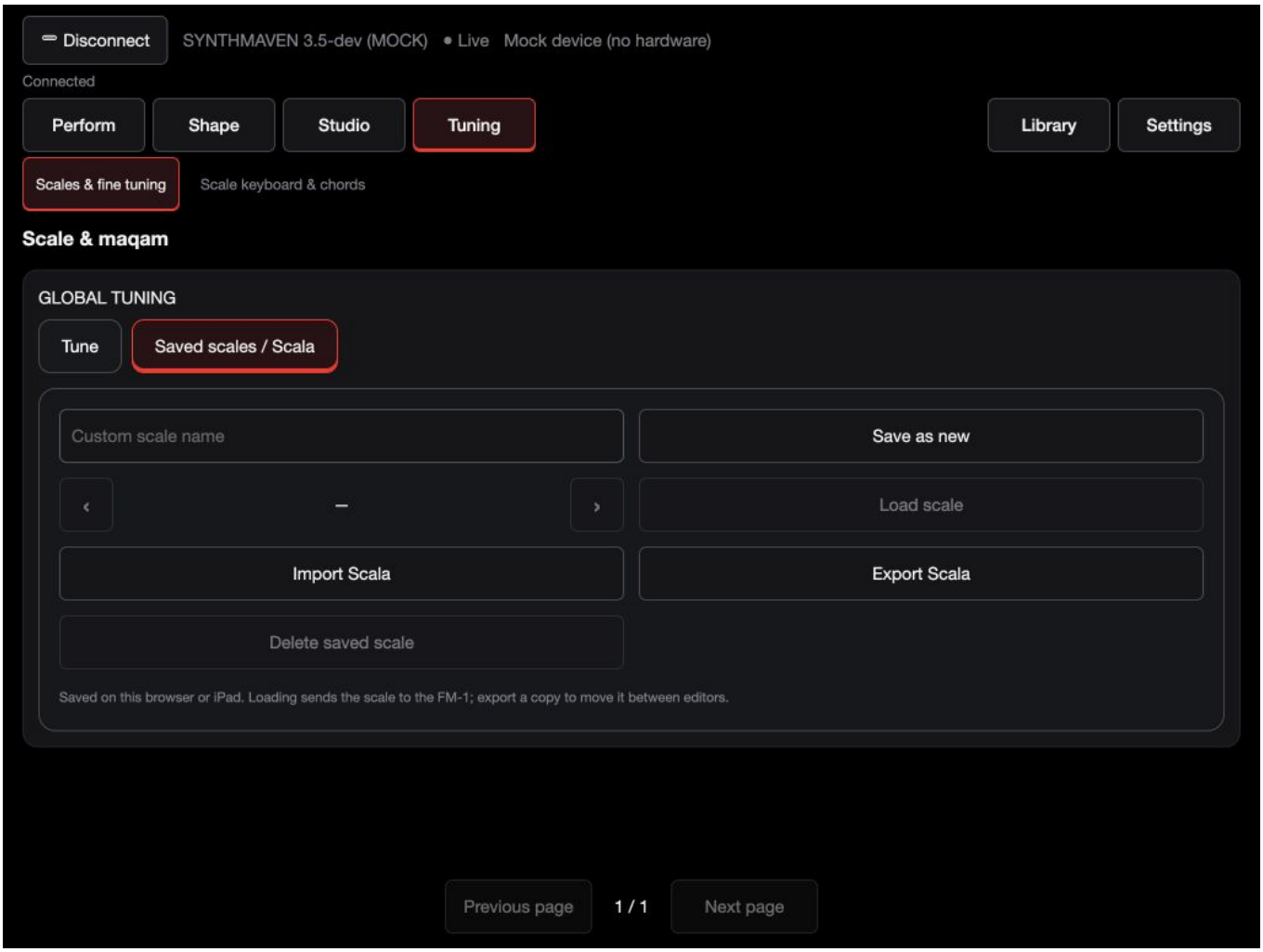
1. Select scale, root and the first affected octave.

2. Enable Master tuning and the required parts.

3. Quantize all constrains notes to the selected scale.

4. Tap a key and drag the wheel in 1-cent steps; double-tap sets -50 cents or resets a tuned key.

Saved scales and Scala



Saved scales and Scala - shared web and iPad layout.

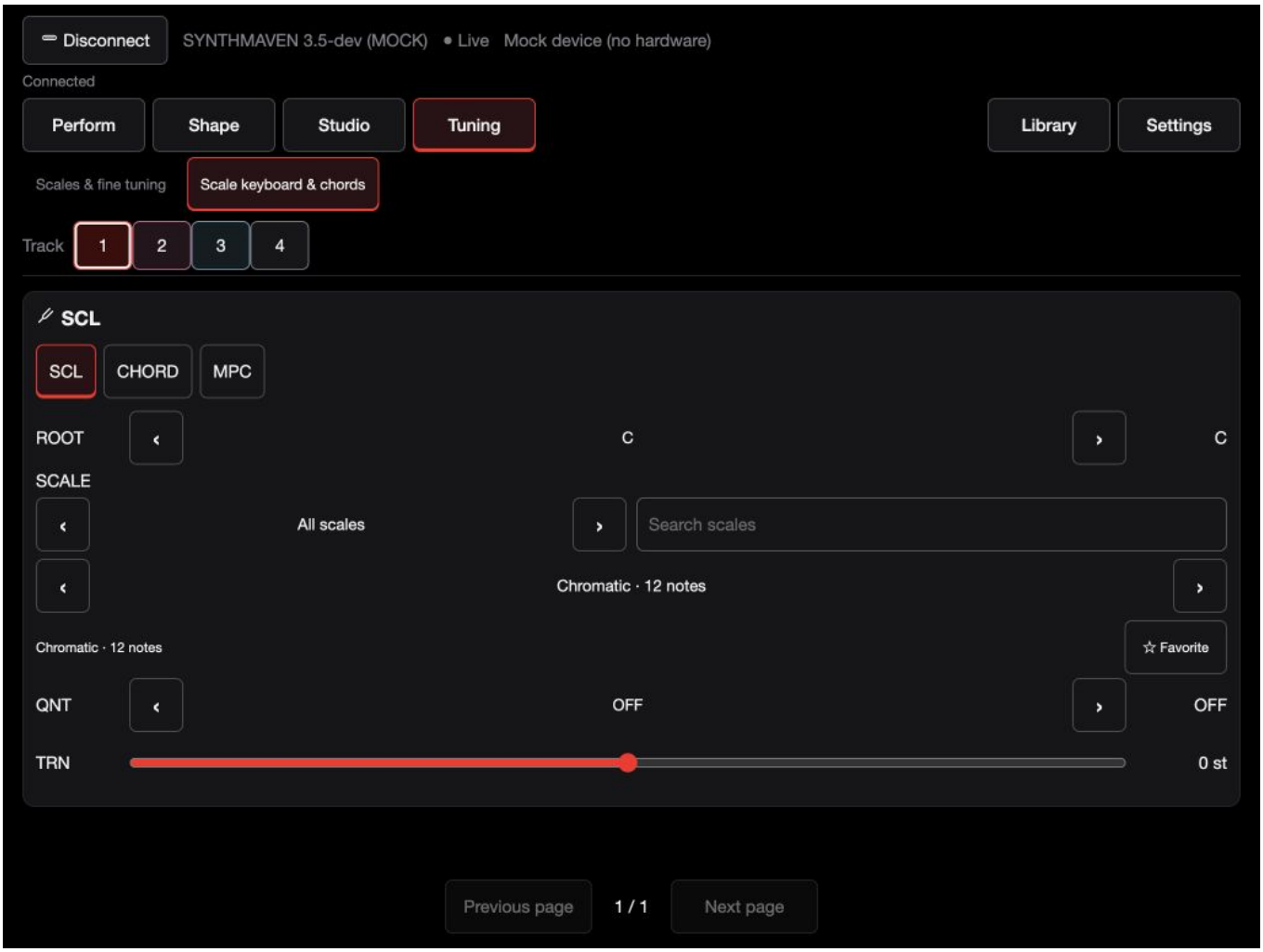
1. Enter a custom-scale name and Save as new.

2. Select a saved scale and Load scale to send it to the FM-1.

3. Import Scala reads a compatible .scl file; Export Scala makes a portable copy.

4. Named scales are local to this browser/app.

Scale keyboard and chords

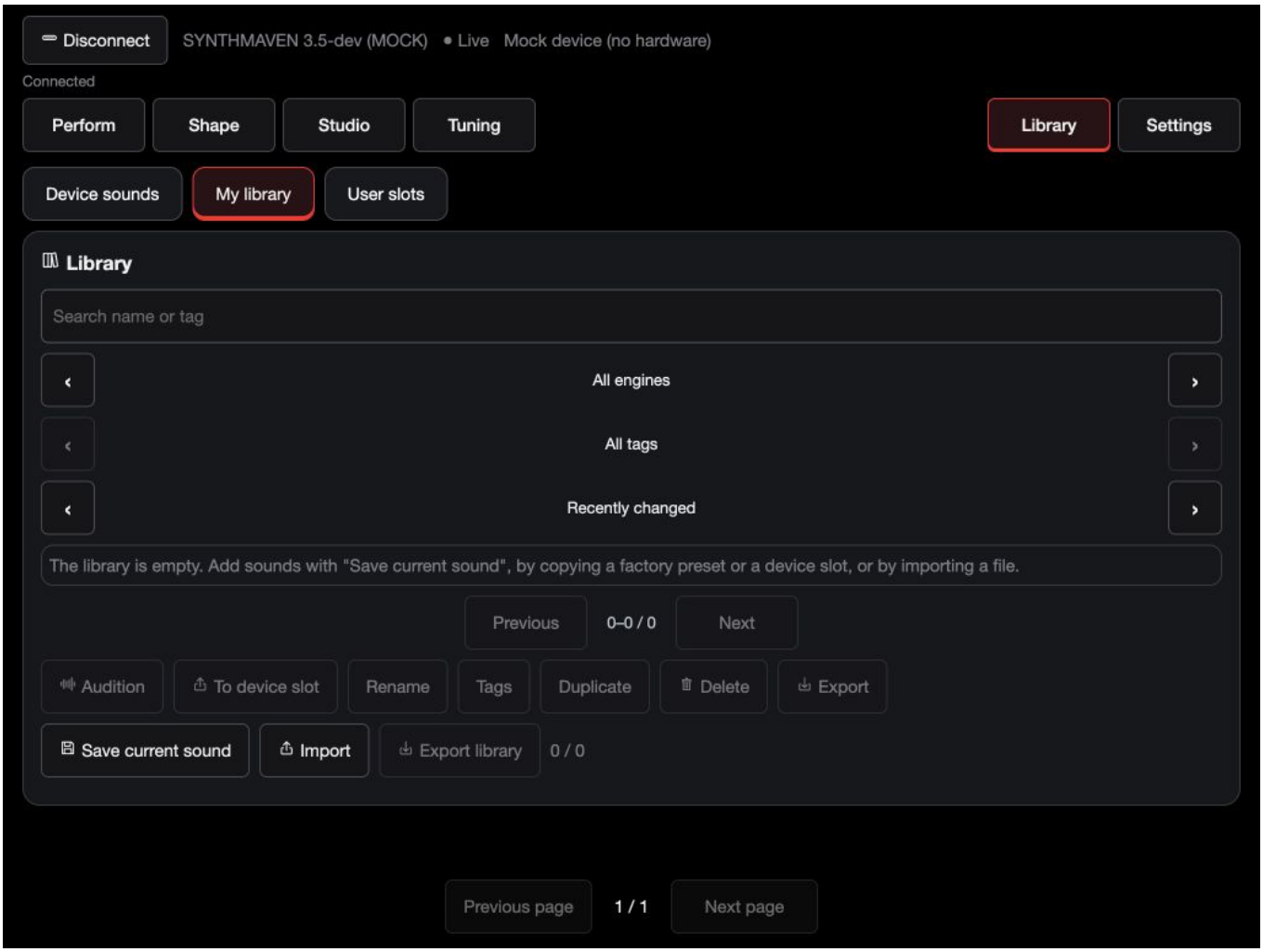


Scale keyboard and chords - shared web and iPad layout.

1. Use the SCL card pages for quantization and transpose, CHORD for voicing and strum, and MPC for scale-degree pads.

2. Master microtuning takes precedence for scale-aware pitch generation when enabled.

Library



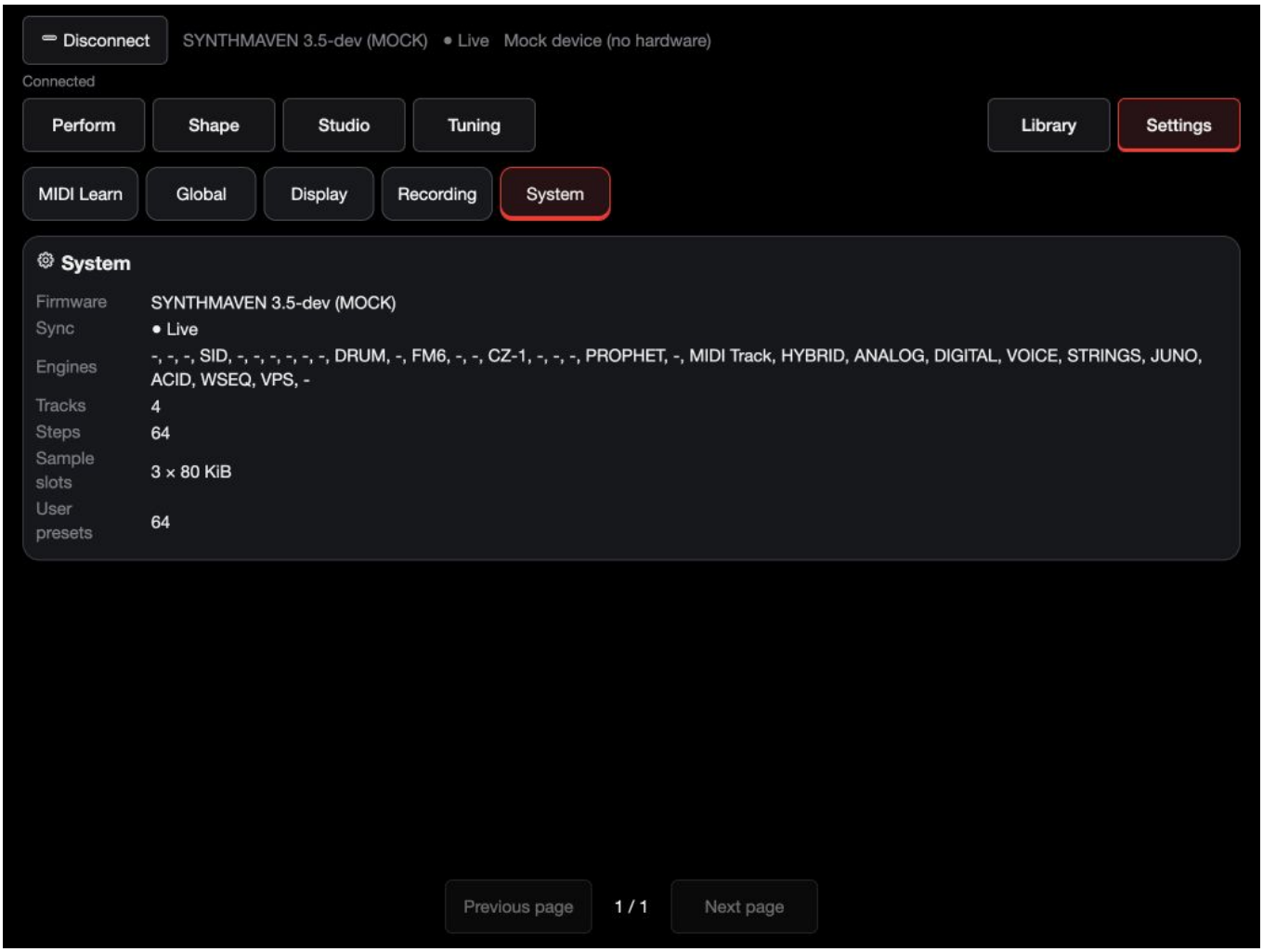
Library - shared web and iPad layout.

1. Use Device sounds to browse factory sounds, My library for local named entries, and User slots for device storage.

2. Save current sound adds a local entry.

3. Export library backs it up; To device slot writes a selected entry to the FM-1.

Settings



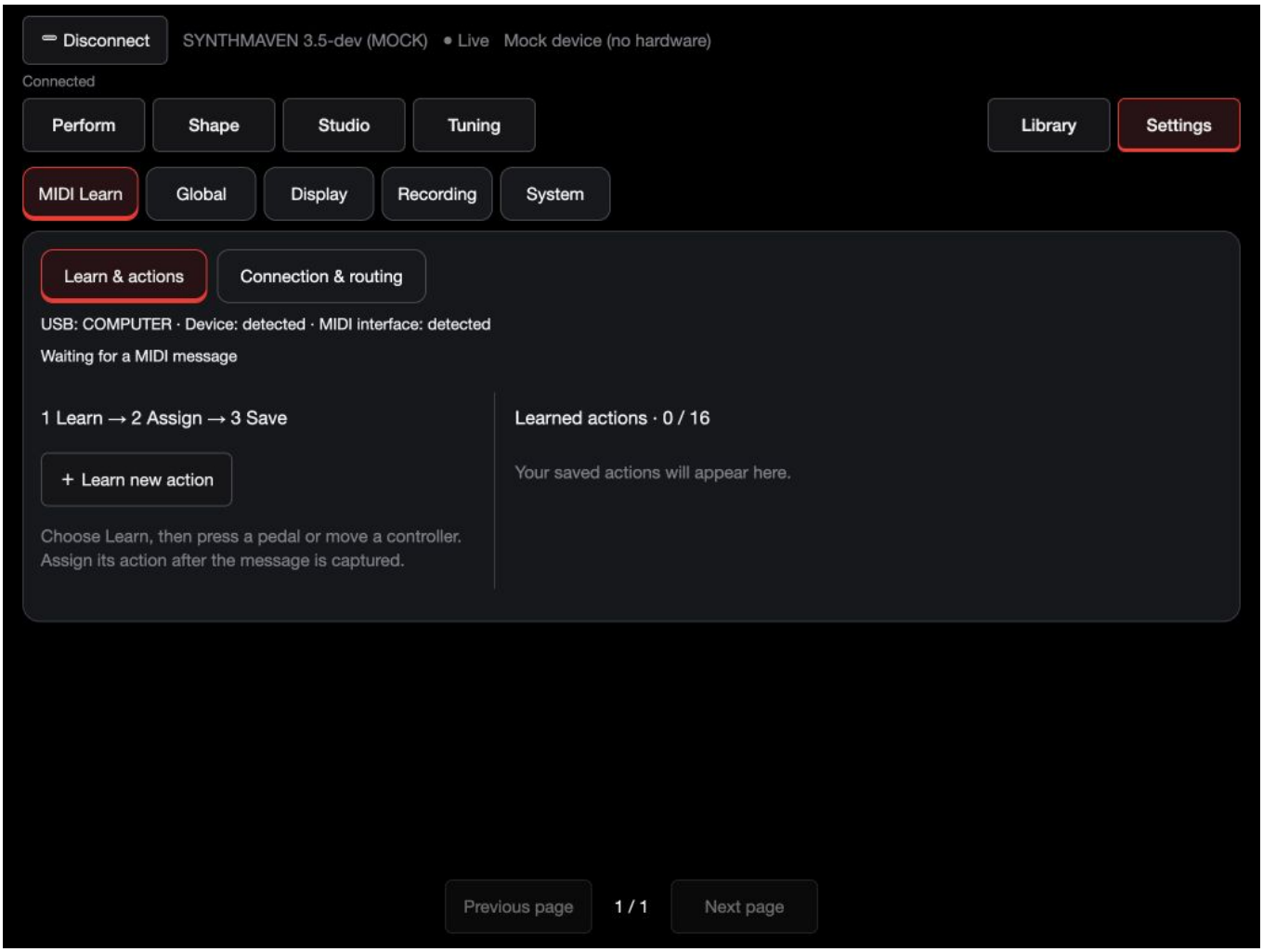
Settings - shared web and iPad layout.

1. Choose Global, Display, Recording or System.

2. Global settings affect device behavior; System identifies the connected build and capabilities.

3. Screen pages expose additional controls without scrolling.

MIDI Learn



MIDI Learn - shared web and iPad layout.

1. Choose Learn new action and operate the controller.

2. After the message is captured, assign the action or parameter and Save action.

3. The learned grid supports Relearn and Unlearn.

4. Set repeat suppression only as needed to ignore duplicate messages without losing deliberate taps.